



جامعة المنصورة كلية الحاسبات والمعلومات



اللائحة الداخلية لمرحلة البكالوريوس بنظام الساعات المعتمدة

٢٠٢٦



جامعة المنصورة **كلية الحاسبات والمعلومات**

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بنظام الساعات المعتمدة

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

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أولاً: مقدمة عامة

كلية الحاسبات والمعلومات - جامعة المنصورة هي واحدة من أول أربع كليات للحاسبات والمعلومات تم أنشائها في الجامعات المصرية بهدف إعداد خريجين متخصصين في مجالات الحاسبات وتقنيات تكنولوجيا المعلومات مؤهلين للعمل والتعامل مع سوق العمل وما يتطلبه من قدرات تنافسية لسد الحاجة المتنامية للمتخصصين في هذه المجالات.

١-١: نبذة عن تاريخ الكلية

- أنشأت الكلية بموجب القرار الجمهوري رقم ٤١٩ لسنة ١٩٩٥.
 - بدأت الدراسة في الكلية في العام الجامعي ١٩٩٦/١٩٩٧ وكانت تضم قسمي نظم المعلومات وعلوم الحاسب.
 - وفي العام ٢٠٠٧ تم تطوير لائحة الكلية وتم إنشاء قسم جديد هو قسم تكنولوجيا المعلومات.
- ولما كان التطور متلاحقاً في مجال تكنولوجيا الحاسبات والمعلومات فقد حرصت الكلية على تحديث وتطوير لائحته الداخلية لتواكب التغيرات التكنولوجية. ولكي تتوافق برامجها الدراسية مع المعايير العالمية فقد تم تصميمها باستخدام نظام الساعات المعتمدة وفقاً للإطار العام المرجعي للمتطلبات الأكاديمية، مما يحقق المرونة للدارسين ويدعم تبني خطط دراسية تتوافق مع التطور التكنولوجي.

٢-١: رؤية الكلية

تحقيق التميز والريادة محلياً وعالمياً في مجال علوم الحاسبات ونظم وتكنولوجيا المعلومات وبناء مجتمع رقمي من خلال التنمية المستدامة والشراكة المجتمعية الفاعلة والتدويل.

٣-١: رسالة الكلية

كلية الحاسبات والمعلومات - جامعة المنصورة هي مؤسسة تعليمية وبحثية وخدمية تتبنى الابتكار تؤهل معرفياً ومهارياً خريجاً قادراً على المنافسة في سوق العمل وخدمة المجتمع طبقاً للمستجدات العالمية مع الالتزام بمبادئ الأصالة.

٤-١: أهداف الكلية

تبنت الكلية مجموعة من الغايات والأهداف الإستراتيجية التي تمثل تعبيراً عن رؤيتها وترجمة حقيقية لرسالتها ونتاجاً للتحليل الرباعي الذي شمل جميع قطاعات الكلية والذي أبرز الواقع

الحقيقي لها. وتمثل الغايات الأهداف الممكنة التي تطمح إليها الكلية. وفيما يلي عرض لغايات الكلية والأهداف الإستراتيجية المرجوة من كل منها.

الغاية الأولى: الارتقاء بالقدرة المؤسسية؛ ووجود نظام من الحوكمة وفقا لمعايير الجودة.

- هدف ١: التنمية المستمرة لقدرات أعضاء هيئة التدريس ومعاونيهم.
- هدف ٢: تنمية الموارد البشرية بالجهاز الإداري ورفع كفاءة الأداء الإداري.
- هدف ٣: تحديث وصيانة المعامل وقاعات التدريس.
- هدف ٤: رفع كفاءة استخدام تكنولوجيا المعلومات بالكلية.
- هدف ٥: دعم الحرية الأكاديمية والحماية الفكرية في إطار منظومة القيم.
- هدف ٦: دعم للحوكمة والمتابعة والمراجعة الداخلية.

الغاية الثانية: الارتقاء بالفعالية التعليمية بالكلية وفقا لمعايير الجودة.

- هدف ١: دعم العملية التعليمية لخريج متميز قادر على المنافسة لتحقيق رؤية ورسالة الكلية.
- هدف ٢: وضع نظام لإدارة العملية التعليمية بالكلية.
- هدف ٣: تطوير التعليم الإلكتروني والتدريب الصيفي ومشاريع التخرج.
- هدف ٤: إنشاء برامج وآليات لدعم الخدمات الطلابية.

الغاية الثالثة: الارتقاء بجودة البحث العلمي والمشروعات البحثية؛ مع الانفتاح على ثقافات العالم ومصادر المعرفة المختلفة.

- هدف ١: وجود خطة بحثية للكلية.
- هدف ٢: تنمية مصادر تمويل العملية البحثية وتنوعها وضمان استدامتها.
- هدف ٣: تحديث البنية التحتية وميكنة الخدمات.
- هدف ٤: الارتقاء بالبحث العلمي بالكلية.
- هدف ٥: دعم التبادل الطلابي وأعضاء هيئة التدريس والتعاون الدولي.

الغاية الرابعة: تنمية المشاركة المجتمعية؛ وتعظيم دور الكلية في خدمة المجتمع وتنمية البيئة.

- هدف ١: زيادة المشاركة الفعالة للكلية في خدمة المجتمع المحلي لخدمة محافظة الدقهلية.
- هدف ٢: آليات لتفعيل أنشطة مركز الخدمة العامة.
- هدف ٣: آليات لتفعيل خدمات الوحدات التي تتبع قطاع خدمة المجتمع.

٥-١: سمات خريجي كلية الحاسبات والمعلومات

يجب أن يكون خريجي كلية الحاسبات والمعلومات قادرين على التالي:

- ١. تطبيق النظريات والمبادئ الأساسية لتطبيقات الحوسبة والمعلومات.
- ٢. دمج وتقييم أدوات وتسهيلات الحوسبة.
- ٣. تطبيق المعرفة في الرياضيات والعلوم.

٤. تصميم نظم الحوسبة، بما يشمل من مكونات وعمليات، لتلبية الاحتياجات المطلوبة ضمن قيود واقعية.
 ٥. استغلال تقنيات ومهارات وأدوات الحوسبة الحديثة اللازمة لممارسات الحوسبة والمعلومات.
 ٦. إظهار المسؤوليات المهنية، والاهتمامات الأخلاقية، والاجتماعية، والثقافية.
 ٧. استخدام ومقارنة وتقييم مجموعة من التقنيات والنظريات والأساليب الرسمية وغير الرسمية لتطوير تطبيقات الحوسبة والمعلومات.
 ٨. الاهتمام والتعامل مع الآثار الفردية والاجتماعية والبيئية والتنظيمية والاقتصادية لتطبيقات الحوسبة والمعلومات.
 ٩. تنفيذ خطط العمل مع الحد الأدنى من الإشراف.
 ١٠. التواصل بشكل فعال.
 ١١. امتلاك المعرفة والمهارات التي تتطلبها صناعة الحوسبة والمعلومات.
 ١٢. الانخراط في التعلم الذاتي مدى الحياة والبحث في مجال الحوسبة والمعلومات.
 ١٣. تلبية متطلبات أصحاب العمل المحتملين.
- بالإضافة الى السمات العامة التي يتمتع بها جميع خريجي كلية الحاسبات والمعلومات مع اختلاف برامجهم الدراسية فإن لكل برنامج دراسي السمات الخاصة التي تميز خريجه.

ثانياً: الأحكام العامة

مادة ١: الأقسام العلمية

تتكون الكلية من الأقسام العلمية التالية:

- ١- قسم علوم الحاسب
- ٢- قسم نظم المعلومات
- ٣- قسم تكنولوجيا المعلومات

ويجوز أن تنشأ بالكلية أقسام أخرى وفقاً لأحكام قانون تنظيم الجامعات.

مادة ٢: الدرجات العلمية

يمنح مجلس الجامعة بناءً على اقتراح مجلس كلية الحاسبات والمعلومات درجة البكالوريوس في الحاسبات والمعلومات وذلك في أحد التخصصات العلمية الآتية:

- ١- علوم الحاسب
- ٢- نظم المعلومات
- ٣- تكنولوجيا المعلومات
- ٤- هندسة البرمجيات
- ٥- المعلوماتية الطبية
- ٦- الذكاء الاصطناعي
- ٧- الشبكات والأمن السيبراني

ويجوز إنشاء أو استحداث تخصصات أو برامج جديدة حسب متطلبات سوف العمل بناءً على طلب مجلس الكلية واعتماد مجلس الجامعة والمجلس الأعلى للجامعات.

مادة ٣: شروط القبول والتحويل

- أ. تقبل كلية الحاسبات والمعلومات الطلاب الحاصلين على الثانوية العامة (شعبة علمي رياضة/ علمي علوم) أو ما يعادلها من الشهادات الأخرى، في نفس العام الدراسي للالتحاق بالكلية، وذلك طبقاً لقواعد القبول المعتمدة من المجلس الأعلى للجامعات ووزارة التعليم العالي.
- ب. على الطلاب الحاصلين على الثانوية العامة علمي علوم اجتياز لمقرر رياضيات ٢ (Math2) الخاص بطلاب علمي رياضة في الثانوية العامة وذلك في ضوء قرار المجلس الأعلى للجامعات في هذا الشأن .
- ج. يجوز قبول الطلاب الوافدين من الحاصلين على الثانوية العامة أو ما يعادلها في كل عام دراسي بحسب ترتيب درجاتهم وفقاً لترشيحات الإدارة العامة لقبول ومنح الطلاب الوافدين بوزارة التعليم العالي. ويحدد مجلس الجامعة بناءً على اقتراح مجلس الكلية مقابل تكلفة الخدمات التعليمية بالإضافة إلى الرسوم والمصروفات الدراسية المقررة بالنسبة لهؤلاء الطلاب.
- د. يمكن قبول الطلاب كمستمعين في مقرر ما - دون الحصول على درجة جامعية - وذلك طبقاً للقواعد التي يحددها مجلس الكلية ويتم اعتمادها من مجلس الجامعة.
- هـ. يجوز لمجلس الكلية قبول طلاب من الحاصلين على درجة البكالوريوس من الكليات الأخرى للدراسة بالكلية وذلك بعد أخذ رأي مجالس الأقسام المختصة وبشرط ألا تقل مدة الدراسة بالكلية عن أربعة فصول دراسية على الأقل، وبشروط قبول، طبقاً للقواعد التي ينظمها المجلس الأعلى للجامعات.
- و. **الطلاب المحولون:** يمكن قبول الطلاب المحولون من كليات أو برامج أخرى (نظام ساعات معتمدة) بعد إجراء مقاصة لما درسه الطالب ويتم احتسابه ضمن متطلبات الحصول على الدرجة العلمية، ولا يتم احتسابها ضمن المعدل التراكمي للطلاب (CGPA). على أن يتم احتساب المعدل التراكمي للطلاب بناءً على ما تم دراسته في البرنامج بالكلية (طبقاً لمتطلبات البرنامج).
- ز. يتم حساب المعدل التراكمي لطلاب محولين لبرنامج يعمل بنظام الساعات المعتمدة، بناءً على ما درسه الطالب من مقررات تحت مظلة هذا البرنامج فقط، ولا يعتد بما درسه في أي برنامج بنظام الساعات المعتمدة في أي جامعة أخرى أو نظام الفصلين الدراسيين سواء بجامعة أو أي جامعة أخرى. وفي جميع الأحوال يتم إجراء مقاصة لما درسه ليتم حسابه ضمن متطلبات الحصول على الدرجة دون احتسابها في حساب المعدل التراكمي للطلاب.

جدول ١: التقديرات المكافئة عند التحويل من النظام الفصلي إلى نظام الساعات المعتمدة.

إلى نظام الساعات المعتمدة		من النظام الفصلي
التقدير	عدد النقاط	النسبة المئوية للدرجة
A+	٤,٠	أكثر من ٩٥٪
A		٩٠٪ إلى أقل من ٩٥٪
A-	٣,٧	٨٥٪ إلى أقل من ٩٠٪
B+	٣,٣	٨٠٪ إلى أقل من ٨٥٪
B	٣,٠	٧٥٪ إلى أقل من ٨٠٪
B-	٢,٧	٧١٪ إلى أقل من ٧٥٪
C+	٢,٣	٦٨٪ إلى أقل من ٧١٪
C	٢,٠	٦٥٪ إلى أقل من ٦٨٪
C-	١,٧	٦٠٪ إلى أقل من ٦٥٪
D+	١,٣	٥٥٪ إلى أقل من ٦٠٪
D	١,٠	٥٠٪ إلى أقل من ٥٥٪
F	٠,٠	أقل من ٥٠٪

مادة ٤: لغة التدريس

الدراسة باللغة الإنجليزية ويمكن تدريس مقررات متطلبات الجامعة باللغة العربية على أن يكون الامتحان بنفس لغة تدريس المقرر.

مادة ٥: نظام الدراسة

أ. تعتمد الدراسة على نظام الساعات المعتمدة وتكون الساعة المعتمدة هي وحدة قياس دراسية لتحديد وزن المقرر الدراسي.

ب. تتكون السنة الدراسية من فصلين نظاميين – فصل الخريف وفصل الربيع – وفصل دراسي صيفي اختياري للطالب، ويتم عقده طبقاً لإمكانيات الكلية وبمقابل مادي يحدده مجلس الكلية ويوافق عليه مجلس الجامعة.

- ج. مدة الفصول النظامية ١٦-١٧ أسبوع تتضمن فترة عقد الامتحانات وطبقا للخريطة الزمنية المعتمدة من المجلس الأعلى للجامعات ومجلس الجامعة.
- د. الفصل الدراسي الصيفي هو فصل مكثف حيث تتضاعف عدد الساعات الدراسية الأسبوعية للمقرر، مدته من ٧-٨ أسابيع تتضمن فترة عقد الامتحانات.
- هـ. المدة القصوى للدراسة في الكلية هي ثماني سنوات دراسية (ستة عشر فصل دراسي نظامي خريف وربيع) مع ملاحظة عدم احتساب فصول إيقاف القيد التي تمت الموافقة عليها من قبل مجلس الكلية ومجلس الجامعة ضمن الفصول المسموح بها.
- و. يتطلب الحصول على البكالوريوس أن يجتاز الطالب بنجاح ١٤٢ ساعة معتمدة.
- ز. تعريف الساعة المعتمدة: أن يحصل الطالب على عدد ساعات معتمدة طبقا لعدد ساعات الاتصال في الأسبوع، وتحسب ساعة الاتصال كما في الجدول رقم ٢:

جدول ٢: قيمة الساعة المعتمدة.

الساعة المعتمدة	مدة الدرس	الدرس
١	ساعة	المحاضرة
١	ساعتين أو ثلاث ساعات	حصص التمارين
١	ساعتين أو ثلاث ساعات	حصص المعامل

مادة ٦: التعليم الهجين

- أ. يجوز لمجلس الكلية بعد أخذ رأي مجلس القسم / المجلس الأكاديمي للبرنامج المختص وحسب طبيعة المقررات الدراسية أن يقرر تدريس مقرر أو أكثر بنمط التعليم الهجين، بحيث تكون الدراسة في المقرر بنسبة تتراوح من ٦٠٪ إلى ٧٠٪ وجها لوجه ونسبة تتراوح من ٣٠٪ إلى ٤٠٪ بنظام التعليم عن بعد، وعلى أن يتم عرض ذلك على مجلس شئون التعليم والطلاب بالجامعة للموافقة عليه ورفعها إلى مجلس الجامعة لاعتماده. وفي جميع الأحوال يجري الامتحان النهائي داخل الحرم الجامعي.
- ب. يجوز لمجلس الكلية بعد أخذ رأي مجلس القسم / المجلس الأكاديمي للبرنامج المختص وحسب طبيعة المقررات الدراسية أن يقرر عقد الامتحانات إلكترونيا على أن يتم عرض ذلك على مجلس شئون التعليم والطلاب بالجامعة للموافقة عليه ورفعها إلى مجلس الجامعة.

مادة ٧: نسب توزيع المواد

توزع الساعات المعتمدة طبقاً للنسب الواردة بالمعايير المرجعية القياسية لقطاع الحاسبات كما في الجدول رقم ٣.

جدول ٣: نسبة توزيع الساعات المعتمدة على المواد.

النسبة وفقاً للمعايير الأكاديمية	النسبة %	عدد الساعات	البند
١٠-٨	٨,٤٥	١٢	متطلبات الجامعة
١٨-١٦	١٦,٩	٢٤	الرياضيات والعلوم الأساسية
٢٨-٢٦	٢٧,٤٦	٣٩	متطلبات الكلية
٣٠-٢٨	٢٩,٥٨	٤٢	متطلبات تخصص (إجبارية)
٥-٣	٢,٨٢	٤	تدريب
٥-٣	٤,٢٢	٦	مشروع
٩٦-٨٤	٨٩,٤٤	١٢٧	مجموع فرعي
١٦-٤	١٠,٥٦	١٥	متطلبات تخصص (اختيارية)
١٠٠	١٠٠	١٤٢	المجموع

* يجوز تعديل تلك النسب في حالة ظهور إصدار جديد للمعايير المرجعية القياسية لقطاع الحاسبات.

مادة ٨: عدد الساعات اللازمة للانتقال بين المستويات المختلفة

- عند التحاق الطالب بالكلية، يقيد الطالب المستجد في المستوى الأول.
- الطالب المحول من كلية مناظرة يتم قيده في المستوى الموازي لعدد الساعات التي اجتازها في الكلية المحول منها بناءً على مقاصة وبشرط ألا يكون قد تجاوز نسبة ٤٠٪ من عدد الساعات عند التحويل.
- يمكن للطالب الانتقال من مستوى لآخر في بداية كل فصل دراسي وذلك بحسب الساعات التي اجتازها كما هو في وارد في الجدول رقم ٤.

د. في المسار العام ينقل الطالب من المستوى الثاني الي المستوى الثالث في أحد البرامج (علوم الحاسب / نظم المعلومات / تكنولوجيا المعلومات) طبقاً لقواعد التشعيب التي يحددها مجلس الكلية.

جدول رقم ٤: المستويات الدراسية.

عدد الساعات المعتمدة التي اجتازها الطالب بنجاح	تعريف الطالب	المستوى الدراسي
من صفر وحتى اقل من ٣٠	Freshman	الأول
من ٣٠ وحتى اقل من ٦٦	Sophomore	الثاني
من ٦٦ وحتى اقل من ٩٩	Junior	الثالث
من ٩٩ وحتى اقل من ١٤٢	Senior	الرابع

مادة ٩: الإرشاد الأكاديمي

أ. يحدد مجلس الكلية لكل مجموعة من الطلاب مرشداً أكاديمياً من أعضاء هيئة التدريس يقوم بمهام الإرشاد الأكاديمي للطالب ويساعده على اختيار المقررات التي يدرسها والتسجيل فيها وتوجيهه طوال فترة الدراسة بالكلية.

ب. رأي المرشد الأكاديمي استشاري، أي أن الطالب هو المسؤول مسؤولية تامة عن المقررات التي يقوم بالتسجيل فيها بناء على رغبته.

مادة ١٠: تسجيل المقررات

- يقوم مجلس الكلية بتحديد الحد الأدنى لعدد الطلاب للتسجيل في أي مقرر طبقاً لإمكانات الكلية.
- قبل بداية كل فصل دراسي بفترة تحددها مجلس الكلية يقوم الطالب بتسجيل المقررات الدراسية التي يختارها.
- يسمح للطالب بدراسة المقررات المختلفة والتسجيل في المستويات الأعلى بناء على قيامه باجتياز المقررات المطلوبة كمتطلبات للمقررات الأعلى ولا يتم تسجيل الطالب في مقرر أعلى إلا إذا نجح في متطلباته ويجوز بناء على موافقة مجلس القسم/ مجلس البرنامج النوعي المعني التجاوز عن هذا الشرط إذا كان الطالب سبق له دراسة المقرر ولم يجتازه أو يكون مسجلاً في المتطلب في نفس الوقت.

■ عدد ساعات التسجيل في الفصول الدراسية المختلفة

أ. بالنسبة للفصول النظامية (خريف وربيع):

● الحد الأقصى للساعات المسجلة:

- ١٨ ساعة معتمدة للطلاب المستجدين من غير المحولين من كليات مناظرة في الفصل الدراسي الأول لالتحاقهم بالكلية.
- ١٨ ساعة معتمدة للطلاب الحاصلين على CGPA في بداية الفصل الدراسي أعلى من أو يساوي ٢ وأقل من ٣.
- ٢١ ساعة معتمدة للطلاب الحاصلين على CGPA في بداية الفصل الدراسي أعلى من أو يساوي ٣.
- ٢١ ساعة معتمدة في حالة تخرج الطالب في ذات العام على أن يتم الاستفادة بها مرة واحدة فقط في فصل دراسي واحد.
- ١٥ ساعة معتمدة للطلاب الحاصلين على CGPA في بداية الفصل الدراسي أعلى من أو يساوي ١ وأقل من ٢.
- ١٢ ساعة معتمدة للطلاب الحاصلين على CGPA في بداية الفصل الدراسي أقل من ١.
- بالإضافة للساعات المذكورة أعلاه يسمح للطلاب الراغبين في تسجيل مقرر واحد إضافي حاصلين به سابقا على تقدير غير مكتمل.
- الحد الأدنى للساعات المعتمدة للتسجيل ١٢ ساعة ويجوز التجاوز عن الحد الأدنى إذا كان عدد الساعات المتبقية لتخرج الطالب المطروحة أقل من ١٢.

ب. بالنسبة للفصل الصيفي

- الفصل الدراسي الصيفي اختياري للطالب. ووفقا للقواعد التي يحددها مجلس الكلية.
- الحد الأقصى للساعات المسجلة للطلاب هو ٩ ساعات معتمدة.
- يحق لمجلس الكلية إضافة شروط إضافية للالتحاق بأي مقرر في الفصل الدراسي الصيفي.
- يحدد مجلس الجامعة بناء على اقتراح مجلس الكلية مقابل الخدمات التعليمية التي يدفعها الطالب مقابل كل ساعة معتمدة في الفصل الصيفي كما يحدد مجلس الجامعة بناء على اقتراح مجلس الكلية المقابل المادي للسادة أعضاء هيئة التدريس ومعاونيهم القائمين بالتدريس خلال الفصل الصيفي.

■ مشروع التخرج

- أ. يسجل الطالب لمقرر المشروع ٦ ساعات معتمدة بعد اجتاز ٦٥٪ من متطلبات التخرج (٩٣ ساعة معتمدة) على الأقل على مدى فصلين دراسيين.
- ب. يتم تسجيل المشروع في فصلين دراسيين نظاميين متتاليين (خريف ثم ربيع) أو (ربيع ثم خريف).

■ التدريب الميداني

- أ. يتم تنظيم تدريب ميداني صيفي لمدة ست أسابيع (بواقع اربع ساعات معتمدة) على أن يكون الطالب قد اجتاز على الأقل ٥٠ ٪ من عدد الساعات المعتمدة اللازمة للتخرج (٧١ ساعة معتمدة).
- ب. يتم تخصيص عضو هيئة تدريس كمسئول عن التدريب مع عدد من أعضاء الهيئة المعاونة وذلك لمتابعة المشاركين في التدريب سنويا ووضع التقييم الخاص بكل منهم طبقا للمعايير التي يتم تحديدها من قبل مجلس الكلية ووفقاً للقواعد الإدارية والمالية المنظمة للتدريب الميداني المعتمدة من مجلس الجامعة.
- ج. مقرر التدريب الميداني لا يحتسب ضمن المجموع التراكمي للطالب وإنما هو من متطلبات التخرج وبالتالي فإن تقدير الطالب به هو (ناجح/راسب). ويتم تسجيله بالسجل الأكاديمي للطالب، وبيان الدرجات والتقدير للخرج.

مادة ١١ : أوقات التسجيل والحذف والإضافة والانسحاب

- أ. يحدد مجلس الكلية الحد الأدنى لعدد الطلاب للتسجيل في كل مقرر.
- ب. مع بداية كل فصل دراسي يقوم الطالب بتسجيل المقررات الدراسية التي يختارها، وذلك في الأوقات التي يحددها مجلس الكلية قبل بدء انتظام الدراسة، طبقا للجدول رقم ٥.
- ج. يجوز للطالب بتوصية من المرشد الأكاديمي أن يحذف أو يضيف مقرا أو أكثر بما لا يخل بالعبء الدراسي المنصوص عليه في المادة ١٠ طبقا للتوقيتات في جدول ٥.

جدول ٥: أوقات التسجيل والحذف والإضافة والانسحاب.

البيان	التوقيت في الفصل النظامي	التوقيت في الفصل الصيفي
تسجيل المقررات	نهاية الأسبوع الثاني	نهاية الأسبوع الأول
الحذف والإضافة (التسجيل المتأخر)	نهاية الأسبوع الرابع	نهاية الأسبوع الثاني
الانسحاب من المقررات	نهاية الأسبوع السابع	نهاية الأسبوع الرابع

مادة ١٢: الانسحاب من المقرر الدراسي

- أ. يجوز للطالب بعد تسجيل المقررات التي اختارها أن ينسحب من مقرر أو أكثر قبل بداية الامتحان النهائي بفترة يحددها مجلس الكلية، بحيث لا يقل عدد الساعات المسجلة للطالب عن الحد الأدنى للتسجيل في الفصل الدراسي الواحد، وفي هذه الحالة لا يعد الطالب راسباً في المقررات التي انسحب منها ويحتسب له تقدير "منسحب" "W"، ويتوجب عليه إعادة المقرر كاملاً دراسة وامتحان.
- ب. إذا انسحب الطالب من مقرر أو أكثر بعد الفترة المحددة لذلك دون عذر قهري يقبله مجلس الكلية يحتسب له تقدير راسب "F" في المقررات التي انسحب منها.

مادة ١٣: الانسحاب من الفصل الدراسي أو إيقاف القيد

- أ. يجوز للطالب إيقاف قيده (الانسحاب الكلي من الفصل الدراسي) بتوصية من المرشد الأكاديمي وموافقة مجلس القسم / المجلس الأكاديمي للبرنامج ومجلس الكلية (وذلك وفقاً للضوابط التي يحددها مجلس الكلية ومجلس الجامعة وبعد أقصى فترة يحددها مجلس الكلية قبل عقد الامتحانات النهائية).
- ب. يطبق قرار مجلس الجامعة بشأن ضوابط وقف القيد وسداد القيمة المحددة بالخدمات المستندية.
- ج. الطالب المستمر الذي لم يحضر- للتسجيل خلال فترة التسجيل والحذف والإضافة في الفصول النظامية يعتبر منسحباً من الفصل الدراسي.
- د. لا يجوز أن يتجاوز عدد الفصول النظامية التي ينسحب منها الطالب عن أربعة فصول دراسية متتالية أو ست فصول منفصلة (حتى ولو كان بعضها متتالي).

مادة ١٤: قواعد الانتظام في الدراسة

يجب أن يلتزم جميع الطلاب بالقواعد الجامعية التالية:

- أ. يتم دفع الرسوم الدراسية والخدمات التعليمية عند بدء التسجيل. ويحدد مجلس الكلية مقابل الخدمات التعليمية ويتم اعتمادها من مجلس الجامعة.
- ب. لا يسمح للطالب بالتسجيل بالمستوى الأعلى أو معرفة نتيجته إلا بعد سداد جميع الرسوم الدراسية للمستوى الأدنى. وعند التخرج لا يتم منح الطالب شهادات التخرج إلا بعد إخلاء طرفه وسداد جميع الرسوم الدراسية كاملة.
- ج. على الطالب أن يخطر إدارة البرنامج بأي تغيير في عنوان مراسلته أو وسيلة الاتصال به.

مادة ١٥: حضور الطالب

يتطلب دخول الطالب الامتحان النهائي تحقيق نسبة حضور لا تقل عن ٧٥٪ من المحاضرات والتمارين في كل مقرر . وإذا تجاوزت نسبة غياب الطالب – دون عذر مقبول – في أحد المقررات ٢٥٪ يكون لمجلس الكلية حرمانه من دخول الامتحان النهائي، ويعتبر راسبًا ويسجل له (محروم) في درجة المقرر. ويعتبر (منسحب) في المقرر إذا قدم عذرًا يقبله مجلس الكلية (في الفترة التي يحددها مجلس الكلية).

مادة ١٦: تقديرات المقررات الدراسية

- أ. نظام الساعات المعتمدة يعتمد على أن الوحدة الأساسية هي المقرر الدراسي وليس السنة ويكون نظام التقييم على أساس التقدير في كل مقرر بنظام النقاط والذي يحدد طبقاً للجدول رقم ٦.
- ب. يتم حساب التقدير لكل مقرر على أساس الدرجات التي يحصل عليها الطالب خلال دراسته لهذا المقرر (الأنشطة – تقييم منتصف الفصل الدراسي – التقييم المستمر- التقييم النهائي) ويوضح الجدول ٦ كيفية حساب التقدير وعدد النقاط المناظر من خلال الدرجات التي تم الحصول عليها.
- ج. يجب على الطالب الحصول على الحد الأدنى من التقدير (D) لاجتياز المقرر الدراسي واستخدام النقاط المناظرة في حساب المعدل الفصلي والتراكمي للطالب طبقاً للجدول ٦.

كيفية حساب المعدل التراكمي المجمع (CGPA)

$$Cumulated\ GPA\ (CGPA) = \frac{\sum_{\forall courses} Grade\ Points * Number\ of\ Credit\ hours}{\sum_{\forall courses} Number\ of\ Credit\ hours}$$

مع التقريب لأقرب علامتين عشريتين.

جدول رقم ٦: تقديرات ونقاط المقررات ذات الساعات المعتمدة.

النقاط	التقدير	النسبة المئوية للدرجة
٤,٠	A+	٩٧٪ فأكثر
	A	٩٣٪ - أقل من ٩٧٪
٣,٧	A-	٨٩٪ - أقل من ٩٣٪
٣,٣	B+	٨٤٪ - أقل من ٨٩٪
٣	B	٨٠٪ - أقل من ٨٤٪
٢,٧	B-	٧٦٪ - أقل من ٨٠٪
٢,٣	C+	٧٣٪ - أقل من ٧٦٪
٢,٠	C	٧٠٪ - أقل من ٧٣٪
١,٧	C-	٦٧٪ - أقل من ٧٠٪
١,٣	D+	٦٤٪ - أقل من ٦٧٪
١,٠	D	٦٠٪ - أقل من ٦٤٪
٠,٠	F	أقل من ٦٠٪

مقررات النجاح والرسوب بدون ساعات معتمدة

يمنح الطالب أحد التقديرات في الجدول رقم ٧.

جدول رقم ٧: تقدير مقررات النجاح بدون ساعات.

التقدير	المعنى
P	ناجح
F	راسب
W	منسحب
Abs	غياب عن حضور الامتحان النهائي بدون عذر مقبول من مجلس الكلية
I	غير مكتمل

مادة ١٧ : نظام الامتحانات

- أ. يتم تحديد توزيع الدرجات على مختلف أنماط التقييم في جدول توصيف المقرر ويجوز لمجلس الكلية تعديل توزيع الدرجات لاختلاف طبيعة البرنامج بناء على طلب مجلس القسم/ مجلس البرنامج النوعي المختص على أن يتم إعلان الطلاب قبل بداية الفصل الدراسي.
- ب. الحد الأدنى للنجاح في المقرر الدراسي ٦٠٪ من مجموع درجات المقرر بشرط أن يحصل الطالب على نسبة ٣٠٪ على الأقل من درجة الامتحان النهائي.

المقررات ذات الساعات المعتمدة:

الدرجة العظمى لكل مقرر (عدا المشروع) هي ١٠٠ درجة توزع على النحو التالي :

■ الأعمال الفصلية

تمثل ٤٠٪ من درجة المقرر توزع على النحو التالي: ١٥٪ امتحان منتصف الفصل الدراسي و ٢٠٪ الامتحانات الأخرى التي يجريها أستاذ المقرر بصفة دورية والتطبيقات العملية والأعمال التي يكلف بها الطلاب أثناء الفصل الدراسي والامتحانات العملية (ينظمها مجلس الكلية في بداية كل فصل دراسي) و ٥٪ الامتحانات الشفوية.

■ قواعد تنظيم الامتحانات الشفوية للمقررات

- يطبق نظام الجلسات الشفهية على المقررات المحدد لها امتحان شفهي وفقا لللائحة الداخلية لأعمال الامتحانات ومكافآتها بالجامعة.
- تشكل لجان امتحانية بمعرفة مجلس قسم/ مجلس إدارة البرنامج من أعضاء هيئة التدريس على أن تكون اللجنة من عضوين على الأقل وثلاثة على الأكثر.

● آلية عقد الامتحانات الشفوية:

- يتم تحضير الأسئلة الامتحانية وكتابتها من خلال اللجنة لقياس مدى تحصيل الطالب.
- يتم تقسيم الطلاب المسجلين للمقرر إلى مجموعات بحيث لا تتعدى خمسة طلاب كل مجموعة جلسة أو عدد المسجلين للامتحان.
- يتم إعلان الطلاب بموعد انعقاد الامتحان الشفوي طبقا لجدول الامتحانات الشفوية بعد اعتماده من مجلس القسم/ مجلس إدارة البرنامج.
- تجتمع اللجنة الامتحانية طبقا للموعد المحدد لمقابلة الطلاب، يختار الطلاب أحد الأسئلة ويتم مناقشتهم من خلال اللجنة الامتحانية.

○ يقوم كل عضو هيئة تدريس باللجنة الامتحانية بوضع درجة لكل طالب (بالجلسة) ثم يتم حساب المتوسط.

○ بنهاية الامتحانات يسلم كشف درجات الطلاب بعد توقيعه من اللجنة الامتحانية إلى رئيس القسم/مدير البرنامج للتوقيع عليه وتسليمه إلى الكنترول.

■ قواعد تنظيم الامتحانات العملية للمقررات

○ يطبق نظام الاختبار العملي على المقررات المحدد لها امتحان عملي.

○ يقوم أستاذ المقرر بوضع الأسئلة لجميع المجموعات مع مراعات وضع أسئلة متكافئة في حالة الاختبار بمواعيد مختلفة.

● آلية عقد الامتحانات العملية:

- يتم تحضير الأسئلة الامتحانية وكتابتها طبقا لمخرجات التعلم.
- يتم تقسيم الطلاب المسجلين للمقرر إلى مجموعات طبقا لكثافة المعامل.
- يتم إعلان الطلاب بموعد انعقاد الامتحان العملي طبقا لجدول الامتحانات العملية بعد اعتماده من مجلس القسم/ مجلس البرنامج النوعي.
- يتم توقيع الطلاب على كشوف حضور الاختبار العملي.
- الطلاب المنعقد اختبارهم في نفس الوقت يكون لهم نفس الأسئلة الامتحانية.
- الطلاب المنعقد اختبارهم بأوقات مختلفة يكون لهم اختبار مكافئ يعتمد أستاذ المقرر ويقرر لهم نفس وقت الإجابة.
- يضع أستاذ المقرر آليات واحدة للتقييم.
- يقوم عضو هيئة تدريس باللجنة الامتحانية بوضع درجة لكل طالب بالجلسة.
- يتم توثيق إجابة الطالب إلكترونيا أو ورقيا.
- بنهاية الامتحان يسلم كشف درجات الطلاب بعد توقيعه من اللجنة الامتحانية إلى رئيس القسم للتوقيع عليه وتسليمه إلى الكنترول مرفق معه إجابات الطلاب.
- يتم تحديد زمن الاختبار.
- يجب أن يقيس الاختبار المهارات العملية والتطبيقية للطلاب.

■ الاختبار النهائي التحريري (فيما عدا مقرر المشروع)

- درجة الامتحان النهائي التحريري تمثل ٦٠٪ من درجة المقرر.
- عدد ساعات الاختبار النهائي لكل مقرر ساعتين.

المقررات بدون ساعات معتمدة

- يمكن أن تنقسم درجات هذه المقررات إلى أعمال فصلية واختبار نهائي تحريري وفي هذه الحالة تتبع نفس القواعد بالنسبة للمقررات ذات الساعات المعتمدة بالنسبة لتوزيع الدرجات.
- عدد ساعات الاختبار النهائي لكل مقرر ساعة واحدة.

مقرر المشروع

- الدرجة العظمى للمشروع هي ٢٠٠ درجة توزع على النحو التالي :
 - ٤٠ ٪ للأعمال الفصلية.
 - ٦٠ ٪ للاختبار النهائي الشفوي (عن طريق لجنة المناقشة) والمستندات المطلوبة للتقييم.
- يتم تحديد توزيع درجات الأعمال الفصلية وآلية عقد الامتحان الشفوي للمشروع وكيفية التقييم من مجلس الكلية.

مقرر التدريب

- يتم تحديد عضو هيئة تدريس مسئول عن متابعة الطلاب.
- يحدد مجلس الكلية القواعد التي يتم بناء عليها تقييم الطلاب.

مادة ١٨ : الإنذار الأكاديمي

يحصل الطالب على إنذار أكاديمي إذا كان حصل في أي فصل دراسي نظامي على معدل تراكمي مجمع (CGPA) أقل من ٢، ولا يسجل في الفصل الدراسي التالي إلا في عدد الساعات المحددة في المادة ١٠، ويفصل الطالب إذا حصل على أربع إنذارات أكاديمية متتالية.

مادة ١٩ : ترتيب الطلاب

يتم ترتيب الطلاب بناء على المجموع الكلي للدرجات (وذلك تطبيقاً لقانون تنظيم الجامعات).

مادة ٢٠ : مرتبة الشرف

يمنح الطالب مرتبة الشرف في حالة اجتيازه للمقررات الدراسية التي درسها بكل مستوى دراسي بتقدير لا يقل عن جيد جداً أي بمعدل تراكمي مجمع (CGPA) لا يقل عن ٣,٠ وبشرط ألا تزيد فترة الدراسة عن أربع سنوات (ثمانية فصول دراسية نظامية) وألا يكون الطالب قد رسب أو تم حرمانه في أي مقرر دراسي خلال دراسته بالكلية أو الكلية المحول منها (إن وجد).

مادة ٢١: شروط المنح

- يمنح مجلس الجامعة بناء على طلب مجلس الكلية درجة البكالوريوس في الحاسبات والمعلومات تخصص (علوم الحاسب / نظم المعلومات / تكنولوجيا المعلومات / هندسة البرمجيات / المعلوماتية الطبية / الذكاء الاصطناعي / الشبكات والأمن السيبراني) متى استوفى الطالب متطلبات الحصول عليها :
- أ. أن يجتاز الطالب بنجاح عدد الساعات المعتمدة المنصوص عليها (بمعدل تراكمي مجمع (CGPA) لا يقل عن ٢.
- ب. يتطلب الحصول على البكالوريوس أن يجتاز الطالب بنجاح ١٤٢ ساعة معتمدة.
- ج. اجتياز ما تنص عليه الجامعة من متطلبات للتخرج.
- د. أن يجتاز الطالب بنجاح جميع المقررات بدون ساعات معتمدة والمنصوص عليها باللائحة.
- هـ. دور التخرج: هو الدور الذي يستكمل فيه الطالب متطلبات التخرج ويتم ذلك من خلال أدوار التخرج الثلاثة وهي دور يناير (الفصل الدراسي الخريف) ودور مايو (الفصل الدراسي الربيع) ودور سبتمبر (الفصل الدراسي الصيفي).

مادة ٢٢: إعادة المقررات**أ. إعادة مقرر رسب فيه سابقا**

- إذا رسب الطالب في مقرر فعليه إعادة دراسته والامتحان فيه مرة أخرى فإذا نجح في المقرر بعد إعادة الدراسة تحتسب له الدرجة الفعلية التي حصل عليها وبما لا يزيد عن ٨٨ (أعلى درجة في B+).
- يحسب معدله التراكمي على هذا الأساس مع احتساب عدد ساعات المقرر مرة واحدة.
- تظهر جميع مرات الإعادة والدرجة (أو التقدير) الحاصل عليه الطالب في كل إعادة في السجل الأكاديمي الخاص بالطالب.

ب. إعادة مقرر نجح فيه سابقا وذلك لرفع معدله التراكمي المجمع لتجنب الفصل

- في حالة حصول الطالب على معدل تراكمي مجمع (CGPA) في بداية الفصل الدراسي أقل من ٢ (الطالب تحت الملاحظة الأكاديمية) يجب عليه رفع معدله.
- يحق للطالب المذكور في النقطة السابقة (تحت الملاحظة الأكاديمية) إعادة أي مقرر سبق وأن نجح فيه (بنقاط أقل من ٢) لرفع معدله التراكمي المجمع لتجنب الفصل. وعليه إعادة دراسته والامتحان فيه مرة أخرى، وفي هذه الحالة يحصل على الدرجة الأعلى من الدرجات الحاصل عليها في جميع مرات الإعادة .

- يجب أن يكون المقرر تابع للمستوى المقيد به الطالب أو تابع لمستوى أقل من المستوى المقيد به الطالب بمستوى واحد.
- لا يوجد عدد أقصى لتلك المقررات وإنما يمكن للطالب (الموضوع تحت الملاحظة الأكاديمية) إعادة أي عدد من المقررات سبق وأن نجح بها (بنقاط أقل من ٢) من أجل رفع معدله التراكمي المجمع (CGPA) إلى ٢.
- يحسب معدله التراكمي على هذا الأساس مع احتساب عدد ساعات المقرر مرة واحدة.
- تظهر جميع مرات الإعادة والدرجة (أو التقدير) الحاصل عليه الطالب في كل إعادة في السجل الأكاديمي الخاص بالطالب.

ج. إعادة مقرر نجح فيه سابقاً وذلك لرفع معدله التراكمي المجمع للتحسين

- يحق للطالب الحاصل على معدل تراكمي أكثر من أو يساوي ٢ التحسين لرفع معدله التراكمي المجمع وذلك فقط في حالة عدم استكمال المتطلبات اللازمة للتخرج.
- يحق للطالب إعادة مقرر سبق وأن نجح فيه (بنقاط أكثر من أو يساوي ٢) لرفع معدله التراكمي المجمع، وعليه إعادة دراسته والامتحان فيه مرة أخرى وفي هذه الحالة يحصل على الدرجة الأعلى من الدرجات الحاصل عليها في جميع مرات الإعادة.
- الحد الأقصى لعدد الساعات المسموح بها للتحسين للمقررات التي حصل فيها الطالب على نقاط أكثر من أو يساوي ٢ هو تسع ساعات سواء كان ذلك لمقرر واحد أو عدد من المقررات.
- يجب أن يكون المقرر تابع للمستوى المقيد به الطالب أو تابع لمستوى أقل أو أكثر من المستوى المقيد به الطالب بمستوى واحد.
- يحسب معدله التراكمي على هذا الأساس مع احتساب عدد ساعات المقرر مرة واحدة.
- تظهر جميع مرات الإعادة والدرجة أو التقدير الحاصل عليه الطالب في كل إعادة في السجل الأكاديمي الخاصة بالطالب.

مادة ٢٣: تقدير "غير مكتمل"

- أ. إذا تقدم الطالب بعذر قهري يقبله مجلس الكلية عن عدم حضور الامتحان النهائي لأي مقرر خلال ثلاث أيام عمل من إجراء الامتحان يحتسب له تقدير غير مكتمل "I" في هذا المقرر بشرط أن يكون حاصله على ٦٠ % على الأقل من درجات الأعمال الفصلية، وألا يكون قد تم حرمانه من دخول الامتحانات النهائية.
- ب. إذا تقدم الطالب بعذر قهري يقبله مجلس الكلية عن عدم حضور الامتحان النهائي لأي مقرر خلال ثلاث أيام عمل من إجراء الامتحان، ولم يتحقق له شرط أن يكون حاصله على ٦٠ % على الأقل من

- درجات الأعمال الفصلية، يحتسب له تقدير "منسحب" "W" في المقرر ويتوجب عليه إعادة المقرر كاملاً دراسة وامتحان.
- ج. يتاح للطالب الحاصل على تقدير "غير مكتمل" أداء الامتحان النهائي فقط، وتحتسب الدرجة النهائية للطالب على أساس الدرجة الحاصل عليها في الامتحان النهائي إضافة إلى الدرجة السابق الحصول عليها في الأعمال الفصلية.
- د. يؤدي الطالب الامتحان النهائي خلال نفس العام الدراسي أو العام الدراسي التالي من احتساب المقرر "غير مكتمل" وإلا يتحول التقدير إلى "منسحب" "W" ويتوجب على الطالب إعادة المقرر كاملاً دراسة وامتحان، دون احتساب الدرجة السابق الحصول عليها في الأعمال الفصلية.
- هـ. يتاح للطالب الحصول على تقدير (غير مكتمل) بحد أقصى ٦ مقررات خلال مدة دراسته بالبرنامج وما زاد على ذلك يحصل فيه الطالب على تقدير "منسحب" "W".

مادة ٢٤: الفصل من البرنامج

- أ. يفصل الطالب في الحالات التالية:
- إذا حصل على إنذار أكاديمي في أربعة فصول دراسية نظامية متتالية وكان الطالب مقيد في المستوى الأول، أو ستة فصول دراسية نظامية متتالية وكان الطالب مقيد في مستوى الثاني، أو عشر فصول دراسية متتالية وكان الطالب في مستوى أعلى من الثاني.
 - إذا تجاوز المدة القصوى للدراسة بالكلية الواردة بالمادة ٨٠، ٨٠ مكرر من اللائحة التنفيذية لقانون تنظيم الجامعات وذلك بعد حذف فصول إيقاف القيد التي تمت الموافقة عليها من مجلس الكلية ومجلس الجامعة.
- ب. الطالب المعرض الفصل من الدراسة لأي سبب من المذكور أعلاه يمكن إتاحة فرصة إضافية ونهائية له للتسجيل في فصلين دراسيين نظاميين متتاليين بالإضافة إلى فصل صيفي وذلك لتحقيق شروط التخرج بشرط أن يكون قد اجتاز ما لا يقل عن ٨٠٪ من إجمالي عدد الساعات اللازمة للتخرج وذلك بعد موافقة مجلس الكلية والجامعة.

مادة ٢٥: قواعد النظام الكودي للمقررات الدراسية

يتكون كود المقرر (Course Code) من مجموعة من الأحرف أقصى اليسار تمثل الرمز الكودي للتخصص أو القسم كما في الجدول رقم ٨، يتبع الأحرف رقم مكون من ثلاث خانات ودلالاتها كالتالي: رقم في خانات المئات يمثل المستوى، يدل الرقم (١) على المستوى الأول، والرقم (٢) على المستوى الثاني، والرقم (٣) على المستوى الثالث، والرقم (٤) على المستوى الرابع. ورقم في كل من خانة الآحاد والعشرات يمثل مسلسل المقرر داخل التخصص.

وتشتمل اللائحة على قوائم المقررات الدراسية المختلفة موضحا بها عدد الساعات المعتمدة لكل مقرر، وما يناظرها من الساعات الفعلية من المحاضرات، وما يدعمها من المعامل والتمارين، والتدريب ومشروع التخرج.

جدول رقم ٩: الرمز الكودي للتخصص.

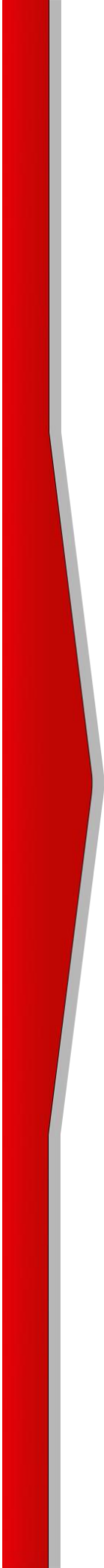
م	التخصص	الرمز الكودي
١	متطلبات الجامعة	UN
٢	العلوم الأساسية	BS
٣	علوم الحاسب	CS
٤	تكنولوجيا المعلومات	IT
٥	نظم المعلومات	IS
٦	هندسة البرمجيات	SE
٧	معلوماتية طبية	MI
٨	الذكاء الاصطناعي	AI
٩	الشبكات والأمن السيبراني	NCS
١٠	تدريب	TR
١١	مشروع	GP

مادة ٢٦: القواعد التأديبية

الطلاب المقيدون بالكلية خاضعون للنظام التأديبي المبين في قانون تنظيم الجامعات ولائحته التنفيذية.

مادة ٢٧: أحكام عامة

- تطبق أحكام قانون تنظيم الجامعات ولائحته التنفيذية وغيره من اللوائح الجامعية الأخرى فيما لم يرد بشأنه نص في هذه اللائحة.
- يسمح للكلية بإضافة أو تعديل قائمة المقررات الاختيارية أو محتويات بعض المقررات بناء على موافقة مجلس الكلية واعتماد مجلس الجامعة.
- يمكن للكلية استخدام الوسائل الإلكترونية في جميع المراسلات المتعلقة بالمسار الأكاديمي والدراسي للطلاب، بما في ذلك الإخطارات والتنبيهات والإنذارات الأكاديمية وأي قرارات أو توجيهات إدارية أو أكاديمية ويعد إرسال أي مراسلة إلى البريد الإلكتروني الأكاديمي الرسمي الخاص بالطلاب بمثابة إشعار رسمي موثق ومعتمد وتترتب عليه كافة الآثار القانونية والأكاديمية المترتبة على الإخطار الورقي



الباب الثاني:

المتطلبات الدراسية للمسار العام

تمهيد :

يدرس الطلاب الملتحقين بكلية الحاسبات والمعلومات المسار العام خلال المستوى الأول والثاني مقررات متطلبات الجامعة ومقررات العلوم الأساسية ومقررات الكلية ويتم تشعب الطلاب المنتقلين إلى المستوى الثالث إلى ثلاث تخصصات (نظم المعلومات وعلوم الحاسب وتكنولوجيا المعلومات) وفيما يلي المتطلبات الدراسية والخطة الدراسية للمستوى الأول والثاني يليها التخصصات الثلاثة.

أولاً: متطلبات الجامعة (١٢ ساعة معتمدة)**أ- متطلبات الجامعة الإجبارية (٩ ساعات معتمدة)**

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Communication Skills, Ethical and Professional Issues	مهارات الاتصال والجوانب الأخلاقية والمهنية	UN101
---	2	2	3	English for Computer Science	لغة إنجليزية لعلوم الحاسب	UN102
---	2	2	3	Technology Entrepreneurship	ريادة الأعمال التكنولوجية	UN201

ب- متطلبات الجامعة الاختيارية (٣ ساعات معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Design Physiology for Computer Science	علم نفس التصميم لعلوم الحاسب	UN104
---	2	2	3	Law for Computer Science	قانون لعلوم الحاسب	UN105
---	2	2	3	Economics for Computer Science	اقتصاد لعلوم الحاسب	UN106
---	2	2	3	Accounting for Computer Science	محاسبة لعلوم الحاسب	UN107
---	2	2	3	Sociology for Computer Science	علم الاجتماع لعلوم الحاسب	UN108

ت- مقررات بدون ساعات معتمدة

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
--	--	1	0	Societal Issues	القضايا المجتمعية	UN103

ثانيا: متطلبات العلوم الأساسية (٢٤ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Mathematics 1 for Computer Science	رياضيات ١ لعلوم الحاسب	BS101
---	2	2	3	Introduction to Computing and Informatics	مقدمة الحاسبات والمعلوماتية	BS102
---	2	2	3	Mathematics 2 for Computer Science	رياضيات ٢ لعلوم الحاسب	BS103
---	2	2	3	Probability and Statistics for Computer Science	احتمالات وتوزيعات إحصائية لعلوم الحاسب	BS104
---	2	2	3	Fundamentals of Electronics and Electrical Circuits	أساسيات الإلكترونيات والدوائر الكهربائية	IT101
---	2	2	3	Data Communication	اتصالات البيانات	IT102
---	2	2	3	Mathematics 3 for Computer Science	رياضيات ٣ لعلوم الحاسب	BS201
---	2	2	3	Operation Research	بحوث العمليات	BS202

ثالثا: متطلبات الكلية (39 ساعة معتمدة)

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Introduction to Programming and Problem Solving	مقدمة للبرمجة وحل المشكلات	CS101
Introduction to Programming and Problem Solving (CS101)	2	2	3	Object-Oriented Design and Programming	تصميم وبرمجة شبيهة التوجه	CS102
Introduction to Programming and Problem Solving (CS101)	2	2	3	Data Structures and Algorithms	هياكل البيانات والخوارزميات	CS201
---	2	2	3	Operating Systems	نظم التشغيل	CS202
Introduction to Programming and Problem Solving (CS101)	2	2	3	Software Engineering	هندسة البرمجيات	CS203
---	2	2	3	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	CS204
Introduction to Programming and Problem Solving (CS101)	2	2	3	Introduction to Web Programming	مقدمة لبرمجة الويب	IS101
Probability and Statistics for Computer Science (BS104)	2	2	3	Data Analysis and Management	تحليل وإدارة البيانات	IS201
Data Structures and Algorithms (CS201)	2	2	3	Introduction to Database Systems	مقدمة لنظم قواعد البيانات	IS202
--	2	2	3	Project Management	إدارة المشروعات	IS203
Fundamentals of Electronics and Electrical Circuits (IT101)	2	2	3	Digital Logic Circuit	دوائر المنطق الرقمية	IT103
Data Communication (IT102)	2	2	3	Computer Networks	شبكات الحاسب	IT201
Computer Networks (IT201)	2	2	3	Internet Technology	تكنولوجيا الإنترنت	IT202

رابعاً: التدريب الصيفي

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
TR201	التدريب الصيفي	Summer Training	4	1	6	اجتياز ٧١ ساعة

خامساً: المشروع

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
GP401	مشروع ١	Project 1	3	1	4	اجتياز ٩٣ ساعة معتمدة
GP402	مشروع ٢	Project 2	3	1	4	GP401

سادسا: الخطة الدراسية المقترحة للمسار العام في المستوى الأول والثاني**١- المستوى الأول (Freshman)**

الفصل الدراسي الأول (Freshman - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS101	رياضيات ١ لعلوم الحاسب	Mathematics 1 for Computer Science	--	3
BS102	مقدمة الحاسبات والمعلوماتية	Introduction to Computing and Informatics	--	3
IT101	أساسيات الإلكترونيات والدوائر الكهربائية	Fundamentals of Electronics and Electrical Circuits	--	3
CS101	مقدمة للبرمجة وحل المشكلات	Introduction to Programming and Problem Solving	--	3
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	--	3
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	--	3
UN103	القضايا المجتمعية	Societal Issues	--	0
المجموع				
18				
الفصل الدراسي الثاني (Freshman – Spring)				
BS103	رياضيات ٢ لعلوم الحاسب	Mathematics 2 for Computer Science	--	3
BS104	احتمالات وتوزيعات إحصائية لعلوم الحاسب	Probability and Statistics for Computer Science	--	3
CS102	تصميم وبرمجة شبيهة التوجه	Object-Oriented Design and Programming	Introduction to Programming and Problem Solving (CS101)	3
IS101	مقدمة لبرمجة الويب	Introduction to Web Programming	Introduction to Programming and Problem Solving (CS101)	3
IT102	اتصالات البيانات	Data Communication	--	3
IT103	دوائر المنطق الرقمية	Digital Logic Circuit	Fundamentals of Electronics and Electrical Circuits (IT101)	3
المجموع				
18				

٢- المستوى الثاني (Sophomore)

الفصل الدراسي الأول (Sophomore - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS201	رياضيات ٣ لعلوم الحاسب	Mathematics 3 for Computer Science	--	3
BS202	بحوث العمليات	Operation Research	--	3
UN201	ريادة الأعمال التكنولوجية	Technology Entrepreneurship	--	3
CS201	هياكل البيانات والخوارزميات	Data Structures and Algorithms	Introduction to Programming and Problem Solving (CS101)	3
IT201	شبكات الحاسب	Computer Networks	Data Communication (IT102)	3
CS203	هندسة البرمجيات	Software Engineering	Introduction to Programming and Problem Solving (CS101)	3
18	المجموع			
الفصل الدراسي الثاني (Sophomore – Spring)				
CS202	نظم التشغيل	Operating Systems	---	3
CS204	أساسيات الذكاء الاصطناعي	Fundamentals of Artificial Intelligence	---	3
IS201	تحليل وإدارة البيانات	Data Analysis and Management	Probability and Statistics for Computer Science (BS104)	3
IS202	مقدمة لنظم قواعد البيانات	Introduction to Database Systems	Data Structures and Algorithms (CS201)	3
IT202	تكنولوجيا الإنترنت	Internet Technology	Computer Networks (IT201)	3
UN10x	اختياري جامعة	University Elective	--	3
18	المجموع			
الفصل الدراسي الصيفي (Summar)				
TR201	التدريب الصيفي	Summar Training	اجتياز ٧١ ساعة	4

برنامج علوم الحاسب

الرؤية:

التميز في إعداد وتأهيل كوادر متخصصة في مجالات علوم الحاسب والريادة في البحث والتطوير والابتكارات العلمية وخلق بيئة بحثية وتعليمية تلبي احتياجات المجتمع.

الرسالة:

يسعى قسم علوم الحاسب إلى إعداد متخصصين قادرين على الإبداع والمنافسة في سوق العمل وإعداد باحثين مؤهلين لإنتاج المعرفة وتطويرها وتوظيفها واستثمارها، وعقد شراكات مع المجتمع الخارجي لتوفير الخدمات والاستشارات الحاسوبية من خلال البرامج الأكاديمية المتميزة والمبادرات النوعية.

الأهداف:

1. إعداد متخصصين مؤهلين أكاديميا وعلميا في ضوء المعايير المحلية والدولية.
2. تطوير البرامج الأكاديمية على النحو الذي يواكب أحدث التطورات في مجالات علوم الحاسب، وتأهيلها للحصول على الاعتماد الأكاديمي.
3. توفير بيئة بحثية محفزة لرفع الكفاءة البحثية للقسم بما يحقق الإبداع في إنتاج المعرفة ونشرها.
4. تنمية قدرات أعضاء هيئة التدريس، والموظفين ومهاراتهم، وتحسين الخدمات المقدمة لهم.
5. بناء شراكة فعالة مع مؤسسات المجتمع، واستهدافها بالخدمات التنموية الفعالة.

سمات خريجي برنامج علوم الحاسب

يمتد برنامج علوم الحاسب على نطاق واسع يشمل دراسة أسسها النظرية والخوارزمية وتطوير الأساليب الحديثة لهندسة البرمجيات لتطوير البرمجيات في مجالات نظم التشغيل وأنظمة الزمن الحقيقي والذكاء الاصطناعي والأنظمة الذكية وتعليم الحاسبات لذلك فإن خريج برنامج علوم الحاسب سيمتلك القدرة على برمجة الحاسبات وفهم مفاهيم لغات الحاسب المختلفة و مترجماتها وبرمجته تطبيقات الموبيل. أيضاً، سيجني معارف في مجالات وتخصصات الذكاء الاصطناعي المتنوعة مثل تعلم الآلة والتعلم العميق والحوسبة المرنة والحوسبة المرئية والذكاء الاصطناعي التوليدي وبناء النماذج اللغوية الكبيرة ومعالجة اللغات الطبيعية والنظم الخبيرة وهندسة المعرفة. إضافة إلى ذلك، فإنه سيحصل على حصة معرفية فيما يخص مجالات المعلوماتية الحيوية والمعلوماتية الطبية والرعايا

الصحية وعلم الجينوم. كما سيتعلم خريج برنامج علوم الحاسب عن تنظيم وبناء الحاسبات والنظم الموزعة والحوسبة عالية الأداء وتطبيقاتها. بالإضافة إلى هذه المعارف التي يكتسبها الخريج، فهناك مجالات متنوعة أخرى سيحصل الخريج على معارف بها مثل تفاعل الإنسان بالحاسب ونظريه الألعاب والخوارزميات المتوازية والموزعة والوكلاء المستقلون والحوسبة الخضراء والذكاء المحيطي والحوسبة المنتشرة وغيرها.

بعد الانتهاء بنجاح من برنامج علوم الحاسب، بالإضافة الى المهارات العامة لخريجي الكلية يجب أن يكون الخريج قادرا على:

١. إظهار المعرفة والكفاءة في المجالات الأساسية لعلوم الحاسب مثل: الخوارزميات وتصميم وتحليل البرمجيات وأساسيات الذكاء الاصطناعي وتطبيقاتها والأنظمة الموزعة.
٢. تطبيق الأسس الرياضية ومبادئ الخوارزميات ونظريات علوم الحاسب في نمذجة وتصميم وتنفيذ وتقييم وتطوير الأنظمة القائمة على الحاسب.
٣. تطبيق معارف الرياضيات والعلوم على مشاكل العالم الحقيقي وكذلك لتحليل وتفسير البيانات.
٤. إظهار المهارات التحليلية اللازمة لتقييم المزايا النسبية للبرامج وأنظمة الحاسوب والنهج الخوارزمية.
٥. فهم وتطبيق مجموعة واسعة من مبادئ وأدوات هندسة البرمجيات، مثل منهجيات التصميم واختيار الخوارزمية واللغة ومكتبات البرامج وتقنية واجهة المستخدم.
٦. فهم وتطبيق مجموعة واسعة من مبادئ وأدوات الذكاء الاصطناعي.
٧. فهم المفاهيم المستخدمة في علوم الحاسب والتي تمكنهم من متابعة المزيد من التعلم، سواء كطلاب دراسات عليا أو بمفردهم .
٨. فهم الخوارزميات وهياكل البيانات، ومعمارية وتنظيم الحاسبات، ومفاهيم لغات البرمجة، والمترجمات، والشبكات، وتعلم الآلة والتعلم العميق، وتفاعل الإنسان والحاسب، وقواعد البيانات، وتحديد متطلبات الحوسبة لحلها.
٩. تصميم وتنفيذ وتقييم الأنظمة، أو العمليات، أو المكونات، أو البرامج القائمة على الحاسب .
١٠. استخدام المعرفة والفهم في نمذجة وتصميم الأنظمة القائمة على الحاسب بطريقة توضح فهم المفاضلة بين خيارات التصميم.

المتطلبات الأكاديمية**أولاً : متطلبات تخصص علوم الحاسب - 57 ساعة معتمدة****١ - متطلبات تخصص علوم الحاسب الإجبارية (42 ساعة معتمدة)**

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
CS301	تطوير برمجيات الموبايل	Mobile Application Development	3	2	2	Software Engineering (CS203)
CS302	تعلم الآلة	Machine Learning	3	2	2	Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)
CS303	تحليل وتصميم الخوارزميات	Analysis and Design of Algorithms	3	2	2	Data Structures and Algorithms (CS201)
CS304	الحوسبة المرنة	Soft Computing	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS305	المعلوماتية الحيوية	Bioinformatics	3	2	2	Data Structures and Algorithms (CS201)
CS306	النظم الموزعة	Distributed Systems	3	2	2	Operating Systems (CS202)
CS307	تفاعل الإنسان مع الحاسب	Human Computer Interaction	3	2	2	Object-Oriented Design and Programming (CS102)
CS308	هندسة البرمجيات متقدم	Advanced Software Engineering	3	2	2	Software Engineering (CS203)
CS309	التعلم العميق	Deep Learning	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS401	معالجة اللغات الطبيعية والتعريب	Natural Language Processing and Arabizations	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS402	الحوسبة فائقة الأداء	High Performance Computing	3	2	2	Operating Systems (CS202)
CS403	بناء المترجمات	Compiler Construction	3	2	2	Object-Oriented Design and Programming (CS102)
IT307	معالجة الصور الرقمية	Digital Image Processing	3	2	2	--
IT309	أمن الحاسبات والشبكات	Computer and Network Security	3	2	2	Computer Networks (IT201)

٢- متطلبات تخصص علوم الحاسب الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
CS310	الحوسبة المرئية	Visual Computing	3	2	2	Digital Image Processing (IT307)
CS311	بناء وتنظيم الحاسبات	Computer Organization and Architecture	3	2	2	Digital Logic Circuit (IT103)
CS312	نظم التشغيل متقدم	Advanced Operating Systems	3	2	2	Operating Systems (CS202)
CS313	الذكاء الاصطناعي التوليدي	Generative Artificial Intelligence	3	2	2	Deep Learning (CS309)
CS314	هندسة برمجيات الزمن الحقيقي	Real-time Software Engineering	3	2	2	Software Engineering (CS203)
CS315	أنظمة الزمن الحقيقي	Real-time Systems	3	2	2	Operating Systems (CS202)
CS316	الذكاء المحيطي	Ambient Intelligence	3	2	2	Human Computer Interaction (CS307)
CS317	البرمجة المنطقية والاستدلال الألى	Logic Programming and Automated Reasoning	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS318	الأنظمة الخبيرة وهندسة المعرفة	Expert Systems and Knowledge Engineering	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS319	مفاهيم لغات البرمجة	Programming Language Concepts	3	2	2	Introduction to Programming and Problem Solving (CS101)
CS320	الحوسبة الطبيعية	Natural Computing	3	2	2	Soft Computing (CS304)

Object-Oriented Design and Programming (CS102), Probability and Statistics for Computer Science (BS104)	2	2	3	Modeling and Simulation	النمذجة والمحاكاة	CS321
Probability and Statistics for Computer Science (BS104)	2	2	3	Game Theory	نظرية الألعاب	CS322
Data Structures and Algorithms (CS201)	2	2	3	Healthcare Informatics	معلوماتية الرعاية الصحية	CS323
Data Structures and Algorithms (CS201)	2	2	3	Biomedical Informatics	المعلوماتية الطبية	CS324
Computer Organization and Architecture (CS311)	2	2	3	Assembly Language	لغة التجميع	CS325
Distributed Systems (CS306)	2	2	3	Distributed Ledgers and Consensus Protocols	الدفاتر الموزعة و بروتوكولات الاجتماع	CS404
--	2	2	3	Selected Topics in Computer Science-1	موضوعات مختاره في علوم الحاسب-١	CS405
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Selected Topics in Artificial Intelligence	موضوعات مختاره في الذكاء الاصطناعي	CS406
--	2	2	3	Selected Topics in Computer Science-2	موضوعات مختاره في علوم الحاسب-٢	CS407
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Distributed Intelligent Automation Systems	أنظمة الأتمتة الموزعة الذكية	CS408
Distributed Systems (CS306)	2	2	3	Parallel and Distributed Algorithms	الخوارزميات المتوازية والموزعة	CS409
Data Structures and Algorithms (CS201)	2	2	3	Algorithmic Graph Theory	نظرية الرسوم	CS410

البيان الحسابية					
CS411	علم الجينوم المتقدم والتحليل الوراثي	Advanced Genomics and Genetic Analysis	3	2	2
CS412	معالجة اللغات المنطوقة	Spoken Language Processing	3	2	2
CS413	معالجة اللغة الطبيعية بالتعلم العميق	Natural Language Processing with Deep Learning	3	2	2
CS414	مقدمة اللغات الرسمية ونظرية الأتمتة	Introduction to Formal Languages and Automata Theory	3	2	2
CS415	اختبار البرمجيات وتوكيد الجودة	Software Testing and Quality Assurance	3	2	2
CS416	هندسة الأنطولوجي	Ontology Engineering	3	2	2
CS417	الوكلاء المستقلون	Autonomous Agents	3	2	2
CS418	التعلم بالتعزيز والتحكم	Reinforcement Learning and Control	3	2	2
CS419	حوسبة البيانات الضخمة عالية الأداء	High- Performance Big Data Computing	3	2	2
CS420	العلوم الاجتماعية الحاسوبية	Computational Social Science	3	2	2
CS421	المقاييس الحيوية	Biometrics	3	2	2
CS422	الحوسبة الخضراء	Green Computing	3	2	2

Operation Research(BS202)	2	2	3	Advanced Operation Research	بحوث عمليات متقدمة	IS411
Data Analysis and Management (IS201)	2	2	3	Decision Support System	نظم دعم اتخاذ القرار	IS412
Introduction to Web Programming (IS101)	2	2	3	Web of Data	شبكة البيانات	IS418
--	2	2	3	Signal Processing	معالجة الإشارات	IT301
Mathematics for Computer Science 3 (BS201)	2	2	3	Computer Graphics	الرسم بالحاسب	IT303
Digital Logic Circuit (IT103)	2	2	3	Internet of Things	إنترنت الأشياء	IT410

ثانياً: الخطة الدراسية لتخصص علوم الحاسب**١- المستوى الثالث تخصص علوم الحاسب (Junior)**

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
CS302	تعلم الآله	Machine Learning	Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)	3
CS303	تحليل وتصميم الخوارزميات	Analysis and Design of Algorithms	Data Structures and Algorithms (CS201)	3
CS304	الحوسبة المرنة	Soft Computing	Fundamentals of Artificial Intelligence (CS204)	3
CS306	النظم الموزعة	Distributed Systems	Operating Systems (CS202)	3
IT307	معالجة الصور الرقمية	Digital Image Processing	---	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
CS301	تطوير برمجيات الموبايل	Mobile Application Development	Software Engineering (CS203)	3
CS305	المعلوماتية الحيوية	Bioinformatics	Data Structures and Algorithms (CS201)	3
CS307	تفاعل الإنسان مع الحاسب	Human Computer Interaction	Object-Oriented Design and Programming (CS102)	3
CS309	التعلم العميق	Deep Learning	Fundamentals of Artificial Intelligence (CS204)	3
XXX	اختياري تخصص	Track-Specific Elective		3
15	المجموع			

٢- المستوى الرابع تخصص علوم الحاسب (Senior)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
CS308	هندسة البرمجيات متقدم	Advanced Software Engineering	Software Engineering (CS203)	3
CS401	معالجة اللغات الطبيعية والتعريب	Natural Language Processing and Arabizations	Fundamentals of Artificial Intelligence (CS204)	3
CS402	الحوسبة فائقة الأداء	High Performance Computing	Operating Systems (CS202)	3
XXX	اختياري تخصص	Track-Specific Elective		3
XXX	اختياري تخصص	Track-Specific Elective		3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
CS403	بناء المترجمات	Compiler Construction	Object-Oriented Design and Programming (CS102)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

برنامج نظم المعلومات

الرؤية:

الريادة والتميز محلياً وعالمياً في مجال نظم المعلومات من خلال إعداد كوادر مؤهلة قادرة على تطوير حلول مبتكرة تدعم التحول الرقمي، وتخدم المجتمع من خلال الشراكات الفاعلة والتنمية المستدامة.

الرسالة :

تقديم تعليم متميز في مجال نظم المعلومات يواكب التطورات العلمية، ويعزز الإبداع والابتكار. إعداد خريجين بمهارات تقنية وشخصية تلبي احتياجات سوق العمل، ودعم أبحاث رائدة وبناء شراكات مجتمعية فعالة لتحقيق التنمية المستدامة وتعزيز التحول الرقمي.

الأهداف:

- تزويد الطلاب بالمعرفة الأساسية والمتقدمة في مجال نظم المعلومات :
 - تقديم مناهج تعليمية تغطي أساسيات نظم المعلومات، بما في ذلك تحليل وتصميم وتطوير النظم.
 - تعليم الطلاب كيفية تطبيق هذه المفاهيم في بيئات العمل الحقيقية.
- تطوير مهارات تقنية وشخصية تلبي احتياجات سوق العمل
 - تنمية قدرات الطلاب في استخدام التقنيات الحديثة والأدوات البرمجية المتقدمة.
 - تعزيز مهارات التواصل، والعمل الجماعي، وحل المشكلات، والقيادة.
- تشجيع البحث العلمي والابتكار
 - تحفيز الطلاب على إجراء بحوث علمية تطبيقية في مجال نظم المعلومات.
 - دعم مشاركتهم في مشاريع بحثية تساهم في تطوير حلول مبتكرة للتحديات المعاصرة.
- تعزيز الشراكات المجتمعية وخدمة المجتمع :
 - تطوير شراكات مع المؤسسات المحلية والإقليمية لتطبيق مشاريع نظم المعلومات التي تخدم المجتمع.
 - تشجيع الطلاب على المشاركة في مبادرات تخدم المجتمع وتعزز التنمية المستدامة.
- الالتزام بالتعليم المستمر والتطوير المهني :
 - توفير فرص للتعليم المستمر والتدريب لمواكبة التطورات السريعة في مجال نظم المعلومات.
 - تشجيع الطلاب والخريجين على السعي نحو التعلم والتطوير الذاتي.

سمات خريجي برنامج نظم المعلومات

تركز نظم المعلومات على دمج حلول نظم المعلومات مع عمليات الأعمال لتلبية احتياجات المعلومات للشركات والمؤسسات الأخرى، وتمكينها من تحقيق أهدافها بفاعلية وكفاءة. يركز منظور هذا التخصص على المعلومات، وينظر إلى التكنولوجيا كأداة لتوليد المعلومات ومعالجتها وتوزيعها .

يجب أن يكون لدى محترفي نظم المعلومات منظور تجاري واسع وواقعي وأن يتمتعوا بمهارات تفكير تحليلي ونقدي قوية. كما يجب أن يظهروا مبادئ أخلاقية قوية وأن يتمتعوا بمهارات جيدة في التواصل بين الأشخاص ومهارات العمل في الفريق وذلك من أجل تصميم وتنفيذ حلول نظم المعلومات التي تعزز الأداء التنظيمي .

بعد الانتهاء بنجاح من برنامج نظم المعلومات، بالإضافة الى المهارات العامة لخريجي الكلية يجب أن يكون الخريج قادرا على:

١. التعرف على المشكلات القابلة لتطبيق نظم المعلومات الحاسوبية عليها، ومعرفة الأدوات اللازمة لحل هذه المشكلات.
٢. فهم أساسيات دورة حياة تطوير النظم، وشبكات المعلومات، وأمن المعلومات، واستخراج البيانات، والتجارة الإلكترونية، ونظم المعلومات الجغرافية، وإدارة الأزمات.
٣. إدارة واستغلال البيانات والمعلومات التنظيمية .تصميم نماذج البيانات والمعلومات، إدارة موارد ومشاريع تطوير نظم المعلومات.
٤. تنفيذ الحلول، بما في ذلك استخدام لغات البرمجة المناسبة، والأنظمة والأدوات المستندة إلى الويب، ومنهجيات التصميم، وأنظمة قواعد البيانات.
٥. تطبيق مبادئ الإدارة الفعالة للمعلومات وتنظيم المعلومات واستخراج المعلومات ومهارات استرجاع المعلومات، على المعلومات بمختلف أنواعها، بما في ذلك النصوص، والصور، والصوت، والفيديو.
٦. معرفة أساسيات تقنيات نظم المعلومات الذكية.
٧. تحديد وتصميم وتنفيذ نظم المعلومات القائمة على الحاسب، وتقييمها من حيث سمات الجودة العامة والمفضالات المحتملة المقدمة في مشكلة معينة.
٨. تطبيق حلول نظم المعلومات على المشكلات الوظيفية والتنظيمية والتشغيلية والإدارية والتنفيذية وخلق فرص منافسة.
٩. وصف خصائص المكونات المختلفة لنظم المعلومات، واستخدام الأدوات والتقنيات المناسبة لتحليل وتصميم وبناء نظم المعلومات.
١٠. إدارة بيانات المؤسسات المختلفة وتحديد أفضل ممارسات حفظ واسترجاع المعلومات.

١١. مساعدة نظم الأعمال المختلفة في اتخاذ القرارات الصحيحة من خلال تحليل البيانات واكتشاف المعرفة في قواعد البيانات وتطبيق نظم دعم القرار الذكية.
١٢. التواصل الفعال بالوسائل الشفوية والمكتوبة والمرئية.
١٣. العمل بفعالية كفرد وعضو في فريق.
١٤. إدارة الوقت بشكل مستقل وفعال.
١٥. إدراك القضايا الأخلاقية الرئيسية التي تؤثر على نظم المعلومات ومسؤولياتهم كمتخصصين في علوم المعلومات والبيانات.

المتطلبات الأكاديمية :**أولاً: متطلبات تخصص نظم المعلومات - 57 ساعة معتمدة****١ - متطلبات تخصص نظم المعلومات الإجبارية (42 ساعة معتمدة)**

المتطلبات السابقة	الساعات المعتمدة	محاضرة	تمارين/عملي	Course	اسم المادة	كود المادة
Software Engineering (CS203)	3	2	2	System Analysis and Design	تحليل وتصميم النظم	IS301
Introduction to Database Systems (IS202)	3	2	2	Advanced Database	قواعد بيانات متقدمة	IS302
--	3	2	2	Enterprise Architecture	معمارية المؤسسات	IS303
Introduction to Web Programming (IS101)	3	2	2	Advanced Web Programming	برمجة الويب المتقدمة	IS304
--	3	2	2	Information Theory	نظرية المعلومات	IS305
System Analysis and Design (IS301)	3	2	2	System Development Methods	طرق تطوير النظم	IS306
Introduction to Database Systems (IS202)	3	2	2	Information Storage and Retrieval	تخزين واسترجاع المعلومات	IS307
Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	3	2	2	Data Mining	التنقيب في البيانات	IS308
System Analysis and Design (IS301)	3	2	2	Information Systems Quality	جودة نظم المعلومات	IS309
Fundamentals of Artificial Intelligence (CS204)	3	2	2	Intelligent Information System	نظم المعلومات الذكية	IS401
Computer Networks (IT201), Advanced Database (IS302)	3	2	2	Information Security and Privacy	امن المعلومات وخصوصياتها	IS402
Information Storage and Retrieval (IS307)	3	2	2	Big Data Processing	معالجة البيانات الضخمة	IS403
Data Analysis and Management (IS201)	3	2	2	Data Science Applications and analysis	تطبيقات علوم البيانات وتحليلها	IS404
Advanced Database (IS302)	3	2	2	Distributed Database Systems	قواعد البيانات الموزعة	IS405

٢- متطلبات تخصص نظم المعلومات الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

كود المادة	اسم المادة	Course	المعتمدة الساعات	محاضرة	تمارين/عملي	المتطلبات السابقة
IS310	اقتصاديات نظم المعلومات	Economics of Information Systems	3	2	2	--
IS311	نظم الأعمال الذكية	Business Intelligent System	3	2	2	Data Analysis and Management (IS201)
IS312	إدارة المخاطر والأزمات	Risk and Crisis Management	3	2	2	Project Management (IS203)
IS313	تقنيات تطبيقات الويب	Web Application Technologies	3	2	2	Introduction to Web Programming (IS101)
IS406	موضوعات مختارة في نظم المعلومات	Selected Topics in Information System	3	2	2	--
IS407	تكامل المؤسسات	Enterprise Integration	3	2	2	Enterprise Architecture (IS303)
IS408	تخطيط موارد المؤسسة	Enterprise Resource Planning	3	2	2	Enterprise Architecture (IS303), Introduction to Database Systems (IS202)
IS409	نظم المعلومات المحددة -المجال	Domain-specific Information Systems	3	2	2	System Analysis and Design(IS301)
IS410	مخازن البيانات	Data Warehouse	3	2	2	Introduction to Database Systems (IS202)
IS411	بحوث عمليات متقدمة	Advanced Operation Research	3	2	2	Operation Research(BS202)
IS412	نظم دعم اتخاذ القرار	Decision Support System	3	2	2	Data Analysis and Management (IS201)
IS413	التمثيل المرئي للبيانات	Data Visualization	3	2	2	Data Analysis and Management (IS201)
IS414	قواعد البيانات المكانية	Spatial Database	3	2	2	Introduction to Database Systems (IS202)
IS415	نظم المعلومات الجغرافية والاستشعار عن بعد	Geographic Information System and Remote Sensing	3	2	2	Spatial Database (IS414)

System Analysis and Design(IS301)	2	2	3	Usability and User Experience (UX)	الاستخدام الكفء وخبرة المستخدم	IS416
Introduction to Web Programming (IS101)	2	2	3	Responsive Website Development and Design	تطوير وتصميم مواقع الويب سريعة الاستجابة	IS417
Introduction to Web Programming (IS101)	2	2	3	Web of Data	شبكة البيانات	IS418
Software Engineering (CS203)	2	2	3	Mobile Application Development	تطوير برمجيات الموبايل	CS301
Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)	2	2	3	Machine Learning	تعلم الآلة	CS302
Data Structures and Algorithms (CS201)	2	2	3	Bioinformatics	المعلوماتية الحيوية	CS305
Object-Oriented Design and Programming (CS102), Probability and Statistics for Computer Science (BS104)	2	2	3	Modeling and Simulation	النمذجة والمحاكاة	CS321
Data Structures and Algorithms (CS201)	2	2	3	Healthcare Informatics	معلوماتية الرعاية الصحية	CS323
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Algorithmic Graph Theory	نظرية الرسوم البيانية الحسابية	CS410
Mathematics 3 for Computer Science (BS201)	2	2	3	Computer Graphics	الرسم بالحاسب	IT303
--	2	2	3	Multimedia	الوسائط المتعددة	IT403
Digital Logic Circuit (IT103)	2	2	3	Internet of Things	إنترنت الأشياء	IT410
Computer Networks (IT201), Multimedia (IT403)	2	2	3	Multimedia Communication	اتصالات الوسائط المتعددة	IT414

ثانياً: الخطة الدراسية لتخصص نظم المعلومات**١ - المستوى الثالث تخصص نظم المعلومات (Junior)**

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
IS301	تحليل وتصميم النظم	System Analysis and Design	Software Engineering (CS203)	3
IS302	قواعد بيانات متقدمة	Advanced Database	Introduction to Database Systems (IS202)	3
IS303	معمارية المؤسسات	Enterprise Architecture	--	3
IS304	برمجة الويب المتقدمة	Advanced Web Programming	Introduction to Web Programming (IS101)	3
IS305	نظرية المعلومات	Information Theory	--	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IS306	طرق تطوير النظم	System Development Methods	System Analysis and Design (IS301)	3
IS307	تخزين واسترجاع المعلومات	Information Storage and Retrieval	Introduction to Database Systems (IS202)	3
IS308	التنقيب في البيانات	Data Mining	Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	3
IS309	جودة نظم المعلومات	Information Systems Quality	System Analysis and Design (IS301)	3
XXX	اختياري تخصص	Track-Specific Elective		3
15	المجموع			

برنامج تكنولوجيا المعلومات

الرؤية:

تحقيق الريادة محلياً وإقليمياً وعالمياً في إعداد كوادر متميزة في مجال تكنولوجيا المعلومات والتطبيقات الرقمية، تسهم في بناء مجتمع رقمي مستدام، ودعم التنمية المعرفية والابتكار التقني.

الرسالة:

يسعى قسم تكنولوجيا المعلومات بكلية الحاسبات والمعلومات – جامعة المنصورة إلى إعداد خريجين متميزين علمياً ومهنيًا في مجالات تكنولوجيا المعلومات والشبكات وأمن المعلومات وأنظمة الأتمتة والتطبيقات المتطورة للذكاء الاصطناعي، قادرين على المنافسة في سوق العمل محلياً وإقليمياً، ودعم البحث العلمي والابتكار، وتقديم حلول تقنية تسهم في خدمة المجتمع وتحقيق التنمية المستدامة، وذلك في إطار من الجودة والالتزام بالمعايير الأكاديمية المحلية والدولية.

الأهداف:

- اكتساب المعرفة الأساسية والمتقدمة في مجالات تكنولوجيا المعلومات والشبكات وأمن المعلومات.
- تطبيق المفاهيم والنظريات العلمية لحل المشكلات التقنية باستخدام أدوات وأساليب حديثة.
- تصميم وتطوير نظم وتطبيقات تقنية تلبى احتياجات المؤسسات والمجتمع.
- مواكبة التطورات التكنولوجية والتعامل مع التقنيات الناشئة بكفاءة.
- العمل بفاعلية ضمن فرق متعددة التخصصات والالتزام بأخلاقيات المهنة والمسؤولية المجتمعية.
- دعم البحث العلمي والابتكار واستخدام المنهجية العلمية في دراسة المشكلات التقنية.
- الاستعداد للتعليم المستمر والدراسات العليا أو الاندماج المباشر في سوق العمل.
- المساهمة في التحول الرقمي بالمؤسسات.
- المشاركة المجتمعية وتعزيز الوعي بقضايا أمن المعلومات.

سمات خريجي برنامج تكنولوجيا المعلومات

يركز قسم تكنولوجيا المعلومات على البنية التحتية للأنظمة والأجهزة الإلكترونية والرقمية وضمان الوقوف أمام المشكلات التي قد تتعرض لها الأنظمة من اختراق وأعطال إلخ. لذلك فإن خريج قسم تكنولوجيا المعلومات سيمتلك القدرة على إنشاء وبرمجة وتقديم الدعم فيما يخص الأجهزة الرقمية

والإلكترونية المختلفة (مثل العمل مع الإنسان الآلي، الأنظمة المضمنة، المعالجات الدقيقة والحاسبات المتنقلة والذكية وغيره). إضافة إلى ذلك، فإنه سيحصل على حصيلة معرفية فيما يخص الشبكات بأنواعها وتقنياتها المختلفة (السلوكية، اللاسلكية، الحوسبة السحابية إلخ). أيضاً، سيجني معارف في مجالات تقنية حديثة مثل أمن الشبكات، الأمن السيبراني، الرؤية بالحاسب وتطبيقاته المختلفة بالإضافة إلى التعامل مع ومعالجة وتحليل أنواع مختلفة من الإشارات التي تستقبلها الأجهزة والأنظمة الرقمية المختلفة (مثل: التعرف والتعامل مع الكلام، الصور المختلفة والوسائط المتعددة). بالإضافة إلى هذه المعارف التي يكتسبها الخريج، فهناك مجالات متنوعة أخرى سيحصل الخريج على معارف بها مثل علم الأدلة الجنائية، التعامل مع الرسوم والرسومات المتحركة، الواقع الافتراضي، تطوير الألعاب، التعليم الإلكتروني، المكتبات الرقمية إلخ.

بعد الانتهاء بنجاح من برنامج تكنولوجيا المعلومات، بالإضافة إلى المهارات العامة لجميع خريجي الكلية يجب أن يكون الخريج قادراً على:

١. تحليل المشكلة، وتحديد وتعريف متطلبات الحوسبة المناسبة لحلها.
٢. تصميم وتنفيذ وتقييم نظام، أو عملية، أو برنامج قائم على الحاسب لتلبية الاحتياجات المطلوبة.
٣. إظهار التفكير النقدي المستقل ومهارات حل المشكلات والعمل بفعالية في فريق لتحقيق هدف مشترك .
٤. التواصل بفعالية مع مجموعة من الجماهير .
٥. الاعتراف بالحاجة والقدرة على الانخراط في التطوير المهني المستمر.
٦. استخدام التقنيات والمهارات والأدوات الحالية اللازمة لممارسة تكنولوجيا المعلومات وفي إنشاء خطة مشروع فعالة.
٧. استخدام وتطبيق المفاهيم والممارسات التقنية الحالية.
٨. معالجة مشاكل تكنولوجيا المعلومات للمنظمات أو الأفراد.
٩. دمج الحلول القائمة على تكنولوجيا المعلومات بشكل فعال في بيئة المستخدم.
١٠. فهم أفضل الممارسات والمعايير وتطبيقها.
١١. فهم الشبكات المختلفة وكيفية برمجتها والتعامل معها مما يخدم المجتمع المدني والمؤسسات المختلفة.
١٢. القدرة على برمجة أجهزة مختلفة للقيام بمهام مختلفة تخدم المجتمع وتواكب التطور التقني العالمي.
١٣. التعامل مع عمليات الاختراق المختلفة للأنظمة والبيانات والشبكات وتقديم حلول عملية مفيدة في هذا الإطار.

١٤. القدرة على معالجة وتحليل أنواع مختلفة من الإشارات الرقمية مثل الكلام والصور والفيديوهات إلخ مما يساعد على إنشاء أنظمة تطبيقية مفيدة في قطاعات مختلفة.
١٥. القدرة على بناء وتطوير أنظمة مبتكرة للتعليم الإلكتروني والمكتبات الرقمية على سبيل المثال والذي بدوره يخدم تطبيق أهداف التحول الرقمي في المجتمع.
١٦. القدرة على إنشاء رسومات رقمية ورسوم متحركة بالإضافة إلى تطوير برمجيات بتقنيات الواقع الافتراضي وتطوير الألعاب حيث يخدم كل ذلك في إنشاء خريج قادر على الابتكار واستغلال هذه المهارات في تطبيقات عملية مختلفة ومفيدة للمجتمع.
١٧. القدرة على استغلال مبادئ وتقنيات تكنولوجيا المعلومات في مجالات حيوية مثل الأدلة الجنائية وتحليل صور الأشعة الطبية.

المتطلبات الأكاديمية:**أولاً: متطلبات تخصص تكنولوجيا المعلومات - 57 ساعة معتمدة****١ - متطلبات تخصص تكنولوجيا المعلومات الإجبارية (42 ساعة معتمدة)**

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
--	2	2	3	Signal Processing	معالجة الإشارات	IT301
--	2	2	3	Pattern Recognition	التعرف على الأنماط	IT302
Mathematics for Computer Science 3 (BS201)	2	2	3	Computer Graphics	الرسم بالحاسب	IT303
Computer Networks (IT201)	2	2	3	Wireless and Sensor Networks	الشبكات اللاسلكية	IT304
Digital Logic Circuit (IT103)	2	2	3	Microprocessors	المعالجات الدقيقة	IT305
--	2	2	3	Intelligent and Quantum Computers	الحاسبات الذكية والكمية	IT306
--	2	2	3	Digital Image Processing	معالجة الصور الرقمية	IT307
Computer Networks (IT201)	2	2	3	Network Programming	برمجة الشبكات	IT308
Computer Networks (IT201)	2	2	3	Computer and Network Security	أمن الحاسبات والشبكات	IT309
Digital Image Processing (IT307)	2	2	3	Computer Vision	الرؤية بالحاسب	IT401
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Robotics 1	علم الإنسان الآلي (1)	IT402
--	2	2	3	Multimedia	الوسائط المتعددة	IT403
Digital Logic Circuit (IT103)	2	2	3	Embedded Systems	الأنظمة المضمنة	IT404
Computer Networks (IT201)	2	2	3	Cyber Security	الأمن السيبراني	IT405

٢- متطلبات تخصص تكنولوجيا المعلومات الاختيارية (يختار الطالب ١٥ ساعة معتمدة)
جدول ١٨: متطلبات اختيارية في تخصص تكنولوجيا المعلومات.

المادة كود	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
IT310	التعرف على الكلام	Speech Recognition	3	2	2	--
IT311	الواقع الافتراضي	Virtual Reality	3	2	2	--
IT312	مقدمة تصميم إمكانية الوصول والشمول	Introduction to Accessibility and Inclusive Design	3	2	2	--
IT313	الأنظمة المتنقلة الذكية	Smart Mobile Systems	3	2	2	Computer Networks (IT201)
IT314	تطوير الألعاب	Games Development	3	2	2	Fundamentals of Artificial Intelligence (CS204)
IT315	شبكات الحاسب ٢	Computer Networks 2	3	2	2	Computer Networks (IT201)
IT406	علوم الأدلة الجنائية	Forensic Science	3	2	2	Cyber Security (IT405)
IT407	موضوعات مختارة في أمن تكنولوجيا المعلومات	Selected Topics in IT-Security	3	2	2	Cyber Security (IT405)
IT408	سلسلة الكتلة	Blockchain	3	2	2	--
IT409	تطبيقات الرؤية بالحاسب	Computer Vision Applications	3	2	2	Computer Vision (IT401)
IT410	إنترنت الأشياء	Internet of Things	3	2	2	Digital Logic Circuit (IT103)
IT411	تحليل الصور الطبية	Medical Image Analysis	3	2	2	Digital Image Processing (IT307)
IT412	الحوسبة السحابية	Cloud Computing	3	2	2	--
IT413	علم الإنسان الآلي (2)	Robotics 2	3	2	2	Robotics 1 (IT402)

Multimedia (IT403)	2	2	3	Multimedia Communication	اتصالات الوسائط المتعددة	IT414
Computer Networks (IT201)	2	2	3	Mobile Computing	الحوسبة المتنقلة	IT415
Introduction to Programming and Problem Solving (CS101)	2	2	3	System-on Chip (SoC) Programming	برمجة أنظمة الرقائق الإلكترونية	IT416
Computer Networks (IT201)	2	2	3	Planning and Designing of Information Networks	تخطيط وتصميم شبكات المعلومات	IT417
Computer Graphics (IT303)	2	2	3	Computer Animation	الرسوم الحاسوبية المتحركة	IT418
Computer Graphics (IT303)	2	2	3	Computer Graphics – 2	الرسم بالحاسب - ٢	IT419
--	2	2	3	E-Learning	التعليم الإلكتروني	IT420
--	2	2	3	Digital Libraries	المكتبات الرقمية	IT421
--	2	2	3	Computer Interfaces	مواجهات الحاسبات	IT422
Digital Image Processing (IT307)	2	2	3	Digital Image Processing – 2	معالجة الصور - ٢	IT423
--	2	2	3	Fault Tolerant Computer Systems	نظم الحاسبات ذات الأعطال المحتملة	IT424
--	2	2	3	Automatic Control	التحكم الآلي	IT425
--	2	2	3	Selected Topics in Information Technology	موضوعات مختارة في تكنولوجيا المعلومات	IT426
--	2	2	3	Selected Topics in Information Technology - 2	موضوعات مختارة في تكنولوجيا المعلومات - ٢	IT427
Introduction to Web Programming (IS101)	2	2	3	Advanced Web Programming	برمجة الويب المتقدمة	IS304
Operation Research (BS202)	2	2	3	Advanced Operation Research	بحوث عمليات متقدمة	IS411
Data Analysis and Management (IS201)	2	2	3	Decision Support System	نظم دعم اتخاذ القرار	IS412

Introduction to Web Programming (IS101)	2	2	3	Responsive Website Development and Design	تطوير وتصميم مواقع الويب سريعة الاستجابة	IS417
Introduction to Web Programming (IS101)	2	2	3	Web of Data	شبكة البيانات	IS418
Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)	2	2	3	Machine Learning	تعلم الآلة	CS302
Data Structures and Algorithms (CS201)	2	2	3	Bioinformatics	المعلوماتية الحيوية	CS305
Operating Systems (CS202)	2	2	3	Real-time Systems	أنظمة الزمن الحقيقي	CS315
Introduction to Programming and Problem Solving (CS101)	2	2	3	Modeling and Simulation	النمذجة والمحاكاة	CS321
Data Structures and Algorithms (CS201)	2	2	3	Healthcare Informatics	معلوماتية الرعاية الصحية	CS323
Data Structures and Algorithms (CS201)	2	2	3	Medical Informatics	المعلوماتية الطبية	CS324
Introduction to Web Programming (IS101)	2	2	3	Computational Social Science	العلوم الاجتماعية الحاسوبية	CS420
Data Structures and Algorithms (CS201)	2	2	3	Biometrics	المقاييس الحيوية	CS421

ثانياً: الخطة الدراسية لتخصص تكنولوجيا المعلومات**٣- المستوى الثالث تخصص تكنولوجيا المعلومات (Junior)**

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
IT301	معالجة الإشارات	Signal Processing	--	3
IT302	التعرف على الأنماط	Pattern Recognition	--	3
IT303	الرسم بالحاسب	Computer Graphics	Mathematics for Computer Science 3 (BS201)	3
IT304	الشبكات اللاسلكية	Wireless and Networks Sensor	Computer Networks (IT201)	3
IT305	المعالجات الدقيقة	Microprocessors	Digital Logic Circuit (IT103)	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IT306	الحاسبات الذكية والكمية	Intelligent and Quantum Computers	--	3
IT307	معالجة الصور الرقمية	Digital Image Processing	--	3
IT308	برمجة الشبكات	Network Programming	Computer Networks (IT201)	3
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
XXX	اختياري تخصص	Track-Specific Elective		3
15	المجموع			

٤- المستوى الرابع تخصص تكنولوجيا المعلومات (Senior)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IT401	الرؤية بالحاسب	Computer Vision	Digital Image Processing (IT307)	3
IT402	علم الإنسان الألى(1)	Robotics 1	Fundamentals of Artificial Intelligence (CS204)	3
IT403	الوسائط المتعددة	Multimedia	--	3
XXX	اختياري تخصص	Track-Specific Elective		3
XXX	اختياري تخصص	Track-Specific Elective		3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IT404	الأنظمة المضمنة	Embedded Systems	Digital Logic Circuit (IT103)	3
IT405	الأمن السيبراني	Cyber Security	Computer Networks (IT201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

الباب الثالث:

المتطلبات الدراسية للبرامج النوعية (برامج خاصة بمصروفات)

برنامج هندسة البرمجيات

الرؤية :

أن يكون برنامج هندسة البرمجيات رائدًا على المستوى المحلي والإقليمي والدولي في إعداد خريجين متميزين في تطوير البرمجيات عالية الجودة، يساهمون في دفع عجلة التكنولوجيا ودعم التحول الرقمي.

الرسالة :

يسعى برنامج هندسة البرمجيات إلى تقديم تعليم عالي الجودة في مجال تصميم وتطوير البرمجيات، من خلال منهج دراسي حديث ومتكامل، وبحث علمي فعال، وشراكات استراتيجية مع القطاع الصناعي، بهدف إعداد خريجين مؤهلين تقنيًا ومهنيًا لتلبية احتياجات سوق العمل المتغيرة ودعم التنمية المستدامة.

الأهداف:

يهدف برنامج هندسة البرمجيات إلى تحقيق رؤية شاملة تواكب التطورات الحديثة في مجال البرمجيات وتلبي احتياجات سوق العمل من الكوادر المؤهلة في هندسة البرمجيات. ومن هذا المنطلق، يسعى البرنامج إلى تحقيق الأهداف الاستراتيجية التالية:

١. إعداد خريجين يتمتعون بقدرات تنافسية عالية: توفير تعليم متميز يركز على تزويد الطلاب بالمعارف والمهارات العملية التي تؤهلهم للتفوق في بيئات العمل المحلية والدولية، ما يساهم في تحسين سمعة البرنامج ويزيد من جاذبيته للطلاب المتميزين.
٢. تطوير المناهج لتلبية احتياجات السوق المستمرة: تحديث المناهج الدراسية بشكل دوري بما يتماشى مع التطورات التكنولوجية السريعة ومتطلبات سوق العمل، مع التركيز على مهارات البرمجة الحديثة، وهندسة الأنظمة، وأدوات إدارة المشاريع البرمجية.
٣. تعزيز البحث العلمي والابتكار: تشجيع البحوث التطبيقية والمشاريع الابتكارية في مجال هندسة البرمجيات، ودعم مشاركة الطلاب وأعضاء هيئة التدريس في الأبحاث التي تساهم في إيجاد حلول للتحديات التقنية والصناعية، مما يساهم في رفع مكانة البرنامج أكاديميًا.
٤. بناء شراكات استراتيجية مع القطاع الصناعي: إقامة علاقات تعاون مثمرة مع مؤسسات صناعية رائدة في مجال البرمجيات وتكنولوجيا المعلومات، بهدف إتاحة فرص تدريب عملي للطلاب، وتعزيز التفاعل بين الجوانب الأكاديمية والعملية، وتسهيل انتقال الخريجين إلى سوق العمل.

٥. **دعم التعلم المستمر والتطوير المهني:** توفير بيئة تدعم التعلم مدى الحياة من خلال تقديم دورات تدريبية وورش عمل للطلاب والخريجين، ما يساعدهم في مواكبة المستجدات التقنية وتطوير مساراتهم المهنية بشكل مستمر.
٦. **ترسيخ قيم المسؤولية المهنية والأخلاقية:** غرس قيم النزاهة والأخلاقيات المهنية في مجال هندسة البرمجيات، وتشجيع الطلاب على احترام خصوصية وأمان المعلومات، والإلمام بتأثيرات الحلول البرمجية على المجتمع والبيئة.
٧. **تعزيز مكانة البرنامج محلياً وعالمياً:** العمل على رفع التصنيف الأكاديمي للبرنامج من خلال تبني معايير الجودة العالمية، وتشجيع التبادل الأكاديمي والبحثي مع جامعات ومؤسسات بحثية دولية، بما يعزز من مكانة البرنامج على المستويين المحلي والدولي.

سمات خريجي برنامج هندسة البرمجيات

يتميز خريجي برنامج هندسة البرمجيات بمعرفة تقنية عميقة تشمل لغات البرمجة وهياكل البيانات والخوارزميات، بالإضافة إلى مهارات عملية في تصميم وتطوير البرمجيات باستخدام الأدوات الحديثة. بالإضافة لامتلاكهم مهارات حل المشكلات، والعمل الجماعي الفعال، والتواصل الجيد والتزامهم بالتعلم المستمر ومواكبة التطورات التكنولوجية. يتميز خريجي هندسة البرمجيات بأنهم منظمون في إدارة الوقت والمسؤوليات، ويتميزون بقدرات قيادية وأخلاقيات مهنية عالية، مما يؤهلهم للاندماج بكفاءة في بيئات العمل المختلفة والمساهمة في تطوير أنظمة برمجية موثوقة وفعالة.

بالإضافة إلى السمات العامة لخريجي كلية الحاسبات والمعلومات فبعد الانتهاء بنجاح من برنامج هندسة البرمجيات، يجب أن يكون الخريج قادراً على:

١. تطبيق المعارف الرياضية والعلمية والهندسية.
٢. تصميم نظم البرمجيات، ومكوناتها، وعملياتها للوفاء بالمتطلبات المرغوبة ضمن القيود الواقعية سواء الاقتصادية، أو البيئية، أو الاجتماعية، أو السياسية، أو الأخلاقية، أو ما يتعلق بالصحة والسلامة، بشكل عملي ومستديم.
٣. تحديد مشاكل هندسة البرمجيات وصياغتها وحلها.
٤. تطبيق مفاهيم ومهارات الرياضيات المتقطعة، والاحتمالات والإحصاء، والعلوم الأخرى ذات الصلة، بشكل مناسب في نظم البرمجيات المعقدة.
٥. استخدام المهارات والتقنيات وأدوات هندسة البرمجيات الحديثة والضرورية لتصميم وبناء البرمجيات.

٦. تحليل نظم البرمجيات، وتصميمها، والتحقق من صحة تصميمها وصحة عملها، وتنفيذها، واستخدامها، وصيانتها.
 ٧. العمل في مجال أو أكثر من المجالات الرئيسة للتطبيقات.
 ٨. إدارة وتسيير عمليات تطوير نظم البرمجيات.
 ٩. فهم المسؤوليات الأخلاقية والمهنية.
 ١٠. التواصل بشكل بناء ومتابعة التعلم مدى الحياة.
 ١١. العمل الجماعي ضمن فريق متعدد التخصصات.
 ١٢. فهم تأثير حلول هندسة البرمجيات على القضايا العالمية، والاقتصادية، والبيئية، والاجتماعية.
- هذه القدرات تجعل خريج برنامج هندسة البرمجيات مؤهلاً للاندماج بفعالية في بيئات العمل المختلفة والمساهمة في تطوير وصيانة الأنظمة البرمجية بفعالية وكفاءة.

المتطلبات الأكاديمية:**أولاً: متطلبات الجامعة (١٢ ساعة معتمدة)****أ- متطلبات الجامعة الإلزامية (٩ ساعات معتمدة)**

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Communication Skills, Ethical and Professional Issues	مهارات الاتصال والجوانب الأخلاقية والمهنية	UN101
---	2	2	3	English for Computer Science	لغة إنجليزية لعلوم الحاسب	UN102
---	2	2	3	Technology Entrepreneurship	ريادة الأعمال التكنولوجية	UN201

ب- متطلبات الجامعة الاختيارية (٣ ساعات معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Design Physiology for Computer Science	علم نفس التصميم لعلوم الحاسب	UN104
---	2	2	3	Law for Computer Science	قانون لعلوم الحاسب	UN105
---	2	2	3	Economics for Computer Science	اقتصاد لعلوم الحاسب	UN106
---	2	2	3	Accounting for Computer Science	محاسبة لعلوم الحاسب	UN107
---	2	2	3	Sociology for Computer Science	علم الاجتماع لعلوم الحاسب	UN108

ت- مقررات بدون ساعات معتمدة

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
--	--	1	0	Societal Issues	القضايا المجتمعية	UN103

ثانياً: متطلبات العلوم الأساسية (٢٤ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Mathematics 1 for Computer Science	رياضيات ١ لعلوم الحاسب	BS101
---	2	2	3	Introduction to Computing and Informatics	مقدمة الحاسبات والمعلوماتية	BS102
---	2	2	3	Mathematics 2 for Computer Science	رياضيات ٢ لعلوم الحاسب	BS103
---	2	2	3	Probability and Statistics for Computer Science	احتمالات وتوزيعات إحصائية	BS104
---	2	2	3	Fundamentals of Electronics and Electrical Circuits	أساسيات الإلكترونيات والدوائر الكهربائية	IT101
---	2	2	3	Data Communication	اتصالات البيانات	IT102
---	2	2	3	Mathematics 3 for Computer Science	رياضيات ٣ لعلوم الحاسب	BS201
---	2	2	3	Operation Research	بحوث العمليات	BS202

ثالثا: متطلبات الكلية (٣٩ ساعة معتمدة)

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Introduction to Programming and Problem Solving	مقدمة للبرمجة وحل المشكلات	CS101
Introduction to Programming and Problem Solving (CS101)	2	2	3	Object-Oriented Design and Programming	تصميم وبرمجة شبيهة التوجه	CS102
Introduction to Programming and Problem Solving (CS101)	2	2	3	Data Structures and Algorithms	هياكل البيانات والخوارزميات	CS201
---	2	2	3	Operating Systems	نظم التشغيل	CS202
Introduction to Programming and Problem Solving (CS101)	2	2	3	Software Engineering	هندسة البرمجيات	CS203
---	2	2	3	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	CS204
Software Engineering (CS203)	2	2	3	Mobile Application Development	تطوير برمجيات الموبايل	CS301
Introduction to Programming and Problem Solving (CS101)	2	2	3	Introduction to Web Programming	مقدمة لبرمجة الويب	IS101
Probability and Statistics for Computer Science (BS104)	2	2	3	Data Analysis and Management	تحليل وإدارة البيانات	IS201
Data Structures and Algorithms (CS201)	2	2	3	Introduction to Database Systems	مقدمة لنظم قواعد البيانات	IS202
--	2	2	3	Project Management	إدارة المشروعات	IS203
Fundamentals of electronics and Electrical Circuits (IT101)	2	2	3	Digital Logic Circuit	دوائر المنطق الرقمية	IT103
Data Communication (IT102)	2	2	3	Computer Networks	شبكات الحاسب	IT201

رابعاً: متطلبات تخصص هندسة البرمجيات - ٥٧ ساعة معتمدة**٣- متطلبات تخصص هندسة البرمجيات الإجبارية (٤٢ ساعة معتمدة)**

المتطلبات السابقة	تدريبات/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Operating Systems (CS202)	2	2	3	Distributed Systems	النظم الموزعة	CS306
Object-Oriented Design and Programming(CS102)	2	2	3	Human Computer Interaction	تفاعل الإنسان مع الحاسب	CS307
Software Engineering (CS203)	2	2	3	Software Testing and Quality Assurance	اختبار البرمجيات وتوكيد الجودة	CS415
--	2	2	3	Enterprise Architecture	معمارية المؤسسات	IS303
Introduction to Web Programming (IS101)	2	2	3	Advanced Web Programming	برمجة الويب المتقدمة	IS304
Software Engineering (CS203)	2	2	3	System Development Methods	طرق تطوير النظم	IS306
Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	2	2	3	Data Mining	التنقيب في البيانات	IS308
Computer Networks (IT201)	2	2	3	Computer and Network Security	أمن الحاسبات والشبكات	IT309
--	2	2	3	Business Process Management	إدارة عمليات الأعمال	SE201
Software Engineering (CS203)	2	2	3	Software Design and Architecture	تصميم ومعمارية البرمجيات	SE301
Software Engineering (CS203)	2	2	3	Software Requirement Analysis	تحليل متطلبات البرمجة	SE302
Software Design and Architecture (SE301)	2	2	3	Software Construction	بناء البرمجيات	SE401
Software Requirement Analysis (SE302)	2	2	3	Cost Estimation of Software Projects' Development and Maintenance	تقدير تكاليف تطوير وصيانة مشاريع البرمجيات	SE402
Software Design and Architecture(SE301)	2	2	3	Open-Source Software Development	تطوير البرمجيات مفتوحة المصدر	SE403

٤- متطلبات تخصص هندسة البرمجيات الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

المتطلبات السابقة	تدريبات/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Software Engineering (CS203)	2	2	3	Real-time Software Engineering	هندسة برمجيات الزمن الحقيقي	CS314
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Expert Systems and Knowledge Engineering	الأنظمة الخبيرة وهندسة المعرفة	CS318
Object-Oriented Design and Programming (CS102), Probability and Statistics for Computer Science (BS104)	2	2	3	Modeling and Simulation	النمذجة والمحاكاة	CS321
Operating Systems (CS202)	2	2	3	High Performance Computing	الحوسبة فائقة الأداء	CS402
Project Management (IS203)	2	2	3	Risk and Crisis Management	إدارة المخاطر والأزمات	IS312
Introduction to Web Programming (IS101)	2	2	3	Web Application Technologies	تقنيات تطبيقات الويب	IS313
Data Analysis and Management (IS201)	2	2	3	Data Science Applications and analysis	تطبيقات علوم البيانات وتحليلها	IS404
Enterprise Architecture (IS303)	2	2	3	Enterprise Integration	تكامل المؤسسات	IS407
Data Analysis and Management (IS201)	2	2	3	Decision Support System	نظم دعم اتخاذ القرار	IS412
--	2	2	3	Cloud Computing	الحوسبة السحابية	IT412
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Games Development	تطوير الألعاب	IT314
Digital Logic Circuit (IT103)	2	2	3	Embedded Systems	الأنظمة المضمنة	IT404
Computer Networks (IT201)	2	2	3	Smart Mobile Systems	الأنظمة المتنقلة الذكية	IT313
Software Construction (SE401)	2	2	3	Software Reengineering	إعادة هندسة البرمجيات	SE404

Software Engineering (CS203), Distributed Systems (CS306)	2	2	3	Global Software Development	تطوير البرمجيات العالمية	SE405
--	2	2	3	Selected Topics in Software Engineering- 1	موضوعات مختارة في هندسة البرمجيات - ١	SE406
--	2	2	3	Selected Topics in Software Engineering -2	موضوعات مختارة في هندسة البرمجيات - ٢	SE407

خامسا : التدريب الصيفي

المادة كود	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/ عملي	المتطلبات السابقة
TR201	التدريب الصيفي	Summer Training	4	1	6	اجتياز ٧١ ساعة

سادسا : المشروع

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/ عملي	المتطلبات السابقة
GP401	مشروع ١	Project 1	3	1	4	اجتياز ٩٣ ساعة معتمدة
GP402	مشروع ٢	Project 2	3	1	4	GP401

سابعاً: الخطة الدراسية المقترحة لبرنامج هندسة البرمجيات

٣- المستوى الأول (Freshman) (٣٦ ساعة معتمدة)

(Freshman - Fall) الفصل الدراسي الأول				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS101	رياضيات ١ لعلوم الحاسب	Mathematics 1 for Computer Science	--	3
BS102	مقدمة الحاسبات والمعلوماتية	Introduction to Computing and Informatics	--	3
IT101	أساسيات الإلكترونيات والدوائر الكهربائية	Fundamentals of Electronics and Electrical Circuits	--	3
CS101	مقدمة للبرمجة وحل المشكلات	Introduction to Programming and Problem Solving	--	3
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	--	3
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	--	3
UN103	القضايا المجتمعية	Societal Issues	--	0
18	المجموع			
(Freshman – Spring) الفصل الدراسي الثاني				
BS103	رياضيات ٢ لعلوم الحاسب	Mathematics 2 for Computer Science	--	3
BS104	احتمالات وتوزيعات إحصائية	Probability and Statistics for Computer Science	--	3
CS102	تصميم وبرمجة شبيهة التوجه	Object-Oriented Design and Programming	Introduction to Programming and Problem Solving (CS101)	3
IS101	مقدمة لبرمجة الويب	Introduction to Web Programming	Introduction to Programming and Problem Solving (CS101)	3
IT102	اتصالات البيانات	Data Communication	--	3
IT103	دوائر المنطق الرقمية	Digital Logic Circuit	Fundamentals of Electronics and Electrical Circuits (IT101)	3
18	المجموع			

٤- المستوى الثاني (Sophomore) (٣٦ ساعة معتمدة)

الفصل الدراسي الأول (Sophomore - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS201	رياضيات ٣ لعلوم الحاسب	Mathematics 3 for Computer Science	--	3
BS202	بحوث العمليات	Operation Research	--	3
UN201	ريادة الأعمال التكنولوجية	Technology Entrepreneurship	--	3
CS201	هياكل البيانات والخوارزميات	Data Structures and Algorithms	Introduction to Programming and Problem Solving (CS101)	3
IT201	شبكات الحاسب	Computer Networks	Data Communication (IT102)	3
CS203	هندسة البرمجيات	Software Engineering	Introduction to Programming and Problem Solving (CS101)	3
18	المجموع			
الفصل الدراسي الثاني (Sophomore – Spring)				
CS202	نظم التشغيل	Operating Systems	---	3
CS204	أساسيات الذكاء الاصطناعي	Fundamentals of Artificial Intelligence	---	3
IS201	تحليل وإدارة البيانات	Data Analysis and Management	Probability and Statistics for Computer Science (BS104)	3
IS202	مقدمة لنظم قواعد البيانات	Introduction to Database Systems	Data Structures and Algorithms (CS201)	3
SE201	إدارة عمليات الأعمال	Business Process Management	--	3
UN10x	اختياري جامعة	University Elective	--	3
18	المجموع			
الفصل الدراسي الصيفي (Summer)				
TR201	التدريب الصيفي	Summer Training	اجتياز ٧١ ساعة	4

٥- المستوى الثالث (Junior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
SE301	تصميم ومعمارية البرمجيات	Software Design and Architecture	Software Engineering (CS203)	3
SE302	تحليل متطلبات البرمجة	Software Requirement Analysis	Software Engineering (CS203)	3
IS303	معمارية المؤسسات	Enterprise Architecture	--	3
IS304	برمجة الويب المتقدمة	Advanced Web Programming	Introduction to Web Programming (IS101)	3
CS306	النظم الموزعة	Distributed Systems	Operating Systems (CS202)	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IS306	طرق تطوير النظم	System Development Methods	Software Engineering (CS203)	3
CS301	تطوير برمجيات الموبايل	Mobile Application Development	Software Engineering (CS203)	3
IS308	التنقيب في البيانات	Data Mining	Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	3
CS307	تفاعل الإنسان مع الحاسب	Human Computer Interaction	Object-Oriented Design and Programming (CS102)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
15	المجموع			

٦- المستوى الرابع (Senior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
SE401	بناء البرمجيات	Software Construction	Software Design and Architecture (SE301)	3
SE402	تقدير تكاليف تطوير وصيانة مشاريع البرمجيات	Cost Estimation of Software Projects' Development and Maintenance	Software Requirement Analysis (SE302)	3
SE403	تطوير البرمجيات مفتوحة المصدر	Open-Source Software Development	Software Design and Architecture (SE301)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
CS415	اختبار البرمجيات وتوكيد الجودة	Software Testing and Quality Assurance	Software Engineering (CS203)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

برنامج المعلوماتية الطبية

الرؤية:

أن يكون برنامج المعلوماتية الطبية مرجعاً أكاديمياً وبحثياً في استخدام التكنولوجيا لتحسين جودة الرعاية الصحية، وتطوير أنظمة صحية ذكية تساهم في تعزيز صحة الأفراد والمجتمعات.

الرسالة:

يعمل برنامج المعلوماتية الطبية على إعداد كوادر متميزة تمتلك المعرفة والمهارات اللازمة لتطوير وتطبيق حلول تقنية فعالة وآمنة، من خلال تعليم حديث ومواكب للتطورات، وبحث علمي موجه لحل المشكلات الصحية، وشراكات مع مؤسسات صحية وبحثية، لدعم تقديم رعاية صحية عالية الجودة.

الأهداف:

يهدف برنامج المعلوماتية الطبية إلى تزويد الطلاب بالمعرفة والمهارات اللازمة لتطبيق التكنولوجيا في المجال الطبي، بما يساهم في تحسين جودة الرعاية الصحية وتسهيل اتخاذ القرارات الطبية المبنية على البيانات. ويعمل البرنامج على تحقيق الأهداف الاستراتيجية التالية:

١. إعداد خريجين مؤهلين في مجال المعلوماتية الطبية: تزويد الطلاب بالمعارف الشاملة والمهارات العملية التي تمكنهم من تقديم حلول تقنية مبتكرة لتحليل وإدارة البيانات الصحية، مما يجعلهم مؤهلين لتلبية احتياجات سوق العمل المتزايدة في قطاع المعلوماتية الصحية محلياً ودولياً.
٢. تطوير مناهج تعليمية تتماشى مع أحدث المستجدات: تحديث وتطوير المناهج الدراسية باستمرار بما يعكس التطورات الحديثة في المعلوماتية الصحية وتحليل البيانات الصحية، مع التركيز على الذكاء الاصطناعي، وتعلم الآلة، وتطبيقات البيانات الكبيرة في القطاع الطبي.
٣. تعزيز البحث العلمي في مجال المعلوماتية الطبية: تشجيع الطلاب وأعضاء هيئة التدريس على الانخراط في البحوث التطبيقية التي تركز على حلول معلوماتية لتحسين الرعاية الصحية وإدارة المعلومات الصحية، وتعزيز بيئة إبداعية تساهم في تطوير تقنيات تساعد على تحسين الخدمات الطبية.
٤. بناء شراكات مع المؤسسات الطبية والمراكز البحثية: تعزيز التعاون مع المستشفيات، وشركات التكنولوجيا الصحية، والمؤسسات البحثية، بهدف تزويد الطلاب بفرص تدريب ميداني في بيئات طبية حقيقية، مما يعزز من مهاراتهم العملية ويساهم في سد الفجوة بين الدراسة الأكاديمية والتطبيق العملي.

٥. دعم التطوير المهني والتعلم المستمر: تهيئة بيئة تعليمية تشجع الطلاب والخريجين على التعلم مدى الحياة من خلال ورش العمل والدورات التدريبية، ما يساعدهم على التكيف مع المستجدات التقنية وتعزيز مساراتهم المهنية في مجال المعلوماتية الطبية.
٦. ترسيخ المسؤولية الأخلاقية والمهنية في التعامل مع البيانات الصحية: غرس قيم أخلاقيات العمل في المعلوماتية الصحية، مع التركيز على أهمية الحفاظ على خصوصية وأمان المعلومات الصحية، وتعريف الطلاب بالتحديات القانونية والأخلاقية المتعلقة بإدارة البيانات الطبية.
٧. تعزيز مكانة البرنامج محليًا ودوليًا: تحسين التصنيف الأكاديمي للبرنامج من خلال تبني معايير الجودة العالمية، وتشجيع التعاون البحثي والتبادل الأكاديمي مع الجامعات الرائدة عالميًا في مجال المعلوماتية الطبية، بما يعزز من سمعة البرنامج ويسهم في توسيع شبكته الدولية.

سمات خريجي برنامج المعلوماتية الطبية

يتميز خريجي برنامج المعلوماتية الطبية بامتلاكهم معرفة شاملة في التقنيات المعلوماتية وتطبيقاتها في المجال الطبي، مع فهم عميق للبيولوجيا والمجالات الصحية المختلفة. بالإضافة لتمتعهم بمهارات تحليلية لحل المشكلات الصحية باستخدام التكنولوجيا، والقدرة على العمل ضمن فرق متعددة التخصصات، والتواصل الفعال مع العاملين في القطاع الصحي. لدى خريجي برنامج المعلوماتية الطبية اهتمام بالتعلم المستمر لمواكبة التطورات في كل من مجالي التكنولوجيا والصحة، ويتميزون بأخلاقيات مهنية عالية، مما يؤهلهم للعمل كخبراء تقنيين في المستشفيات والصحة العامة.

بالإضافة إلى السمات العامة لخريجي كلية الحاسبات والمعلومات فبعد الانتهاء بنجاح من برنامج المعلوماتية الطبية ، يجب أن يكون الخريج قادرًا على:

١. جمع وتنظيم وتحليل البيانات الصحية لاستخلاص رؤى مفيدة تساعد في تحسين الرعاية الصحية.
٢. استخدام أدوات وتقنيات تحليل البيانات لتفسير المعلومات الطبية بدقة.
٣. تطوير نظم إدارة المعلومات الصحية (HIS) التي تسهل الوصول إلى البيانات وتحسين العمليات داخل المؤسسات الطبية.
٤. تصميم وتطوير قواعد بيانات صحية آمنة وفعالة لحفظ واسترجاع المعلومات الطبية.
٥. تطبيق التكنولوجيا لتحسين إجراءات الرعاية الصحية وتسهيل التواصل بين مقدمي الخدمات الصحية.
٦. تنفيذ أنظمة الصحة الإلكترونية (eHealth) والتطبيب عن بعد (telemedicine) لدعم تقديم الرعاية الصحية عن بُعد.
٧. فهم والالتزام باللوائح والمعايير الصحية المحلية والدولية المتعلقة بإدارة البيانات الصحية وحمايتها.

٨. ضمان سرية وأمان المعلومات الصحية الحساسة وحمايتها من الاختراق.
 ٩. تطوير أنظمة دعم اتخاذ القرار السريري (CDSS) لمساعدة الأطباء في تشخيص الأمراض واتخاذ القرارات العلاجية.
 ١٠. استخدام الذكاء الاصطناعي وتعلم الآلة لتحليل البيانات الصحية وتقديم توصيات علاجية مستندة إلى البيانات.
 ١١. قيادة وتنسيق مشاريع تقنية المعلومات الصحية لضمان تنفيذها بنجاح وفي الوقت المحدد.
 ١٢. العمل بفعالية مع فرق متعددة التخصصات لضمان تحقيق أهداف المشاريع الصحية.
 ١٣. التواصل بفعالية مع العاملين في المجال الصحي لفهم احتياجاتهم التقنية وتوفير الحلول المناسبة.
 ١٤. التعاون مع فرق تقنية وطبية لتطوير وتحسين الأنظمة الصحية.
 ١٥. متابعة التطورات الحديثة في مجال المعلوماتية الصحية وتطبيقها لتحسين الأداء والكفاءة.
 ١٦. الالتزام بالتعلم المستمر لتطوير المهارات والبقاء على اطلاع بأحدث التقنيات والممارسات.
- هذه القدرات تُمكن خريجي برنامج المعلوماتية الطبية من تقديم إسهامات مهمة في تحسين نظم الرعاية الصحية، وتعزيز جودة الخدمات الصحية، وتحقيق نتائج أفضل للمرضى.

المتطلبات الأكاديمية**أولاً: متطلبات الجامعة (١٢ ساعة معتمدة)****أ- متطلبات الجامعة الإلزامية (٩ ساعات معتمدة)**

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	3	2	2	---
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	3	2	2	---
UN202	أسس البيولوجيا	Fundamental of Biology	3	2	2	---

ب- متطلبات الجامعة الاختيارية (٣ ساعات معتمدة)

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
UN105	قانون لعلوم الحاسب	Law for Computer Science	3	2	2	---
UN106	اقتصاد لعلوم الحاسب	Economics for Computer Science	3	2	2	---
UN107	محاسبة لعلوم الحاسب	Accounting for Computer Science	3	2	2	---
UN108	علم الاجتماع لعلوم الحاسب	Sociology for Computer Science	3	2	2	---
UN109	مبادئ الإدارة	Fundamental of Management	3	2	2	---

ت- مقررات بدون ساعات معتمدة

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
UN103	القضايا المجتمعية	Societal Issues	0	1	--	--

ثانياً: متطلبات العلوم الأساسية (٢٤ ساعة معتمدة)

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Mathematics 1 for Computer Science	رياضيات ١ لعلوم الحاسب	BS101
---	2	2	3	Introduction to Computing and Informatics	مقدمة الحاسبات والمعلوماتية	BS102
---	2	2	3	Mathematics 2 for Computer Science	رياضيات ٢ لعلوم الحاسب	BS103
---	2	2	3	Probability and Statistics for Computer Science	احتمالات وتوزيعات إحصائية لعلوم الحاسب	BS104
---	2	2	3	Fundamentals of Electronics and Electrical Circuits	أساسيات الإلكترونيات والدوائر الكهربائية	IT101
---	2	2	3	Data Communication	اتصالات البيانات	IT102
---	2	2	3	Mathematics 3 for Computer Science	رياضيات ٣ لعلوم الحاسب	BS201
---	2	2	3	Pathology and Epidemiology	علم الأمراض والأوبئة	BS203

ثالثا: متطلبات الكلية (٣٩ ساعة معتمدة)

المتطلبات السابقة	الساعات المعتمدة	محاضرة	تمارين/عملي	Course	اسم المادة	كود المادة
---	3	2	2	Introduction to Programming and Problem Solving	مقدمة للبرمجة وحل المشكلات	CS101
Introduction to Programming and Problem Solving (CS101)	3	2	2	Object-Oriented Design and Programming	تصميم وبرمجة شبيهة التوجه	CS102
Introduction to Programming and Problem Solving (CS101)	3	2	2	Data Structures and Algorithms	هياكل البيانات والخوارزميات	CS201
---	3	2	2	Operating Systems	نظم التشغيل	CS202
Introduction to Programming and Problem Solving (CS101)	3	2	2	Software Engineering	هندسة البرمجيات	CS203
---	3	2	2	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	CS204
Introduction to Programming and Problem Solving (CS101)	3	2	2	Introduction to Web Programming	مقدمة لبرمجة الويب	IS101
Probability and Statistics for Computer Science (BS104)	3	2	2	Data Analysis and Management	تحليل وإدارة البيانات	IS201
Data Structures and Algorithms (CS201)	3	2	2	Introduction to Database Systems	مقدمة لنظم قواعد البيانات	IS202
--	3	2	2	Project Management	إدارة المشروعات	IS203
Fundamentals of Electronics and Electrical Circuits (IT101)	3	2	2	Digital Logic Circuit	دوائر المنطق الرقمية	IT103
Data Communication (IT102)	3	2	2	Computer Networks	شبكات الحاسب	IT201
--	3	2	2	Signal Processing	معالجة الإشارات	IT301

رابعاً: متطلبات تخصص المعلوماتية الطبية - ٥٧ ساعة معتمدة
٣- متطلبات تخصص المعلوماتية الطبية الإجبارية (٤٢ ساعة معتمدة)

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Data Structures and Algorithms (CS201)	2	2	3	Bioinformatics	المعلوماتية الحيوية	CS305
Data Structures and Algorithms (CS201)	2	2	3	Healthcare Informatics	معلوماتية الرعاية الصحية	CS323
Data Structures and Algorithms (CS201)	2	2	3	Biomedical Informatics	المعلوماتية الطبية	CS324
Bioinformatics (CS305)	2	2	3	Advanced Genomics and Genetic Analysis	علم الجينوم المتقدم والتحليل الوراثي	CS411
Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	2	2	3	Data Mining	التنقيب في البيانات	IS308
Data Analysis and Management (IS201)	2	2	3	Decision Support System	نظم دعم اتخاذ القرار	IS412
--	2	2	3	Digital Image Processing	معالجة الصور الرقمية	IT307
Computer Networks (IT201)	2	2	3	Computer and Network Security	أمن الحاسبات والشبكات	IT309
Digital Logic Circuit (IT103)	2	2	3	Internet of Things	إنترنت الأشياء	IT410
Digital Image Processing (IT307)	2	2	3	Medical Image Analysis	تحليل الصور الطبية	IT411
Fundamental of Biology (UN202)	2	2	3	Genetics	علم الجينات	MI201
Introduction to Database Systems (IS202)	2	2	3	E-Health Record	السجلات الصحية الإلكترونية	MI301
--	2	2	3	Hospital Management Systems	نظام إدارة المستشفيات	MI302
Healthcare Informatics (CS323)	2	2	3	Health Information Systems	نظم المعلومات الصحية	MI303

٤- متطلبات تخصص المعلوماتية الطبية الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/عملي	المتطلبات السابقة
BS207	بحوث العمليات	Operation Research	3	2	2	---
CS318	الأنظمة الخبيرة وهندسة المعرفة	Expert Systems and Knowledge Engineering	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS416	هندسة الأنطولوجي	Ontology Engineering	3	2	2	Fundamentals of Artificial Intelligence (CS204)
CS421	المقاييس الحيوية	Biometrics	3	2	2	Data Structures and Algorithms (CS201)
IS312	إدارة المخاطر والأزمات	Risk and Crisis Management	3	2	2	Project Management (IS203)
IS404	تطبيقات علوم البيانات وتحليلها	Data Science Applications and analysis	3	2	2	Data Analysis and Management (IS201)
IT302	التعرف على الأنماط	Pattern Recognition	3	2	2	--
IT311	الواقع الافتراضي	Virtual Reality	3	2	2	--
IT401	الرؤية بالحاسب	Computer Vision	3	2	2	Digital Image Processing (IT307)
IT403	الوسائط المتعددة	Multimedia	3	2	2	--
IT404	الأنظمة المضمنة	Embedded Systems	3	2	2	Digital Logic Circuit (IT103)
IT408	سلسلة الكتلة	Blockchain	3	2	2	--
IT412	الحوسبة السحابية	Cloud Computing	3	2	2	--
IT416	برمجة أنظمة الرقائق الإلكترونية	System-on Chip (SoC) Programming	3	2	2	Introduction to Programming and Problem Solving (CS101)
IT423	معالجة الصور – ٢	Digital Image Processing – 2	3	2	2	Digital Image Processing (IT307)
MI304	إدارة الجودة في المستشفيات	Quality Management in Hospitals	3	2	2	Hospital Management Systems(MI)
MI305	العلاج عن بعد والصحة المتنقلة	Telemedicine and Mobile Health	3	2	2	--

MI401	موضوعات مختارة في المعلوماتية الطبية ١ -	Selected Topics in Medical Informatics - 1	3	2	2	---
MI402	موضوعات مختارة في المعلوماتية الطبية ٢ -	Selected Topics in Medical Informatics- 2	3	2	2	---

خامسا : التدريب الصيفي

جدول ٢٤: التدريب الصيفي.

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/ عملي	المتطلبات السابقة
TR201	التدريب الصيفي	Summer Training	4	1	6	اجتياز ٧١ ساعة

سادسا : المشروع

جدول ٢٥: المشروع

كود المادة	اسم المادة	Course	الساعات المعتمدة	محاضرة	تمارين/ عملي	المتطلبات السابقة
GP401	مشروع ١	Project 1	3	1	4	اجتياز ٩٣ ساعة معتمدة
GP402	مشروع ٢	Project 2	3	1	4	GP401

سابعاً: الخطة الدراسية المقترحة لبرنامج المعلوماتية الطبية**١- المستوى الأول (Freshman) (٣٦ ساعة معتمدة)**

(Freshman - Fall) الفصل الدراسي الأول				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS101	رياضيات ١ لعلوم الحاسب	Mathematics 1 for Computer Science	--	3
BS102	مقدمة الحاسبات والمعلوماتية	Introduction to Computing and Informatics	--	3
IT101	أساسيات الإلكترونيات والدوائر الكهربائية	Fundamentals of Electronics and Electrical Circuits	--	3
CS101	مقدمة للبرمجة وحل المشكلات	Introduction to Programming and Problem Solving	--	3
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	--	3
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	--	3
UN103	القضايا المجتمعية	Societal Issues	--	0
18	المجموع			
(Freshman – Spring) الفصل الدراسي الثاني				
BS103	رياضيات ٢ لعلوم الحاسب	Mathematics 2 for Computer Science	--	3
BS104	احتمالات وتوزيعات إحصائية لعلوم الحاسب	Probability and Statistics for Computer Science	--	3
CS102	تصميم وبرمجة شئية التوجه	Object-Oriented Design and Programming	Introduction to Programming and Problem Solving (CS101)	3
IS101	مقدمة لبرمجة الويب	Introduction to Web Programming	Introduction to Programming and Problem Solving (CS101)	3
IT102	اتصالات البيانات	Data Communication	--	3
IT103	دوائر المنطق الرقمية	Digital Logic Circuit	Fundamentals of Electronics and Electrical Circuits (IT101)	3
18	المجموع			

٢- المستوى الثاني (Sophomore) (٣٦ ساعة معتمدة)

الفصل الدراسي الأول (Sophomore - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS201	رياضيات ٣ لعلوم الحاسب	Mathematics 3 for Computer Science	--	3
UN202	أسس البيولوجيا	Fundamental of Biology	---	3
BS203	علم الأمراض والأوبئة	Pathology and epidemiology	---	3
CS201	هياكل البيانات والخوارزميات	Data Structures and Algorithms	Introduction to Programming and Problem Solving (CS101)	3
IT201	شبكات الحاسب	Computer Networks	Data Communication (IT102)	3
CS203	هندسة البرمجيات	Software Engineering	Introduction to Programming and Problem Solving (CS101)	3
18	المجموع			
الفصل الدراسي الثاني (Sophomore – Spring)				
CS202	نظم التشغيل	Operating Systems	---	3
CS204	أساسيات الذكاء الاصطناعي	Fundamentals of Artificial Intelligence	---	3
IS201	تحليل وإدارة البيانات	Data Analysis and Management	Probability and Statistics for Computer Science (BS104)	3
IS202	مقدمة لنظم قواعد البيانات	Introduction to Database Systems	Data Structures and Algorithms (CS201)	3
MI201	علم الجينات	Genetics	Fundamental of Biology (UN202)	3
UN10x	اختياري جامعة	University Elective	--	3
18	المجموع			
الفصل الدراسي الصيفي (Summer)				
TR201	التدريب الصيفي	Summer Training	اجتياز ٧١ ساعة	4

٣- المستوى الثالث (Junior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
IT301	معالجة الإشارات	Signal Processing	--	3
CS323	معلوماتية الرعاية الصحية	Healthcare Informatics	Data Structures and Algorithms (CS201)	3
CS324	المعلوماتية الطبية	Biomedical Informatics	Data Structures and Algorithms (CS201)	3
MI301	السجلات الصحية الإلكترونية	E-Health Record	Introduction to Database Systems (IS202)	3
MI302	نظام إدارة المستشفيات	Hospital Management Systems	--	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
MI303	نظم المعلومات الصحية	Health Information Systems	Healthcare Informatics (CS323)	3
CS305	المعلوماتية الحيوية	Bioinformatics	Data Structures and Algorithms (CS201)	3
IT307	معالجة الصور الرقمية	Digital Image Processing	---	3
IS308	التنقيب في البيانات	Data Mining	Data Analysis and Management (IS201), Introduction to Database Systems (IS202)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
15	المجموع			

٤- المستوى الرابع (Senior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IT410	إنترنت الأشياء	Internet of Things	Digital Logic Circuit (IT103)	3
CS411	علم الجينوم المتقدم والتحليل الوراثي	Advanced Genomics and Genetic Analysis	----	3
IS412	نظم دعم اتخاذ القرار	Decision Support System	Data Analysis and Management (IS201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
IT411	تحليل الصور الطبية	Medical Image Analysis	Digital Image Processing (IT307)	3
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

برنامج الذكاء الاصطناعي

الرؤية:

أن يصبح برنامج الذكاء الاصطناعي مركزًا للتميز والابتكار، وأن يسهم في قيادة التغيير الرقمي على المستويين المحلي والإقليمي والدولي عبر تخريج أجيال قادرة على ابتكار وتطوير تقنيات ذكية مستدامة.

الرسالة:

يهدف برنامج الذكاء الاصطناعي إلى تقديم تعليم متطور في مجالات الذكاء الاصطناعي، وتعزيز بيئة بحثية تدعم الابتكار والإبداع، وتأسيس شركات استراتيجية مع القطاع الصناعي، بهدف إعداد خريجين يمتلكون القدرة على تطوير تقنيات متقدمة تدعم التحول الرقمي وتحل التحديات المجتمعية.

الأهداف:

يهدف برنامج الذكاء الاصطناعي إلى إعداد كوادر مؤهلة قادرة على قيادة وتطوير تطبيقات الذكاء الاصطناعي وتحليل البيانات الضخمة بما يلي الاحتياجات المتنامية في مختلف القطاعات، ويعزز من تنافسية المجتمع الرقمي. ويسعى البرنامج إلى تحقيق الأهداف الاستراتيجية التالية:

1. إعداد خريجين يمتلكون مهارات الذكاء الاصطناعي الحديثة: تقديم تعليم عالي الجودة يركز على تنمية مهارات البرمجة، وتحليل البيانات، وتطوير الخوارزميات الذكية، بهدف تأهيل الطلاب للابتكار وتقديم حلول تقنية متقدمة في مجالات متعددة.
2. تطوير المناهج لتواكب التطورات السريعة في الذكاء الاصطناعي: تحديث المناهج بشكل مستمر لتشمل أحدث الابتكارات والتقنيات في مجالات تعلم الآلة، والرؤية الحاسوبية، ومعالجة اللغة الطبيعية، والروبوتات الذكية، بما يضمن تزويد الطلاب بأحدث المعارف والتقنيات التي تواكب التطورات العالمية.
3. تعزيز البحث العلمي والابتكار في الذكاء الاصطناعي: دعم الأبحاث التطبيقية والمشاريع الابتكارية في مجال الذكاء الاصطناعي، وتشجيع الطلاب وأعضاء هيئة التدريس على المساهمة في تطوير حلول ذكية تلبي احتياجات المجتمع وتساهم في مواجهة التحديات التقنية المعاصرة.
4. إقامة شراكات استراتيجية مع مؤسسات التكنولوجيا والصناعة: بناء علاقات تعاون مع شركات التكنولوجيا ومؤسسات الصناعة، بهدف توفير فرص تدريبية ومشاريع مشتركة للطلاب في بيئات عمل حقيقية، وتعزيز التفاعل بين الجوانب الأكاديمية والعملية في تطبيقات الذكاء الاصطناعي.
5. دعم التعلم المستمر والتطوير المهني للطلاب والخريجين: إنشاء بيئة تعليمية تشجع الطلاب على التعلم مدى الحياة من خلال ورش العمل والدورات التدريبية المتخصصة،

لمساعدتهم في البقاء على اطلاع دائم بأحدث التقنيات، وتعزيز فرصهم في التطور المهني المستمر.

٦. **ترسيخ المسؤولية الأخلاقية والمهنية في تطبيقات الذكاء الاصطناعي:** تعزيز وعي الطلاب بالتحديات الأخلاقية والاجتماعية المرتبطة بتطبيقات الذكاء الاصطناعي، مع التركيز على تطوير تقنيات تحترم الخصوصية وتلتزم بمعايير السلامة والشفافية.

٧. **تحقيق مكانة متميزة للبرنامج محليًا ودوليًا:** العمل على رفع تصنيف البرنامج من خلال تبني معايير الجودة العالمية، وتشجيع التعاون الأكاديمي والبحثي مع الجامعات والمؤسسات البحثية العالمية الرائدة في مجال الذكاء الاصطناعي، بما يعزز من سمعة البرنامج ويدعم قدراته في جذب الطلاب والباحثين المتميزين.

سمات خريجي برنامج الذكاء الاصطناعي

خريجي برنامج الذكاء الاصطناعي يتميزون بقدرة فائقة على فهم وتطبيق تقنيات الذكاء الاصطناعي والتعلم الآلي ومعالجة البيانات الضخمة. يمتلكون مهارات برمجية متقدمة ومعرفة عميقة بالخوارزميات ونماذج التعلم العميق والشبكات العصبية. يتسمون بمهارات تحليلية لحل المشكلات المعقدة وابتكار حلول ذكية، ولديهم قدرة على العمل الجماعي والتعاون مع فرق متعددة التخصصات. يتميزون أيضًا بالرغبة في التعلم المستمر ومواكبة التطورات السريعة في مجال الذكاء الاصطناعي، فضلاً عن التزامهم بأخلاقيات مهنية عالية، مما يؤهلهم للابتكار في مختلف الصناعات والمساهمة في تطوير تقنيات مستقبلية.

بالإضافة إلى السمات العامة لخريجي كلية الحاسبات والمعلومات فبعد الانتهاء بنجاح من برنامج الذكاء الاصطناعي، يجب أن يكون الخريج قادراً على:

١. تصميم وتنفيذ الحلول التكنولوجية المناسبة للمشكلات العصرية المتعلقة بمجال الذكاء الاصطناعي.
٢. المنافسة محلياً وإقليمياً على الوظائف المتخصصة في هذا المجال.
٣. الالتزام بأخلاقيات المهنة خلال الممارسة العلمية في مجال الذكاء الاصطناعي.
٤. تطبيق التفكير النقدي والإبداعي لحل المشكلات التقنية المعقدة بفعالية.
٥. استخدام تقنيات الذكاء الاصطناعي لتحسين العمليات والأداء في مختلف الصناعات.
٦. التعاون بفعالية مع فرق متعددة التخصصات في مشاريع الذكاء الاصطناعي.
٧. متابعة التطورات الحديثة في مجال الذكاء الاصطناعي وتطبيقها في العمل.
٨. الابتكار في تطوير حلول جديدة وتقنيات متقدمة لتلبية احتياجات السوق.
٩. فهم التحديات الأخلاقية والاجتماعية المتعلقة بتطبيقات الذكاء الاصطناعي والعمل على معالجتها بفعالية.

هذه القدرات تمكن خريجي برنامج الذكاء الاصطناعي من أن يكونوا مهنيين مبتكرين وفعالين في مجالات متعددة، مما يساعدهم على تقديم إسهامات قيمة في تطوير تقنيات المستقبل.

المتطلبات الأكاديميةأولاً: متطلبات الجامعة (١٢ ساعة معتمدة)أ- متطلبات الجامعة الإلزامية (٩ ساعات معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Communication Skills, Ethical and Professional Issues	مهارات الاتصال والجوانب الأخلاقية والمهنية	UN101
---	2	2	3	English for Computer Science	لغة إنجليزية لعلوم الحاسب	UN102
---	2	2	3	Technology Entrepreneurship	ريادة الأعمال التكنولوجية	UN201

ب- متطلبات الجامعة الاختيارية (٣ ساعات معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Design Physiology for Computer Science	علم نفس التصميم لعلوم الحاسب	UN104
---	2	2	3	Law for Computer Science	قانون لعلوم الحاسب	UN105
---	2	2	3	Economics for Computer Science	اقتصاد لعلوم الحاسب	UN106
---	2	2	3	Accounting for Computer Science	محاسبة لعلوم الحاسب	UN107
---	2	2	3	Sociology for Computer Science	علم الاجتماع لعلوم الحاسب	UN108

ت- مقررات بدون ساعات معتمدة

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
--	--	1	0	Societal Issues	القضايا المجتمعية	UN103

ثانياً: متطلبات العلوم الأساسية (٢٤ ساعة معتمدة)

جدول ٢٩: متطلبات العلوم الأساسية (إجبارية) لبرنامج الذكاء الاصطناعي

المتطلبات السابقة	تمارين/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Mathematics 1 for Computer Science	رياضيات ١ لعلوم الحاسب	BS101
---	2	2	3	Introduction to Computing and Informatics	مقدمة الحاسبات والمعلوماتية	BS102
---	2	2	3	Mathematics 2 for Computer Science	رياضيات ٢ لعلوم الحاسب	BS103
---	2	2	3	Probability and Statistics for Computer Science	احتمالات وتوزيعات إحصائية لعلوم الحاسب	BS104
---	2	2	3	Mathematics 3 for Computer Science	رياضيات ٣ لعلوم الحاسب	BS201
---	2	2	3	Operation Research	بحوث العمليات	BS202
---	2	2	3	Fundamentals of Electronics and Electrical Circuits	أساسيات الإلكترونيات والدوائر الكهربائية	IT101
---	2	2	3	Data Communication	اتصالات البيانات	IT102

ثالثا: متطلبات الكلية (٣٩ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Introduction to Programming and Problem Solving	مقدمة للبرمجة وحل المشكلات	CS101
Introduction to Programming and Problem Solving (CS101)	2	2	3	Object-Oriented Design and Programming	تصميم وبرمجة شيئية التوجه	CS102
Introduction to Programming and Problem Solving (CS101)	2	2	3	Data Structures and Algorithms	هياكل البيانات والخوارزميات	CS201
---	2	2	3	Operating Systems	نظم التشغيل	CS202
Introduction to Programming and Problem Solving (CS101)	2	2	3	Software Engineering	هندسة البرمجيات	CS203
---	2	2	3	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	CS204
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Logic Programming and Automated Reasoning	البرمجة المنطقية والاستدلال الألي	CS317
Introduction to Programming and Problem Solving (CS101)	2	2	3	Introduction to Web Programming	مقدمة لبرمجة الويب	IS101
Probability and Statistics for Computer Science (BS104)	2	2	3	Data Analysis and Management	تحليل وإدارة البيانات	IS201
Data Structures and Algorithms (CS201)	2	2	3	Introduction to Database Systems	مقدمة لنظم قواعد البيانات	IS202
--	2	2	3	Project Management	إدارة المشروعات	IS203
Fundamentals of Electronics and Electrical Circuits (IT101)	2	2	3	Digital Logic Circuit	دوائر المنطق الرقمية	IT103
Data Communication (IT102)	2	2	3	Computer Networks	شبكات الحاسب	IT201

رابعاً: متطلبات تخصص الذكاء الاصطناعي - ٥٧ ساعة معتمدة**١ - متطلبات تخصص الذكاء الاصطناعي الإجبارية (٤٢ ساعة معتمدة)**

المتطلبات السابقة	تمايز/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Fundamentals of Artificial Intelligence (CS205), Probability and Statistics for Computer Science (BS105)	2	2	3	Machine Learning	تعلم الاله	CS302
Data Structures and Algorithms (CS201)	2	2	3	Analysis and Design of Algorithms	تحليل وتصميم الخوارزميات	CS303
Object-Oriented Design and Programming (CS102)	2	2	3	Human Computer Interaction	تفاعل الإنسان مع الحاسب	CS307
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Deep Learning	التعلم العميق	CS309
Deep Learning (CS309)	2	2	3	Generative Artificial Intelligence	الذكاء الاصطناعي التوليدي	CS313
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Natural Language Processing and Arabizations	معالجة اللغات الطبيعية والتعريب	CS401
Data Analysis and Management (IS201)	2	2	3	Data Science Applications and analysis	تطبيقات علوم البيانات وتحليلها	IS404
--	2	2	3	Digital Image Processing	معالجة الصور الرقمية	IT307
Computer Networks (IT201)	2	2	3	Computer and Network Security	أمن الحاسبات والشبكات	IT309
--	2	2	3	Speech Recognition	التعرف على الكلام	IT310
--	2	2	3	Virtual Reality	الواقع الافتراضي	IT311
Digital Image Processing (IT307)	2	2	3	Computer Vision	الرؤية بالحاسب	IT401
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Robotics 1	علم الإنسان الآلي (1)	IT402
Digital Logic Circuit (IT103)	2	2	3	Internet of Things	إنترنت الأشياء	IT410

٢- متطلبات تخصص الذكاء الاصطناعي الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Human Computer Interaction (CS307)	2	2	3	Ambient Intelligence	الذكاء المحيطي	CS316
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Expert Systems and Knowledge Engineering	الأنظمة الخبيرة وهندسة المعرفة	CS318
Object-Oriented Design and programming (CS102), Probability and Statistics for Computer Science(BS105)	2	2	3	Modeling and Simulation	النمذجة والمحاكاة	CS321
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Distributed Intelligent Automation Systems	أنظمة الأتمتة الموزعة الذكية	CS408
--	2	2	3	Introduction to Formal Languages and Automata Theory	مقدمة اللغات الرسمية ونظرية الأتمتة	CS414
Fundamentals of Artificial Intelligence (CS205)	2	2	3	Autonomous Agents	الوكلاء المستقلون	CS417
Machine Learning (CS302)	2	2	3	Reinforcement Learning and Control	التعلم بالتعزيز والتحكم	CS418
Operation Research(BS202)	2	2	3	Advanced Operation Research	بحوث عمليات متقدمة	IS411
Data Analysis and Management (IS201)	2	2	3	Decision Support System	نظم دعم اتخاذ القرار	IS412
--	2	2	3	Pattern Recognition	التعرف على الأنماط	IT302
--	2	2	3	Intelligent and Quantum Computers	الحاسبات الذكية والكمية	IT306
--	2	2	3	Virtual Reality	الواقع الافتراضي	IT311
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Games Development	تطوير الألعاب	IT314
--	2	2	3	Blockchain	سلسلة الكتلة	IT408

Computer Vision (IT401)	2	2	3	Computer Vision Applications	تطبيقات الرؤية بالحاسب	IT409
Robotics 1 (IT402)	2	2	3	Robotics 2	علم الإنسان الألى (2)	IT413
--	2	2	3	Automatic Control	التحكم الآلى	IT425
----	2	2	3	Selected Topic in Artificial Intelligence - 1	موضوعات مختارة في الذكاء الاصطناعي ١ -	AI401
----	2	2	3	Selected Topic in Artificial Intelligence - 2	موضوعات مختارة في الذكاء الاصطناعي ٢ -	AI402

خامسا : التدريب الصيفي

جدول ٣٣: التدريب الصيفي

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
اجتياز ٧١ ساعة	6	1	4	Summer Training	التدريب الصيفي	TR201

سادسا : المشروع

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
اجتياز ٩٣ ساعة معتمدة	4	1	3	Project 1	مشروع ١	GP401
GP401	4	1	3	Project 2	مشروع ٢	GP402

سابعاً: الخطة الدراسية المقترحة لبرنامج الذكاء الاصطناعي**١- المستوى الأول (Freshman) (٣٦ ساعة معتمدة)**

الفصل الدراسي الأول (Freshman - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS101	رياضيات ١ لعلوم الحاسب	Mathematics 1 for Computer Science	--	3
BS102	مقدمة الحاسبات والمعلوماتية	Introduction to Computing and Informatics	--	3
IT101	أساسيات الإلكترونيات والدوائر الكهربائية	Fundamentals of Electronics and Electrical Circuits	--	3
CS101	مقدمة للبرمجة وحل المشكلات	Introduction to Programming and Problem Solving	--	3
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	--	3
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	--	3
UN103	القضايا المجتمعية	Societal Issues	--	0
18	المجموع			
الفصل الدراسي الثاني (Freshman – Spring)				
BS103	رياضيات ٢	Mathematics 2 for Computer Science	--	3
BS104	احتمالات وتوزيعات إحصائية	Probability and Statistics for Computer Science	--	3
CS102	تصميم وبرمجة شبيهة التوجه	Object-Oriented Design and Programming	Introduction to Programming and Problem Solving (CS101)	3
IS101	مقدمة لبرمجة الويب	Introduction to Web Programming	Introduction to Programming and Problem Solving (CS101)	3
IT102	اتصالات البيانات	Data Communication	--	3
IT103	دوائر المنطق الرقمية	Digital Logic Circuit	Fundamentals of Electronics and Electrical Circuits (IT101)	3
18	المجموع			

٢- المستوى الثاني (Sophomore) (٣٦ ساعة معتمدة)

الفصل الدراسي الأول (Sophomore - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS201	رياضيات ٣ لعلوم الحاسب	Mathematics 3 for Computer Science	--	3
BS202	بحوث العمليات	Operation Research	--	3
UN201	ريادة الأعمال التكنولوجية	Technology Entrepreneurship	--	3
CS201	هياكل البيانات والخوارزميات	Data Structures and Algorithms	Introduction to Programming and Problem Solving (CS101)	3
IT201	شبكات الحاسب	Computer Networks	Data Communication (IT102)	3
CS203	هندسة البرمجيات	Software Engineering	Introduction to Programming and Problem Solving (CS101)	3
18	المجموع			
الفصل الدراسي الثاني (Sophomore – Spring)				
CS202	نظم التشغيل	Operating Systems	---	3
CS204	أساسيات الذكاء الاصطناعي	Fundamentals of Artificial Intelligence	---	3
IS201	تحليل وإدارة البيانات	Data Analysis and Management	Probability and Statistics for Computer Science (BS104)	3
IS202	مقدمة لنظم قواعد البيانات	Introduction to Database Systems	Data Structures and Algorithms (CS201)	3
IT311	الواقع الافتراضي	Virtual Reality	--	3
UN10x	اختياري جامعة	University Elective	--	3
18	المجموع			
الفصل الدراسي الصيفي (Summer)				
TR201	التدريب الصيفي	Summer Training	اجتياز ٧١ ساعة	4

٣- المستوى الثالث (Junior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
CS302	تعلم الاله	Machine Learning	Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)	3
CS303	تحليل وتصميم الخوارزميات	Analysis and Design of Algorithms	Data Structures and Algorithms (CS201)	3
IT310	التعرف على الكلام	Speech Recognition	----	3
CS317	البرمجة المنطقية والاستدلال الألى	Logic Programming and Automated Reasoning	Fundamentals of Artificial Intelligence (CS205)	3
IT402	علم الإنسان الآلى(1)	Robotics 1	Fundamentals of Artificial Intelligence (CS204)	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
CS309	التعلم العميق	Deep Learning	Fundamentals of Artificial Intelligence (CS204)	3
CS307	تفاعل الإنسان مع الحاسب	Human Computer Interaction	Object-Oriented Design and Programming (CS102)	3
IT307	معالجة الصور الرقمية	Digital Image Processing	---	3
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
15	المجموع			

٤- المستوى الرابع (Senior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
CS401	معالجة اللغات الطبيعية والتعريب	Natural Language Processing and Arabizations	Fundamentals of Artificial Intelligence (CS204)	3
IT410	إنترنت الأشياء	Internet of Things	Digital Logic Circuit (IT103)	3
IT401	الرؤية بالحاسب	Computer Vision	Digital Image Processing (IT307)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
CS313	الذكاء الاصطناعي التوليدي	Generative Artificial Intelligence	Deep Learning (CS309)	3
IS404	تطبيقات علوم البيانات وتحليلها	Data Science Applications and analysis	Data Analysis and Management (IS201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

برنامج الشبكات والأمن السيبراني

الرؤية:

أن يكون برنامج الشبكات والأمن السيبراني منبرًا رياديًا محليًا ودوليًا في إعداد كوادر مؤهلة ومتميزة في إدارة الشبكات وتأمين الأنظمة الرقمية، بما يساهم في دعم التحول الرقمي لمجتمع آمن ومستدام.

الرسالة:

يهدف برنامج الشبكات والأمن السيبراني إلى تقديم تعليم متطور وشامل في مجال الشبكات وأمن المعلومات، من خلال توفير بيئة تعليمية وبحثية متكاملة، وإقامة شراكات مع قطاعات الصناعة والأمن الرقمي. يركز البرنامج على تزويد الطلاب بالمهارات التقنية والمعرفة الأكاديمية التي تؤهلهم لمواجهة التحديات السيبرانية وحماية البنية التحتية الرقمية، بما يواكب احتياجات سوق العمل المتغيرة.

الأهداف:

يهدف برنامج الشبكات والأمن السيبراني إلى إعداد خريجين مؤهلين يمتلكون المعرفة والمهارات التقنية اللازمة لتصميم وإدارة وتأمين شبكات المعلومات، وذلك عبر توفير تعليم عالي الجودة في مجال الشبكات وأمن المعلومات. يسعى البرنامج إلى تمكين الطلاب من حماية الأنظمة الرقمية والتصدي للتحديات والتهديدات السيبرانية المتزايدة، وذلك بما يواكب التطورات التقنية الحديثة ويستجيب لاحتياجات سوق العمل في القطاعين الحكومي والخاص. ويسعى البرنامج إلى تحقيق الأهداف الاستراتيجية التالية:

١. تأهيل خريجين متخصصين في أمن الشبكات والمعلومات: إعداد خريجين يمتلكون معرفة تقنية متقدمة ومهارات تطبيقية في تصميم وتأمين الشبكات، والتصدي للهجمات السيبرانية، بحيث يكونون مؤهلين لتلبية احتياجات سوق العمل في مجال الأمن السيبراني على المستويين المحلي والعالمي.

٢. تطوير مناهج تعليمية مرنة ومحدثة تواكب التغيرات في مجال الأمن السيبراني: تحديث المناهج الدراسية باستمرار لتشمل أحدث التطورات في مجال الشبكات والتشفير وأمن المعلومات، مع توفير فرص تعلم عملي وتمارين ميدانية تعزز المهارات التقنية وتتيح للطلاب تطبيق معارفهم في بيئات عملية حقيقية.
٣. تعزيز ثقافة البحث العلمي والابتكار في مجال الشبكات والأمن السيبراني: تشجيع البحوث التطبيقية والمشاريع الابتكارية الهادفة إلى إيجاد حلول لمشكلات وتحديات الأمن السيبراني المتطورة، ودعم مبادرات الطلاب وأعضاء هيئة التدريس في هذا المجال لتعزيز دور البرنامج كمركز للبحث والابتكار.
٤. إقامة شراكات استراتيجية مع المؤسسات الرائدة في قطاع الأمن السيبراني: التعاون مع شركات التكنولوجيا والقطاع الحكومي لتقديم فرص تدريب عملي وتوظيف للخريجين، وتبادل الخبرات في تطوير حلول تقنية جديدة، بما يعزز من فرص الطلاب في الدخول إلى سوق العمل بأعلى مستوى من الجاهزية.
٥. غرس القيم الأخلاقية والمسؤولية المهنية في الأمن السيبراني: توعية الطلاب بأهمية أخلاقيات المهنة والمسؤولية المجتمعية في حماية البيانات وخصوصية الأفراد، مع التركيز على الالتزام بالمعايير القانونية والمعايير الأخلاقية في ممارسة عملهم.
٦. تعزيز مكانة البرنامج كمركز تعليمي وبحثي متميز في الأمن السيبراني: لسعي للحصول على اعتمادات أكاديمية دولية، وتعزيز التصنيف الأكاديمي للبرنامج من خلال تفعيل التعاون الأكاديمي والتبادل الطلابي والبحثي مع الجامعات والمؤسسات العالمية الرائدة في مجالي الشبكات والأمن السيبراني.
٧. دعم التطوير المهني للطلاب والخريجين: توفير برامج تعليم مستمر وتطوير مهني للخريجين تمكّنهم من مواكبة التحديثات المتسارعة في الأمن السيبراني، وتعزيز من قدرتهم على التكيف مع التغيرات السريعة في التقنيات والحلول الأمنية الرقمية.

سمات خريجي برنامج الشبكات والأمن السيبراني

يستهدف برنامج الشبكات والأمن السيبراني القدرة على التعامل مع البيانات المختلفة وتحليلها وتشفيرها وحمايتها منذ البدء في تصميم وتطوير الأنظمة والتطبيقات أو أثناء نقلها على الإنترنت أو في أماكن تخزينها. أيضاً، فالأمن السيبراني يستهدف الحفاظ على سلامة البيانات وعدم تعديلها إلى من خلال الأطراف المصرح لها بذلك وبالصلاحيات المسموح بها لكل طرف وتسجيل كل ما يتم من إجراءات على هذه الأنظمة والبيانات من قبل كل طرف لضمان تتبع مسؤولية كل طرف عن الإجراءات التي يقوم بها. أيضاً، يستهدف المجال التأكد من توافر البيانات والأنظمة عند الحاجة إلى الوصول إليها مع القدرة على ضمان استمرارية العمل على الأنظمة والبيانات أثناء وبعد الكوارث مع الاستعداد لكشف الحوادث والانتهاكات التي تستهدفهم من خلال مجموعة متنوعة من أساليب اكتشاف الأدلة الجنائية الرقمية والتعامل مع هذه الانتهاكات وفقاً للقوانين واللوائح الحاكمة.

بالإضافة إلى السمات العامة لخريجي كلية الحاسبات والمعلومات فبعد الانتهاء بنجاح من برنامج الشبكات والأمن السيبراني، يجب أن يكون الخريج قادراً على:

١. معرفة أنواع البيانات المختلفة من صور وفيديوهات وملفات وغيره وطرق التعامل معها وتخزينها وصور نقلها المختلفة من مكان لآخر.
٢. التعرف على طرق التحليل والتشفير المختلفة للبيانات وفقاً لنوع البيانات المراد تأمينها وشكل التأمين المطلوب لها.
٣. الإلمام بمفاهيم أمن البيانات والشبكات المطلوبة (مثل: السرية، السلامة، التوافر، المصادقة وغيرها) أثناء تصميم الأنظمة المختلفة للحفاظ عليها أثناء تطوير الأنظمة المختلفة أو التعامل مع البيانات وتخزينها أو الحاجة لنقلها على وسائل نقل البيانات الرقمية المختلفة.
٤. الإلمام بسبل اكتشاف الحوادث الرقمية والانتهاكات التي قد تتعرض لها الأنظمة والبيانات المختلفة والتعرف على أساليب اكتشاف الأدلة الجنائية الرقمية في حال حدوث أي من هذه الانتهاكات.
٥. التعرف على الأساليب المختلفة لضمان استمرارية العمل والوصول للبيانات أثناء وبعد الكوارث.
٦. الإلمام بالمعايير والمسؤوليات الأخلاقية والقانونية للمهنة.

هذه القدرات تجعل خريج برنامج الشبكات والأمن السيبراني مؤهلاً لتقديم المساعدة في بيئات العمل المختلفة والمساهمة في تطوير أنظمة آمنة والحفاظ على البيانات المختلفة والتصدي للحوادث الرقمية المختلفة والتعامل معها وفقاً للقوانين واللوائح المنوطة بذلك.

المتطلبات الأكاديمية**أولاً: متطلبات الجامعة (١٢ ساعة معتمدة)****أ- متطلبات الجامعة الإلزامية (٩ ساعات معتمدة)**

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Communication Skills, Ethical and Professional Issues	مهارات الاتصال والجوانب الأخلاقية والمهنية	UN101
---	2	2	3	English for Computer Science	لغة إنجليزية لعلوم الحاسب	UN102
---	2	2	3	Technology Entrepreneurship	ريادة الأعمال التكنولوجية	UN201

ب- متطلبات الجامعة الاختيارية (٣ ساعات معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Law for Computer Science	قانون لعلوم الحاسب	UN105
---	2	2	3	Economics for Computer Science	اقتصاد لعلوم الحاسب	UN106
---	2	2	3	Accounting for Computer Science	محاسبة لعلوم الحاسب	UN107
---	2	2	3	Sociology for Computer Science	علم الاجتماع لعلوم الحاسب	UN108

ت- مقررات بدون ساعات معتمدة

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
--	--	1	0	Societal Issues	القضايا المجتمعية	UN103

ثانيا: متطلبات العلوم الأساسية (٢٤ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Mathematics 1 for Computer Science	رياضيات ١ لعلوم الحاسب	BS101
---	2	2	3	Introduction to Computing and Informatics	مقدمة الحاسبات والمعلوماتية	BS102
---	2	2	3	Mathematics 2 for Computer Science	رياضيات ٢ لعلوم الحاسب	BS103
---	2	2	3	Probability and Statistics for Computer Science	احتمالات وتوزيعات إحصائية	BS104
---	2	2	3	Mathematics 3 for Computer Science	رياضيات ٣ لعلوم الحاسب	BS201
---	2	2	3	Operation Research	بحوث العمليات	BS202
---	2	2	3	Fundamentals of Electronics and Electrical Circuits	أساسيات الإلكترونيات والدوائر الكهربائية	IT101
---	2	2	3	Data Communication	اتصالات البيانات	IT102

ثالثا: متطلبات الكلية (٣٩ ساعة معتمدة)

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
---	2	2	3	Introduction to Programming and Problem Solving	مقدمة للبرمجة وحل المشكلات	CS101
Introduction to Programming and Problem Solving (CS101)	2	2	3	Object-Oriented Design and Programming	تصميم وبرمجة شبيهة التوجه	CS102
Introduction to Programming and Problem Solving (CS101)	2	2	3	Data Structures and Algorithms	هياكل البيانات والخوارزميات	CS201
---	2	2	3	Operating Systems	نظم التشغيل	CS202
Introduction to Programming and Problem Solving (CS101)	2	2	3	Software Engineering	هندسة البرمجيات	CS203
---	2	2	3	Fundamentals of Artificial Intelligence	أساسيات الذكاء الاصطناعي	CS204
Introduction to Programming and Problem Solving (CS101)	2	2	3	Introduction to Web Programming	مقدمة لبرمجة الويب	IS101
Probability and Statistics for Computer Science (BS104)	2	2	3	Data Analysis and Management	تحليل وإدارة البيانات	IS201
Data Structures and Algorithms (CS201)	2	2	3	Introduction to Database Systems	مقدمة لنظم قواعد البيانات	IS202
--	2	2	3	Project Management	إدارة المشروعات	IS203
Fundamentals of Electronics and Electrical Circuits (IT101)	2	2	3	Digital Logic Circuit	دوائر المنطق الرقمية	IT103
Data Communication (IT102)	2	2	3	Computer Networks	شبكات الحاسب	IT201
Computer Networks (IT201)	2	2	3	Cyber Security	الأمن السيبراني	IT405

رابعاً: متطلبات تخصص الشبكات والأمن السيبراني - ٥٧ ساعة معتمدة**١ - متطلبات تخصص الشبكات والأمن السيبراني الإجبارية (٤٢ ساعة معتمدة)**

المتطلبات السابقة	السماعات المعتمدة	محاضرة	تمارين/عملي	Course	اسم المادة	كود المادة
Data Structures and Algorithms (CS201)	3	2	2	Analysis and Design of Algorithms	تحليل وتصميم الخوارزميات	CS303
--	3	2	2	Pattern Recognition	التعرف على الأنماط	IT302
Computer Networks (IT201)	3	2	2	Wireless and Sensor Networks	الشبكات اللاسلكية	IT304
Computer Networks (IT201)	3	2	2	Network Programming	برمجة الشبكات	IT308
Computer Networks (IT201)	3	2	2	Computer and Network Security	أمن الحاسبات والشبكات	IT309
Cyber Security (IT405)	3	2	2	Forensic Science	علوم الأدلة الجنائية	IT406
--	3	2	2	Cloud Computing	الحوسبة السحابية	IT412
Cyber Security (IT405), Mathematics 3 for Computer Science (BS201)	3	2	2	Cryptography	التشفير	NCS301
Operating Systems (CS202)	3	2	2	Operating Systems Security	أمن أنظمة التشغيل	NCS302
Object-Oriented Design and Programming (CS102)	3	2	2	Secure Software Development	تطوير البرمجيات الآمن	NCS303
Wireless and Sensor Networks (IT304), Cyber Security (IT405)	3	2	2	Security of Distributed Systems	أمن الأنظمة الموزعة	NCS304
Computer Networks (IT201)	3	2	2	Wireless Networks and Mobile Security	أمن الشبكات اللاسلكية والمحمول	NCS401
Cryptography (NCS301)	3	2	2	Cryptanalysis	تحليل الشفرات	NCS404
---	3	2	2	Cyber Security Risk Management	إدارة مخاطر الأمن السيبراني	NCS405

٢- متطلبات تخصص الأمن السيبراني الاختيارية (يختار الطالب ١٥ ساعة معتمدة)

المتطلبات السابقة	تفاريق/عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
Fundamentals of Artificial Intelligence (CS204), Probability and Statistics for Computer Science (BS104)	2	2	3	Machine Learning	تعلم الآلة	CS302
Fundamentals of Artificial Intelligence (CS204)	2	2	3	Deep Learning	التعلم العميق	CS309
Operating Systems (CS202)	2	2	3	Real-time Systems	أنظمة الزمن الحقيقي	CS315
---	2	2	3	Signal Processing	معالجة الإشارات	IT301
---	2	2	3	Digital Image Processing	معالجة الصور الرقمية	IT307
Computer Networks (IT201)	2	2	3	Computer Networks 2	شبكات الحاسب ٢	IT315
Digital Image Processing (IT307)	2	2	3	Computer Vision	الرؤية بالحاسب	IT401
Digital Logic Circuit (IT103)	2	2	3	Embedded Systems	الأنظمة المضمنة	IT404
---	2	2	3	Blockchain	سلسلة الكتلة	IT408
Digital Logic Circuit (IT103)	2	2	3	Internet of Things	إنترنت الأشياء	IT410
Computer Networks (IT201)	2	2	3	Mobile Computing	الحوسبة المتنقلة	IT415
Cyber Security (IT405)	2	2	3	Social Engineering	الهندسة الاجتماعية	NCS305
Cyber Security (IT405)	2	2	3	Information Security Management	إدارة أمن المعلومات	NCS306
Computer networks (IT201), Cryptography (NCS301)	2	2	3	Social Networks Computing	حوسبة الشبكات الاجتماعية	NCS307
Cyber Security (IT405)	2	2	3	Human Security	الأمن الإنساني	NCS308
Cyber Security (IT405)	2	2	3	Cyber Law and Cyber Security	القانون السيبراني والأمن السيبراني	NCS309

Computer Networks (IT201)	2	2	3	Grid Computing	الحوسبة الشبكية	NCS310
Object-Oriented Design and Programming (CS102), Cyber Security (IT405)	2	2	3	Proactive Computer Security	الأمن الاستباقي للكمبيوتر	NCS311
Software Engineering (CS203), Secure Software Development (NCS303)	2	2	3	Software Security Engineering	هندسة أمن البرمجيات	NCS312
Cryptography (NCS301)	2	2	3	Advanced Cryptography	التشفير المتقدم	NCS313
Internet of Things (IT410), Cyber Security (IT405)	2	2	3	Cyber Security for Internet of Things	الأمن السيبراني لإنترنت الأشياء	NCS314
---	2	2	3	Incident Response, Disaster Recovery, and Business Continuity Systems	أنظمة الاستجابة للحوادث، التعافي من الكوارث واستمرارية الأعمال	NCS315
(Cyber Security IT405)	2	2	3	Selected Topics in Cyber Security -1	موضوعات مختارة في الأمن السيبراني 1	NCS316
(Cyber Security IT405)	2	2	3	Selected Topics in Cyber Security -2	موضوعات مختارة في الأمن السيبراني 2	NCS317
Computer Networks (IT201)	2	2	3	Selected Topics in Networks Technology-1	موضوعات مختارة في تكنولوجيا الشبكات-1	NCS318
Computer Networks (IT201)	2	2	3	Selected Topics in Networks Technology-2	موضوعات مختارة في تكنولوجيا الشبكات-2	NCS319
--	2	2	3	Data Integrity and Authentication	سلامة البيانات والمصادقة	NCS402
Cryptography (NCS301)	2	2	3	Computer Security and Privacy	أمن وخصوصية الحاسب	NCS403
Cloud Computing (IT412), Cryptography (NCS301)	2	2	3	Cloud Computing Security	أمن الحوسبة السحابية	NCS406
Social Networks Computing (NCS307)	2	2	3	Social Networks Analytics	تحليلات الشبكات الاجتماعية	NCS407
Computer Security and Privacy (NCS403)	2	2	3	Enterprise Server Management and Security	إدارة خوادم المؤسسات وتأمينها	NCS408

Computer Security and Privacy (NCS403)	2	2	3	Reverse-Engineering Malware: Malware Analysis Tools and Techniques	الهندسة العكسية للبرمجيات الضارة: أدوات وأساليب تحليل البرامج الضارة	NCS409
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خامساً: التدريب الصيفي

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
اجتياز ٧١ ساعة	6	1	4	Summer Training	التدريب الصيفي	TR201

سادساً: المشروع

المتطلبات السابقة	تمارين/ عملي	محاضرة	الساعات المعتمدة	Course	اسم المادة	كود المادة
اجتياز ٩٣ ساعة معتمدة	4	1	3	Project 1	مشروع ١	GP401
GP401	4	1	3	Project 2	مشروع ٢	GP402

سابعاً: الخطة الدراسية المقترحة لبرنامج الشبكات والأمن السيبراني**المستوى الأول (Freshman) (٣٦ ساعة معتمدة)**

الفصل الدراسي الأول (Freshman - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS101	رياضيات ١ لعلوم الحاسب	Mathematics 1 for Computer Science	--	3
BS102	مقدمة الحاسبات والمعلوماتية	Introduction to Computing and Informatics	--	3
IT101	أساسيات الإلكترونيات والدوائر الكهربائية	Fundamentals of electronics and electrical circuits	--	3
CS101	مقدمة للبرمجة وحل المشكلات	Introduction to Programming and Problem Solving	--	3
UN101	مهارات الاتصال والجوانب الأخلاقية والمهنية	Communication Skills, Ethical and Professional Issues	--	3
UN102	لغة إنجليزية لعلوم الحاسب	English for Computer Science	--	3
UN103	القضايا المجتمعية	Societal Issues	--	0
18	المجموع			
الفصل الدراسي الثاني (Freshman – Spring)				
BS103	رياضيات ٢ لعلوم الحاسب	Mathematics 2 for Computer Science	--	3
BS104	احتمالات وتوزيعات إحصائية لعلوم الحاسب	Probability and Statistics for Computer Science	--	3
CS102	تصميم وبرمجة شبيهة التوجه	Object-Oriented Design and Programming	Introduction to Programming and Problem Solving (CS101)	3
IS101	مقدمة لبرمجة الويب	Introduction to Web Programming	Introduction to Programming and Problem Solving (CS101)	3
IT102	اتصالات البيانات	Data Communication	--	3
IT103	دوائر المنطق الرقمية	Digital Logic Circuit	Fundamentals of Electronics and Electrical Circuits (IT101)	3
18	المجموع			

المستوى الثاني (Sophomore) (٣٦ ساعة معتمدة)

الفصل الدراسي الأول (Sophomore - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
BS201	رياضيات ٣ لعلوم الحاسب	Mathematics 3 for Computer Science	--	3
BS202	بحوث العمليات	Operation Research	--	3
UN201	ريادة الأعمال التكنولوجية	Technology Entrepreneurship	--	3
CS201	هياكل البيانات والخوارزميات	Data Structures and Algorithms	Introduction to Programming and Problem Solving (CS101)	3
CS203	هندسة البرمجيات	Software Engineering	Introduction to Programming and Problem Solving (CS101)	3
IT201	شبكات الحاسب	Computer Networks	Data Communication (IT102)	3
18	المجموع			
الفصل الدراسي الثاني (Sophomore – Spring)				
CS202	نظم التشغيل	Operating Systems	---	3
CS204	أساسيات الذكاء الاصطناعي	Fundamentals of Artificial Intelligence	---	3
IS201	تحليل وإدارة البيانات	Data Analysis and Management	Probability and Statistics for Computer Science (BS104)	3
IS201	مقدمة لنظم قواعد البيانات	Introduction to Database Systems	Data Structures and Algorithms (CS201)	3
IT405	الأمن السيبراني	Cyber Security	Computer Networks (IT201)	3
UN10x	اختياري جامعة	University Elective	--	3
18	المجموع			
الفصل الدراسي الصيفي (Summer)				
TR201	التدريب الصيفي	Summer Training	اجتياز ٧١ ساعة	4

المستوى الثالث (Junior) (٣٣ ساعة معتمدة)

الفصل الدراسي الأول (Junior - Fall)				
الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IS203	إدارة المشروعات	Project Management	--	3
IT302	التعرف على الأنماط	Pattern Recognition	--	3
IT304	الشبكات اللاسلكية	Wireless and Sensor Networks	Computer Networks (IT201)	3
CS303	تحليل وتصميم الخوارزميات	Analysis and Design of Algorithms	Data Structures and Algorithms (CS201)	3
NCS301	التشفير	Cryptography	Cyber Security (IT405), Mathematics 3 for Computer Science (BS201)	3
NCS303	تطوير البرمجيات الآمن	Secure Software Development	Object-Oriented Design and Programming (CS102)	3
18	المجموع			
الفصل الدراسي الثاني (Junior – Spring)				
NCS302	أمن أنظمة التشغيل	Operating Systems Security	Operating systems (CS202)	3
IT308	برمجة الشبكات	Network Programming	Computer Networks (IT201)	3
IT309	أمن الحاسبات والشبكات	Computer and Network Security	Computer Networks (IT201)	3
IT406	علوم الأدلة الجنائية	Forensic Science	Cyber Security (IT405)	3
XXX	اختياري تخصص	Track-Specific Elective		3
15	المجموع			

المستوى الرابع (Senior) (٣٣ ساعة معتمدة)

الكود	اسم المادة	Course Name	المتطلب السابق	عدد الساعات
IT412	الحوسبة السحابية	Cloud Computing	--	3
NCS304	أمن الأنظمة الموزعة	Security of Distributed Systems	Wireless and Sensor Networks (IT304), Cyber Security (IT405)	3
NCS401	أمن الشبكات اللاسلكية والمحمول	Wireless Networks and Mobile Security	Computer Networks (IT201)	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP401	مشروع ١	Graduation Project 1		3
18	المجموع			
NCS404	تحليل الشفرات	Cryptanalysis	Cryptography (NCS301)	3
NCS405	إدارة مخاطر الأمن السيبراني	Cyber Security Risk Management	---	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
XXX	اختياري تخصص	Track-Specific Elective	XXX	3
GP402	مشروع ٢	Graduation Project 2	Graduation Project 1 (GP401)	3
15	المجموع			

الباب الرابع:

محتوى المقررات الدراسية

UN101	Communication Skills, Ethical and Professional Issues			3	
Prerequisites	---				
Number of weekly Contact Hours					
Lecture	Tutorial		Laboratory		
2	2		---		
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
Communication skills allow students to build and refine their communication skills and help them improve and perfect their interpersonal and presentation skills, as well as their oral communication with diverse audiences, including listening, presenting, interviewing, conducting meetings, negotiation, and interpersonal communication, in addition to intercultural communication—written communication through letters and business/research reports. For the creative thinking component, students will learn to understand meta-cognition (thinking about thinking). The course includes Edward de Bono's CoRT (cognitive research trust) program of learning, thinking, vertical and lateral thinking approaches, and Creative thinking tools like Brainstorming, Tony Buzan's Mind mapping, and Edward de Bono's Six Thinking Hats, using multi-media exploration into important intra- and interpersonal skills required for Emotional Intelligence. The course also introduces ethical and professional issues in computer science and software development. Professional and ethical responsibilities in computer science and software development. Privacy and personal data in on-line world. An overview of ethics and plagiarism policy. Intellectual property rights and protection. Legal perspective, copyright laws, patents, trade secrets, trademarks, cheating and plagiarism, piracy, federal regulations, software protection, public domain, fair use, and use of open-source software. In addition to the ACM code.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

UN102	English for Computer Science				3
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course should reflect the stylistic variety that advanced learners have to be able to deal with; The course gives practice in specific points of grammar to consolidate and extend learner’s existing knowledge; analysis of syntax; comprehension; Skimming and scanning exercises develop the learners skills; comprehension questions interpretation and implication; the activities and games used develop listening; speaking and writing skills through a communicative; functional approach; with suggested topics for discussion and exercises in summary writing and composition. Also, this course aims to give the student the basic rudiments of report writing, the rationale for report writing, the structure of reports, and such details as physical appearance and linguistic style. In addition to writing reports; student will also be given supplementary exercises, as necessary, to enhance their general writing skills.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

UN103		Societal Issues				0	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
1			---			---	
Required SWL		---		Equivalent ECTS		---	
Course Content							
The university board determines the content of this course for all the university's students.							
Taught in Program(s)							
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☐ Exam	☐ Assessment	☐ Exam	☐ Assessment	☒ Exam	
					100		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

UN104		Design Physiology for Computer Science		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces students to the physiological and cognitive aspects of human-centered design. The course will explore how human perception, cognition, and behavior influence design decisions and how design can be used to enhance user experience and interaction. In the first section of the course, students will learn about the basic concepts and theories in design physiology. The section will cover topics such as visual perception, attention, memory, and decision-making. Students will also learn about the research methods and techniques used in design physiology. In the second section of the course, students will learn about the application of design physiology to user experience and interaction. The section will cover user-centered design, usability testing, and accessibility. Students will also learn about the design of interfaces, products, and environments that consider human physiology and cognition. In the third section of the course, students will learn about the social and cultural aspects of design physiology. The section will cover topics such as user diversity, cultural differences, and the ethical implications of design decisions. Throughout the course, students will learn how to apply design physiology concepts and techniques to design and evaluate user interfaces, products, and environments. The course will also cover case studies of technology firms and the design decisions they made to enhance user experience and interaction.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

UN105		Law for Computer Science			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		2			---	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides students with an introduction to the legal principles and issues relevant to the development, distribution, and use of software and digital products. The course will explore the legal framework that governs intellectual property, privacy, and security in the digital age. Students will learn about the legal implications of software development, including the creation and distribution of open-source software and proprietary software. The course will begin by examining the basics of the legal system, including the role of courts, legal precedent, and the hierarchy of laws. Students will then explore the legal principles that govern intellectual property, including patents, trademarks, and copyrights. The course will also cover the legal framework governing the use and distribution of open-source software, including licenses such as the GPL and the MIT license. Throughout the course, students will be encouraged to think critically about the legal issues arising in software development and digital product contexts. Students will develop the skills to analyze legal issues and apply legal principles to real-world scenarios. The course will be a mix of lectures, discussions, and case studies, and students will be expected to participate actively in class discussions and complete assignments that require independent research and analysis.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

UN106		Economic for Computer Science		3							
Prerequisites		---									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		2		---							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course introduces students to fundamental concepts in microeconomics and macroeconomics. The course will explore how economic principles apply to the technology industry and how economic factors influence decision-making in the technology sector. In the microeconomics section, students will learn about the principles of supply and demand, the role of markets in allocating resources, and the economic behavior of firms. The section will also cover topics such as pricing strategies, market structures, and the economics of information. In the macroeconomics section, students will explore the broader economic environment in which technology firms operate. This section will cover topics such as economic growth, inflation, monetary policy, fiscal policy, and international trade. Throughout the course, students will learn how to analyze economic data and apply economic principles to real-world scenarios. The course will also cover case studies of technology firms and the economic factors that influenced their success or failure. By the end of the course, students will have a solid understanding of the economic principles that shape the technology industry and the ability to apply those principles to their work as computer Science.											
Taught in Program(s)											
As major in Program(s) Code(s): ---											
As minor in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

UN107		Accounting for Computer Science			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		2			---	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces students to the principles of financial accounting. The course will cover the basic accounting concepts and techniques used by businesses to record, measure, and report financial data. In the first section of the course, students will learn about the accounting cycle and the preparation of financial statements. The section will cover topics such as debit and credit entries, journal entries, and the trial balance. Students will also learn how to prepare financial statements, including the balance sheet, income statement, and statement of cash flows. In the second section of the course, students will learn about financial analysis and decision-making. The section will cover topics such as ratio analysis, cash flow analysis, and budgeting. Students will also learn how to use financial information to make business decisions. In the third section of the course, students will learn about accounting for partnerships and corporations. The section will cover topics such as stock issuance, dividends, and stockholders' equity. Throughout the course, students will learn how to use accounting software to record and analyze financial data. The course will also cover case studies of technology firms and the accounting principles that influenced their success or failure. By the end of the course, students will have a solid understanding of basic accounting principles and the ability to apply them to their work as computer Science. They will also be able to analyze and interpret financial data for business decision-making.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

UN108		Sociology for Computer Science		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces students to the study of human society and social behavior. The course will explore how social structures, institutions, and processes influence and are influenced by technology and the technology industry. In the first section of the course, students will learn about the basic concepts and theories in sociology. The section will cover topics such as socialization, culture, social stratification, and social change. Students will also learn about the research methods and techniques used in sociology. In the second section of the course, students will learn about the impact of technology on society and social behavior. The section will cover topics such as the digital divide, social media, online communities, and the social implications of emerging technologies such as AI and blockchain. In the third section of the course, students will learn about social institutions and organizations in the technology industry. The section will cover topics such as corporate culture, diversity and inclusion, and the social responsibility of technology firms. Throughout the course, students will learn how to apply sociological concepts and theories to analyze and understand the social and cultural context of the technology industry. The course will also cover case studies of technology firms and the social issues they face. By the end of the course, students will have a solid understanding of the basic concepts and theories in sociology and the ability to apply those concepts to the analysis of the technology industry. They will also be able to identify and analyze the social and cultural implications of technology and the technology industry.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

UN109		Fundamental of Management		3			
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		2		---			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course provides an introduction to the fundamental principles and practices of management. It covers the essential functions of management, including planning, organizing, leading, and controlling. Students will explore key management concepts, theories, and tools for effective decision-making and leadership across various organizational settings. The course also emphasizes the development of managerial skills necessary to address contemporary challenges in the business environment. The course covers the following topics: • The basic concepts and functions of management. • The roles and responsibilities of managers. • Planning, organizing, leading, and controlling organizational resources. • Various management theories and their applications. • The impact of the external environment on management practices. • Problem-solving and decision-making skills in a managerial context.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): MI							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

UN201	Technology Entrepreneurship			3	
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
In this course, students will explore how entrepreneurs match promising technology with customer needs to launch successful new businesses. Using real-world examples, they will apply critical thinking to commercialize technologies and learn about the venture creation process from founders, funders, and industry experts.					
The course covers the following topics:					
• A systematic approach to technology entrepreneurship. • How to generate new use scenarios by matching customer needs with promising technology seeds. • How to align business and operating models. • How to evaluate a technology for readiness and market fit. • How to position opportunities to secure funding.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

UN202	Fundamental of Biology			3	
Prerequisites	---				
Number of weekly Contact Hours					
Lecture	Tutorial		Laboratory		
2	2		---		
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
This course is designed to provide medical informatics specialists with a foundational understanding of key biological concepts and principles essential to effectively working with biomedical data and health information systems. The course covers the basics of cellular biology, genetics, physiology, and human anatomy, with a focus on their relevance to medical informatics. Emphasis is placed on understanding the biological data types encountered in medical informatics and how this knowledge can be applied to improve healthcare technology and decision-making processes. The course covers the following topics: • The fundamental concepts of biology. • Knowledge of cellular structure, function, and biological processes. • The basics of human genetics and genomics. • Human anatomy and physiology with a focus on clinical relevance. • The types of biological data. • Apply biological concepts to health information systems.					
Taught in Program(s)					
As major in Program(s) Code(s): MI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

BS101		Mathematics 1 for Computer Science			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		2			---	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces elementary discrete mathematics and its applications to computer science. It emphasizes mathematical definitions, proofs, tools, and methods. Topics include an introduction to logic, methods of proof, induction, well-ordering, set theory, functions, sequences, summations, matrices, elements of number theory, relations, introduction to graphs and trees, integer congruences, asymptotic notation, counting principles, and recursive definitions and relations.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

BS102		Introduction to Computing and Informatics				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
The course goal is to introduce the fundamental concepts of computer science and informatics. As an overview of the discipline, the course covers a breadth of topics, including algorithmic foundations; hardware issues such as number systems and computer architectures; and software issues such as operating systems, programming languages, database systems, networks, and security. Also, this course introduces fundamental concepts underlying software development, basic algorithms, basic software engineering principles, programming skills, simple data types, statement sequencing, input/output, control structures, array and string data structures, functions, testing, and debugging.							
Taught in Program(s)							
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

BS103		Mathematics 2 for Computer Science		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The main objective of this course is to provide students with pre-calculus review, Sets, Real-valued function, the continuity and the differentiability of a real function, Techniques of differentiation, Derivatives of the trigonometric functions, Implicit differentiation, linear approximations, and differentials. Applications of the derivative: Extreme of functions, optimization problems, velocity, and acceleration. Integrals: Indefinite integrals, change of variables, definite integrals, the fundamental theorem of calculus, and numerical integration. Applications of definite integrals: Areas, solids of revolution, arc length and surfaces of revolution, work, moments, and centers of mass. Transcendental functions: Derivative of inverse function, natural logarithm function, exponential functions, inverse trigonometric functions, hyperbolic and inverse hyperbolic functions, indeterminate forms, and rules. Understanding techniques of integration: Integration by parts, trigonometric integrals and substitutions, integrals of rational functions, quadratic expressions, tables of integrals, improper integrals. Simulation will be performed using mathematical software such as MATLAB.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

BS104		Probability and Statistics for Computer Science			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		2			---	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides an elementary introduction to probability and statistics. Topics include introduction to probability and its theory, combinatorics, probability models, Joint distributions, covariance, correlation, covariance, independence, Central limit theorem, contingency table, tree diagram, conditional probability, Bayesian theorem, random variables, discrete and continuous probability distributions, concepts of statistics, data types, measurement levels, sampling methods, descriptive statistics, frequency tables and graphical representations. Topics may also include statistical estimation and testing, confidence intervals, and an introduction to regression coefficients and analysis.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

BS201		Mathematics 3 for Computer Science		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The main objective of this course is to provide students with understanding of linear algebra, Matrix Algebra, system of linear equations, vector spaces, vector algebra, Orthogonalization and Linear transformation, Eigen values and Eigen vector, Diagonalization, function of a matrix, ordinary differential equation, linear and non-liner differential equation, solution of a differential equations, high order differential equations, applications of first order differential equations, special functions, partial differential equations, numerical analysis, complex variables. In this course on Linear Algebra, we look at what linear algebra is and how it relates to vectors and matrices. Then we look at what vectors and matrices are, how to work with them, including the knotty problem of eigenvalues and eigenvectors, and how to use them to solve problems. Finally, we look at how to use these to do fun things with datasets - like how to rotate images of faces and how to extract eigenvectors to look at how the PageRank algorithm works. Simulation will be performed using mathematical software such as MATLAB.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

BS202		Operation Research		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The main objective of this course is to provide students with an introduction to Operations Research and Decision Support software, theory, and applications. This course covers mathematical and engineering methods for studying optimization problems in Business and Management, Economics, Computer Science, and related fields. This course introduces frameworks and ideas about various types of optimization problems in the business world. In particular, we focus on how to formulate real business problems into mathematical models that computers can solve. The use of professional software packages and languages in operations research and mathematical programming (computer-based modeling) is stressed throughout the course. One of these packages will be studied in more detail, including its model specification, model verification, alternative solvers, and output display. The course also emphasizes the computational aspects of simulation models, using both procedural and general-purpose simulation languages. Issues related to data structures and statistical analyses of simulation output are also considered. The content contains introduction to OR Models, Graphical Method for LPP, Convex sets, Simplex Method, Big M Method, Two Phase, Multiple solutions of LPP, Unbounded solution of LPP, Infeasible solution of LPP, Weak Duality Theorem, More Duality Theorems, Primal-Dual relationship of solutions, Dual Simplex Method, Sensitivity Analysis, Integer Programming Goal Programming, Multi-Objective Programming, Dynamic Programming, Transportation Problem, Assignment Problem, Processing n Jobs on Two Machines, Processing n Jobs through Three Machines, Processing two jobs through m machines, Processing n jobs through m machines, Two Person Zero-Sum Game, Theorems of Game Theory, Solution of Mixed Strategy Games, Linear Programming method for solving games, Case studies and Exercises.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)

	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5		60
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-	-		-		-	

BS203		Pathology and Epidemiology		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		2		---	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course provides an in-depth exploration of pathology and epidemiology, focusing on their applications in medical informatics. It covers the principles and methods of disease pathology and epidemiology, emphasizing how these fields contribute to the understanding, analysis, and management of health data. Students will learn about the mechanisms of disease, patterns of disease distribution, and the use of epidemiological methods to improve public health through data analysis and informatics tools. The course covers the following topics: • The fundamental concepts of pathology and epidemiology. • The mechanisms and progression of diseases. • The patterns and determinants of health and disease in populations. • Applying epidemiological methods to health data. • Integrate knowledge of pathology and epidemiology into medical informatics applications. • Use informatics tools to analyze and interpret epidemiological data for public health decision-making.					
Taught in Program(s)					
As major in Program(s) Code(s): MI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS101		Introduction to Programming and Problem Solving			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides a fundamental concept underlying software development. Basic software engineering principles and programming skills. Simple data types, statement sequencing, input/output, control structures, looping, recursion, array and string data structures, functions, basic algorithms, testing, and debugging. Modeling and problem-solving skills applicable to programming at this level. The course will be taught using a structured programming approach.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS102		Object-Oriented Design and Programming		3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course covers the basic concepts of object-oriented programming and their relationships to the principles of software engineering. Emphasis is placed on the programming structures of interfaces, classes, objects, abstraction, inheritance, and polymorphism, and how these concepts and structures aid in the development of extensible software systems. In addition, concepts of exceptions and exception handlers are covered, along with basic file handling and programming a desktop UI as object-oriented classes/components. Practical experience is gained in the design, construction, and debugging of systems in a strongly typed object-oriented language. This course provides the programming skills needed to develop software in other courses and provides the context for more advanced modules on software engineering (e.g., the Unified Modeling Language) and project management.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS201		Data Structures and Algorithms			3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces fundamental data structures, algorithms, and abstract data types using a mix of programming and theory. Main topics include data structures such as arrays, lists, linked lists, stacks, queues, hash tables, heaps, priority queues, graphs, and trees. It introduces algorithms used for list manipulation, graph searches, sorting, searching, and tree traversals. Also, it introduces analyzing and managing the complexity associated with data structures and their operations.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS202		Operating Systems				3
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces computer systems as they are relevant to application programmers today, with an emphasis on operating systems principles. This course covers many concepts related to real-world Operating Systems. Topics include operating system design, system call interfaces, system-level I/O, processes, inter-process communication (IPC), threads, CPU scheduling, synchronization techniques, deadlock, process management, memory management, file systems, virtualization, and distributed operating systems.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS203		Software Engineering		3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The course objective is to introduce students to the basic principles of software engineering. Topics will include: • Introduction to software engineering and its importance. • Software process and its common models. • Software requirements analysis. • Introduction to software design • Introduction to agile. • Agile-case studies. • System models. • Business models. • Introduction to UI design patterns. • Introduction to software testing and quality assurance. • Introduction to software evolution.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS204		Fundamentals of Artificial Intelligence				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
Overview of the Artificial Intelligence concepts, terminology, and application areas of problem solving; problem spaces; search techniques; knowledge representation; programming languages for Artificial Intelligence (AI); games; predicate logic; knowledge-based systems; dealing with uncertainty, probabilistic reasoning, planning and learning techniques; advanced search methods such as genetic algorithms, reactive systems; machine learning, natural language understanding; computer vision; robotics.							
Taught in Program(s)							
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS301	Mobile Application Development			3	
Prerequisites	Software Engineering (CS203)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS	5.14	
Course Content					
Students learn to develop mobile applications for smartphones and tablets. Students are introduced to the current mobile operating systems and mobile application development environments. Students design mobile applications with multiple user interfaces and system components. Students develop applications for two popular mobile devices. Students create mobile applications that use multi-threading to access online servers. Students design and build a variety of apps throughout the course to reinforce the concepts taught and help them practice what they are learning. Upon successful completion of this subject, students should be able to:					
- Recognize the capabilities and limitations of mobile devices. - Design mobile applications with multiple user interfaces and system components. - Develop mobile applications on a popular mobile development platform. - Evaluate the performance of mobile applications against pre-set criteria.					
Taught in Program(s)					
As major in Program(s) Code(s): CS					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS302		Machine Learning			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204) Probability and Statistics for Computer Science (BS104)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Algorithms involved in machine learning and their applications to different problems arising in computer vision, natural language processing, and/or robotics. The topics include, but are not limited to, supervised learning, unsupervised learning, reinforcement learning, ensemble methods, machine learning system design, and experimental evaluation of learning algorithms. Design and implementation of a technical project applied to real-world datasets (images, text, and signals). The algorithms include linear regression, logistic regression, SVMs, decision trees, random forests, dimensionality reduction, gradient boosting machines, bagging and boosting ensembles, and probabilistic methods.						
Taught in Program(s)						
As major in Program(s) Code(s): CS and AI						
As minor in Program(s) Code(s): IS and IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS303		Analysis and Design of Algorithms		3	
Prerequisites		Data Structures and Algorithms (CS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course emphasizes understanding data structures and algorithms from an analytical perspective rather than an implementation standpoint. It covers different methods for constructing algorithms using fundamental strategies such as divide-and-conquer, recursion, greedy, and dynamic programming. Also, it covers methods for analyzing algorithms mathematically for correctness and efficiency (i.e., running time and space usage), such as recursion trees, the master method, and the iteration method. The course starts with definitions of algorithmic efficiency, discusses powerful paradigms for algorithm design, introduces analysis tools and techniques for algorithms, presents strategies for designing algorithms, introduces graph theory, and defines the theory of NP-completeness to understand intractable problems. Also, this course synthesizes efficient algorithms for common computer design situations and fields, such as sorting and searching lower bounds, minimum spanning trees, shortest paths and flows in networks, and artificial intelligence (especially in natural language processing).					
Taught in Program(s)					
As major in Program(s) Code(s): CS and AI					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS304		Soft Computing			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Genetic Algorithms – Chromosomes – Population – Fitness functions – Crossover – Mutation – Binary bit chromosomes – Floating point array chromosomes – Schema theory – Fuzzy logic – Fuzzy systems – Fuzzy operators – Fuzzy rule-based systems – Basis of Neural networks Biological neurons - Various learning techniques such as McCulloch-Pitts, Perceptron, Hebbian, and ADALINE - Self-organizing map (SOM) – Hybrid systems – Hybrid Neuro Fuzzy networks - Training of NNs with GAs – Evolution of fuzzy rule-based systems – Genetic programming – Immune systems – Evolution strategy.						
Taught in Program(s)						
As major in Program(s) Code(s): CS						
As minor in Program(s) Code(s): SE						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS305		Bioinformatics		3	
Prerequisites		Data Structures and Algorithms (CS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course covers the algorithmic and foundational basics of computational biology, combining theory with practice. We cover both foundational topics in computational biology and current research frontiers. We study fundamental techniques and recent advances in the field, and we work directly with current large-scale biological datasets. Genomes: Biological sequence analysis, hidden Markov models, gene finding, comparative genomics, RNA structure, sequence alignment, and hashing. Networks: Gene expression, clustering/classification, EM / Gibbs sampling, motifs, Bayesian networks, microRNAs, regulatory genomics, epigenomics. Evolution: Gene/species trees, phylogenomic, coalescent, personal genomics, population genomics, human ancestry, recent selection, disease mapping.					
Taught in Program(s)					
As major in Program(s) Code(s): CS and MI					
As minor in Program(s) Code(s): IS and IT					
Targeted Sustainable Development Goals:					
☒ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS306		Distributed Systems			3	
Prerequisites		Operating Systems (CS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The course covers fundamental remote and distributed processing models, algorithms for distributed systems, inter-process communication, and how to handle synchronization, consistency, replication, fault tolerance, and security in a distributed system. Also, it covers various middleware models and the basics of different distributed computing paradigms, including Clusters, Grid, Edge, Fog, and Cloud, the technologies behind them, and how to develop algorithms and applications on them. Some distributed systems with specific architectures and requirements are introduced. Also, the practical implementation of distributed systems algorithms and models, such as stream-based processing algorithms and MapReduce, is covered.						
Taught in Program(s)						
As major in Program(s) Code(s): CS						
As minor in Program(s) Code(s): SE						
Targeted Sustainable Development Goals:						
☒ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS307		Human Computer Interaction		3	
Prerequisites		Object-Oriented Design and Programming (CS102)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course overviews the major types of interfaces, introduces the future generation of Human Computer Interaction, and focuses on the practical issues of creating interactive systems and products from a human-centered perspective. It covers methodologies, techniques, and technologies for designing high-quality interactive systems, products, and services, as well as techniques for reflecting on design throughout the development of the interactive system. Survey of human-computer interaction concepts, theory, and practice. Basic components of human-computer interaction. Interdisciplinary underpinnings. Informed and critical evaluation of computer-based technology. User-oriented perspective, rather than system-oriented, with two thrusts: human (cognitive, social) and technological (input/output, interaction styles, devices). Design guidelines, evaluation methods, participatory design, and communication between users and system developers. Design and implementation of object-oriented graphical user interfaces (GUI) and two-dimensional computer graphics systems. It can also cover 3D graphics systems and animation basics. Camera Control is also one of the 3D graphics components that can be covered.					
Taught in Program(s)					
As major in Program(s) Code(s): CS and SE					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☒ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS308		Advanced Software Engineering			3	
Prerequisites		Software Engineering (CS203)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The course objective is to introduce students to the advanced topics in software Engineering. Topics will include, but not be limited to: Software architecture – Architectural styles – Service-oriented architectures – Component-based architecture - Study of design patterns - Software quality metrics – Reviews – Refactoring – Software Configuration management – Software evolution and maintenance– reverse engineering.						
Taught in Program(s)						
As major in Program(s) Code(s): CS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☒ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS309		Deep Learning		3									
Prerequisites		Fundamentals of Artificial Intelligence (CS204)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
Algorithms and techniques in Deep Learning and its applications in perception and Artificial Intelligence problems (image classification, detection, and segmentation; image captioning; visual question answering, automatic game playing). The topics include neural network designs and architectures; biological and artificial neural networks; learning algorithms (backpropagation, stochastic gradient descent); regularization; various deep learning architectures; and the use of deep learning for image classification, object detection, segmentation, and more. The deep learning models include multi-layer perceptrons, convolutional neural networks, recurrent neural networks, long short-term memory networks, memory networks, generative adversarial networks, autoencoders, Boltzmann machines, deep belief networks, and Transformers.													
Taught in Program(s)													
As major in Program(s) Code(s): CS and AI													
As minor in Program(s) Code(s): NCS													
Targeted Sustainable Development Goals:													
☒ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

CS310		Visual Computing			3	
Prerequisites		Digital Image Processing (IT307)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides a comprehensive exploration of the principles and algorithms involved in visual computing, empowering students to understand and analysis visual data. It introduces common mathematical methods and algorithms for processing images, videos, and 3D models. Visual computing introduces a visual computing environment that facilitates human-computer interaction and enables the analysis and manipulation of vast amounts of data, including videos, 3D sequences, and other visual entities. The student delves into the algorithms that enable computing systems to comprehend and interpret visual objects, supporting human-computer vision interaction. Students will also gain knowledge of how artificial intelligence techniques can be utilized for visual computing tasks such as object detection, semantic segmentation, image generation, and image captioning.						
Taught in Program(s)						
As major in Program(s) Code(s): CS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS311		Computer Organization and Architecture		3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course explores modern computer architectures, focusing on instruction sets and the organization of processors, controllers, memories, devices, and communication links. The course includes the following topics: • Assembly-level organization: Basic organization of the von Neumann machine; control unit, instruction fetch, decode, and execution; instruction sets and types (data manipulation, control, I/O); assembly/machine language programming; instruction format; addressing modes; subroutine call and return mechanisms; I/O and interrupts. • Memory systems: storage systems and their technology; coding, data compression, and data integrity; memory hierarchy; main memory organization and operations; latency, cycle time, bandwidth, and interleaving; cache memories (address mapping, block size, replacement, and store policy); virtual memory (page table, TLB). • Interfacing and communication: I/O fundamentals, interrupt structures, buses. • Functional organization and Parallel Organizations: introduction to instruction-level parallelism (ILP); multiprocessor architecture: introduction to SIMD, MIMD, VLIW, EPIC; shared memory systems; cache coherence. • Performance enhancements: RISC architecture, branch prediction, prefetching, and scalability. • Contemporary architectures: Hand-held devices; embedded systems. • Parallel computer paradigms – Architecture-oriented programming – Power and energy aware computing – Tools and simulation for computer design and performance enhancement.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam

20	15	5	60
Student Activities Breakdown (Distribution for Study Blocks)			
Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

CS312		Advanced Operating Systems			3	
Prerequisites		Operating Systems (CS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The course covers advanced concepts and design principles of modern operating systems (OSes), both from theory and practical aspects, spanning the following aspects: virtualization, virtual machines/containers, memory virtualization, concurrency, synchronization, concurrency control, transactions, persistence and storage, multiprocessor scheduling, multicore scheduling, OSes of internet of things.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS313		Generative Artificial Intelligence				3	
Prerequisites		Deep Learning (CS309)					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course covers the mathematical and computational foundations of generative AI modeling and Large Language Models (LLMs), as well as real-world use cases (Text, Images, Video, and Music Content Generation) across manufacturing, healthcare, finance, etc. Specific topics include variational autoencoders, generative adversarial networks, autoregressive models, normalizing flow models, information lattice learning, neural text decoding, prompt programming, data augmentation, synthesis, and personalization and detection of generated content. Exploration of practical LLM (Chatbots, Language Translation, Content Generation, Virtual Assistants). Exploration of ethical concerns and responsible AI development and deployment guidelines.							
Taught in Program(s)							
As major in Program(s) Code(s): AI							
As minor in Program(s) Code(s): CS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS314		Real-time Software Engineering		3	
Prerequisites		Software Engineering (CS203)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course gives general definitions and architectural frameworks for designing, analyzing, and building real-time software. It discusses modern software engineering methodologies and tools for system development and design, scheduling, concurrency control, execution time prediction, fault tolerance, validation, safety, and reliability. In addition, the course covers the different types of programming, software errors, and software life cycles, and how a multitasking structure benefits a real-time system design. Software design patterns and pattern-based approaches are illustrated for building real-time software, and a focus is on a Model-Based software development process based on the MDA (Model-Driven Architecture) approach for designing mission-critical and safety-critical real-time systems. The course also shows how real-time extensions of UML and the MARTE profile are used with other tools such as Simulink, AADL, and SPARK to express requirements and design as models that use formal contracts.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS and SE					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS315		Real-time Systems		3	
Prerequisites		Operating Systems (CS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces the theory, algorithms, and protocols, as well as the mechanisms and implementations, of real-time computer systems. Fundamentals of real-time systems and computing and their types, hardware for embedded real-time systems, various types of real-time tasks and task-servers, and the various real-time scheduling algorithms working on them, schedulable analysis, real-time synchronization, real-time operating system kernels, and real-time programming languages (Ada, RT Java, C/POSIX), and Validation of embedded real-time systems. Design and analysis of real-time resource management problems (e.g., priority inversion) and the associated running-time problems (e.g., scheduling, synchronization), and the resource management algorithms (e.g., scheduling, synchronization) are overviewed, along with their implementations in production operating system kernels, experimental studies of those implementations, and real-time application development in association with Real-time Database. Also, important reliability, dependability, and fault-tolerance issues in real-time systems software are covered. Also, the main concepts of mixed-critical real-time systems, which include both real-time and non-real-time tasks, are covered, and algorithms for scheduling them are presented.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS, IT, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS316		Ambient Intelligence		3							
Prerequisites		Human Computer Interaction (CS307)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course will provide an overview of a new vision for Human-Computer Interaction (HCI) in which intelligent, intuitive interfaces in everyday objects surround people. This course focuses on creating environments that are aware of their context, can understand and interpret user behavior, and can respond intelligently to provide personalized, contextually relevant services. It aims to enhance people's daily lives by making technology less obtrusive and more integrated into the environment. The course includes an introduction to ambient intelligence, Intelligence Augmentation and Software Agents, Location-Based and Context-Aware Systems, Interfaces with Common Sense, Ubiquitous Computing, User Modeling, Personalization and Recommender Systems, Tangible and Ambient Interfaces, Augmented Reality/Mixed Realities, The Meaning of Things, Smart Objects, and intelligent environments.											
Taught in Program(s)											
As major in Program(s) Code(s): ---											
As minor in Program(s) Code(s): CS and AI											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

CS317		Logic Programming and Automated Reasoning			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to familiarize students with the logic programming paradigm, its programming techniques, and applications. It provides students with applications of mathematical logic to computer science, the implementation of declarative concepts, first-order logic principles, and programming in logic. It enables students to formalize natural language sentences and convert them into logical models. Also, it allows students to apply reasoning procedures for several logic languages. Prolog may be taught as a logic programming language in this course.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS318		Expert Systems and Knowledge Engineering		3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Techniques for the construction of expert systems, including computer inference and knowledge acquisition; knowledge representation schemes; conceptual data analysis; plausible reasoning techniques; validation and measurement methods; production-rule programming. Topics include Knowledge representations and mappings, approaches and issues (e.g., predicate logic, fuzzy logic, weak and strong slot and filler structures), knowledge acquisition, the frame problem, symbolic reasoning under uncertainty (non-monotonic reasoning, augmenting a problem Solver), statistical reasoning, building knowledge-based systems, the Semantic web, and knowledge-based ontologies. Also, it covers types of applications and architectures of expert systems, the relationships of expert systems to AI and to knowledge-based systems, inference engines and their types, the generation of explanations, and dealing with uncertainty—probabilistic reasoning based on Bayesian belief networks. Project work involves hands-on experience with prevalent expert system shells (e.g., Prover9/Mace4, CLIPS, FuzzyCLIPS, Matlab, HuginLite).					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS, SE, MI, and AI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS319		Programming Language Concepts			3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The purpose of this course is to study fundamental concepts in programming languages and the major tools and techniques for implementing them. Syntactic and semantic specification of programming languages. Basic programming language concepts such as variables, binding, block structure, scope, data types, and type checking. Data and data abstraction: Primitive and composite types, user-defined types, abstract data types. Implementation issues. Control and control abstraction: Control structures, subprograms, and parameter passing, parallel processes. Language structure and problem models: Imperative, functional, logic-based, and object-oriented languages. Programming language history.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS320		Natural Computing		3	
Prerequisites		Soft Computing (CS304)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
The lectures cover the following subjects: • Computational aspects of animal behavior and of biological, chemical, or physical systems. • The basics of Genetic Algorithms: selection, recombination and mutation, fitness, and objective functions • Variants of GAs: different types of crossovers and mutation, of selection and replacement. Inversion and other operators, crowding, niching, island, and cellular models • Theory: the schema theorem and its flaws; selection takeover times; statistical mechanics approaches as a theoretical basis for studying GA issues • Hybrid algorithms, memetic algorithms • Pareto optimization • Ant Colony Optimization: Basic method for the traveling salesperson problem, local search, application to bin packing, tuning, convergence issues, and complexity. • Swarm intelligence, particle swarms, differential evolution. • Greedy randomized adaptive search procedure • DNA computing, molecular computing, and membrane computing. • Applications such as engineering optimization, scheduling and timetabling, data mining, and neural net design. • Comparisons among metaheuristic algorithms, no-free-lunch theorems • Experimental issues: design and analysis of sets of experiments.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60

Student Activities Breakdown (Distribution for Study Blocks)			
Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

CS321		Modeling and Simulation		3	
Prerequisites		Object-Oriented Design and Programming (CS102) Probability and Statistics for Computer Science (BS104)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces simulation and modelling. It presents the concepts of system, model, simulation, and simulation components, in association with Program organization and logic steps in a simulation study. The course may cover the following main topics:					
• Statistical models, pdf, CDF: Statistical models in simulation, Input probability distribution functions for discrete systems, Continuous and empirical distribution functions, Problem solving on statistical models in simulation. • Queueing systems and single server queue: Characteristics and performance measures of queueing systems, analysis, simulation, and computer representation of a single server queueing system. • Random numbers and random variates: Generation and testing of random numbers, issues and challenges in congruential generators, Generation of random variates, problem solving on random numbers and variates. • Manufacturing systems. Simulation and issues of manufacturing and material handling systems, Modelling of system randomness, model verification, validation, and credibility, Problem solving and case studies on simulation of manufacturing systems. • Monte Carlo Simulation: Basics of Monte Carlo simulation, Inventory control simulation using Monte Carlo technique, Monte Carlo simulation of queueing problems, problem solving on Monte Carlo Simulation.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS, CS, SE, and AI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					

Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

CS322		Game Theory		3							
Prerequisites		Probability and Statistics for Computer Science (BS104)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course covers the mathematical theory of strategic interactions between self-interested agents. It provides a range of models for representing strategic interactions and, associated with these, a family of solution concepts that attempt to characterize the rational outcomes of games. This course provides ways to incorporate artificial intelligence methods and algorithms into game design. The course includes: Strategy games – rationality – dominance relations – bargaining theory – noncooperative games – cooperative games – games with incomplete information – repeated games and evolutionary stable strategies – representation of games – decision making and coordinating action – tactical and strategic AI – case studies will be used to illustrate the implementation of game theory to real-world problems along with game-theoretic settings.											
Taught in Program(s)											
As major in Program(s) Code(s): ---											
As minor in Program(s) Code(s): CS											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

CS323		Healthcare Informatics		3							
Prerequisites		Data Structures and Algorithms (CS201)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course aims to provide intensive input in the areas of healthcare informatics, medical software engineering, and pharmaceutical modelling. By the end of this course, students will learn machine learning and software engineering technologies to analyze data trends and predict pandemic behavior of any disease progression, examine patient data during surgical planning, and build prognostic and survival models, etc. This course seeks to close the gap between computer science and medicine, combining the design and problem-solving skills of computer science with medical and biological sciences to advance health care treatment, including diagnosis, monitoring, and therapy. This course also covers clinical informatics, health information exchange, telehealth, mobile health application development, personalized healthcare, health informatics standards, and regulations.											
Taught in Program(s)											
As major in Program(s) Code(s): MI											
As minor in Program(s) Code(s): IS, CS, and IT											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

CS324		Biomedical Informatics		3	
Prerequisites		Data Structures and Algorithms (CS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course focuses on the foster of computer science principles in the field of biomedicine. It combines the disciplines of biology, medicine, computer science, and predictive analysis to address the challenges and opportunities in biomedical research. The course covers various topics such as introduction to medical informatics - acquisition, storage, and use of biomedical data - medical imaging systems - computer-aided diagnostics (CAD) - Natural language processing in healthcare and biomedicine - ethics in biomedical informatics: Users, Standards, and Outcomes. Students gain knowledge and skills in areas such as predictive analytics, artificial intelligence-based clinical decision-making, genomics, and medical imaging. The course may include lectures, laboratory work, research projects, and practical exercises to provide students with hands-on experience in applying informatics principles to real-world biomedical scenarios.					
Taught in Program(s)					
As major in Program(s) Code(s): MI					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS325		Assembly Language		3									
Prerequisites		Computer Organization and Architecture (CS311)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
The course objectives are to learn about the hardware of the personal computer and to understand machine-language code and the hexadecimal format. understand the steps involved in assembling, linking, and executing a program, write programs in assembly language to handle the keyboard and screen, perform arithmetic, convert between ASCII and binary formats, perform table searches and sorts, and handle disk I/O, trace machine execution as an aid in program debugging, write macro instructions to facilitate faster coding, and Link separately assembled programs into one executable program.													
Taught in Program(s)													
As major in Program(s) Code(s): ---													
As minor in Program(s) Code(s): CS													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

CS401		Natural Language Processing and Arabizations		3			
Prerequisites		Fundamentals of Artificial Intelligence (CS204)					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		---		2			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course covers basic topics in natural language processing (NLP), including word and sentence tokenization, syntax analysis, parsing, part-of-speech tagging, information extraction, and machine translation. Underlying theory from probability, statistics, and machine learning for the field. Fundamental algorithms like n-gram language modeling, naive Bayes, and maximum entropy classifiers; sequence models like Hidden Markov Models. Familiarize students with the special characteristics of the Arabic language and the most important algorithms and data structures that are commonly used to solve many Arabic NLP problems.							
Taught in Program(s)							
As major in Program(s) Code(s): CS and AI							
As minor in Program(s) Code(s):---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)		Oral Exam (OE)		Final (FE)	
		☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20			15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS402		High Performance Computing		3	
Prerequisites		Operating Systems (CS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The course introduces the architectures and programming models of modern parallel computing systems, particularly those implemented for HPC. It presents Moore’s Law, Flynn's taxonomy, shared-memory and distributed-memory architectures, CPU design for parallel computing: functional units, instruction set, pipelining, branch prediction, ILP (superscalar, VLIW, SIMD instructions), and multithreading. Caches. The course also provides an overview of modern HPC architectures (e.g., many-core GPGPUs, multi-core CPUs) and Programming paradigms and languages for parallel computation, including shared-memory multiprocessing (e.g., OpenMP) and many-core architectures (e.g., GPGPUs: operations, features, and programming). The programming techniques encountered cover the main algorithm design and analysis ideas for three major classes of machines: for multicore and manycore shared memory machines, via the work-span model; for distributed memory machines like clusters and supercomputers, via network models; and for sequential or parallel machines with deep memory hierarchies (e.g., caches). These techniques were applied to fundamental problems, such as sorting, tree and graph search, and linear algebra, among others. Programming models to be used include OpenMP, MPI, and possibly others. In addition, the course provides an overview of applying modern high-performance computing across clusters, grids, the cloud, Edge, and Fog computing as distributed HPC architectures capable of processing large amounts of data (i.e., Big Data) with high performance.					
Taught in Program(s)					
As major in Program(s) Code(s): CS					
As minor in Program(s) Code(s): SE					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS403		Compiler Construction		3	
Prerequisites		Object-Oriented Design and Programming (CS102)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS	
				5.14	
Course Content					
The course includes the theory, the design, and the implementation of a large language translator system: lexical analysis, syntactic analysis, parsing, syntax-directed translation, syntax trees, declarations, types, and symbol management, run-time environments, storage organization, parameter passing, dynamic storage allocation, intermediate language, code generation, and optimization. Programming projects using selected programming languages to enhance practical aspects. Detailed topics of the course may include: • Compiler Structure: Analysis and Synthesis Model of Compilation. Basic concepts related to Compilers, such as interpreters, simple One-Pass Compilers, preprocessors, macros, symbol tables, and error handlers. • Lexical Analysis: Its role, Specification and Recognition of tokens, Input Buffer, Finite Automata relevant to compiler construction, syntactic specification of languages. • Syntax Analysis: Its role, Basic parsing techniques: Problem of Left Recursion, Left Factoring, Ambiguous Grammar, Top-down parsing, Bottom-up parsing. • Semantic Analysis: Static & Dynamic Checks, Typical Semantic errors, Scoping, Type Checking; Syntax directed definitions (SDD) & Translation (SDT), Attribute Types. • Symbol Table Design: Function of Symbol Table, Information provided by Symbol Table, Attributes and Data Structures for Symbol Table. • Run-time storage management. • Intermediate Code Generator: High-level and Low-level intermediate representation, Syntax tree & DAG representations, Three-address code, Quadruples, Triples, SDT for intermediate code. • Code Generator: Factors affecting a code generator, Target Language, Basic blocks and flow graphs, Dynamic programming code-generation algorithm. • Code Optimization: Need and criteria of Code Optimization, Basic optimization techniques. • Case Studies of some compilers, like the C compiler, C++ compiler.					
Taught in Program(s)					
As major in Program(s) Code(s): CS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	

Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-	-		-		-	

CS404		Distributed Ledgers and Consensus Protocols			3	
Prerequisites		Distributed Systems (CS306)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
In this course, students learn the key technical elements of the blockchain (or, more generally, the distributed ledger) technology and its advanced features, such as smart contracts, variations (e.g., permissioned versus permissionless and private blockchains), and available blockchain platforms. Students also learn the following issues: security, efficiency, and the scalability of the technology. Cybercurrency (e.g., Bitcoin) and other typical applications in areas such as finance will also be introduced.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS405	Selected Topics in Computer Science-1				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to introduce selected topics in computer science.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS406		Selected Topics in Artificial Intelligence			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Artificial Intelligence and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS and IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS407		Selected Topics in Computer Science-2				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course aims to introduce current trends in computer science.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): CS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education	
☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation		☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth	
☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		☒ GOAL 11: Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		☐ GOAL 15: Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions	
☒ GOAL 17: Partnerships to achieve the Goal							
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)		Oral Exam (OE)		Final (FE)	
		☐ Assessment		☒ Exam		☐ Assessment	
		☒ Exam		☐ Assessment		☒ Exam	
20		15		5		60	
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS408		Distributed Intelligent Automation Systems		3							
Prerequisites		Fundamentals of Artificial Intelligence (CS204)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course addresses the problem of how to design future automation systems in the Internet of Things era, when all sensors and actuators are connected to wired and wireless networks and equipped with their own embedded microcontrollers (aka intelligent sensors/actuators). It discusses motivations for designing automation systems in a distributed manner in application areas such as advanced manufacturing and critical infrastructure. Multi-Agent Systems. The main benefit of using such an automation architecture is related to flexibility and adaptability requirements: it is easier to reconfigure a system that consists of intelligent building blocks in a plug-and-play way than to design and program a new configuration. The main tool used in this course will be the emerging automation architecture of distributed function blocks. It will be presented alongside a variety of modern computing technologies, such as human-computer interaction, model-based design, and artificial intelligence.											
Taught in Program(s)											
As major in Program(s) Code(s): ---											
As minor in Program(s) Code(s): CS											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20		15			5			60			
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)		Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)			
-		-			-			-			

CS409		Parallel and Distributed Algorithms			3	
Prerequisites		Distributed Systems (CS306)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The course objectives are to learn parallel and distributed algorithms development techniques for shared-memory and message-passing models; to study the main classes of parallel algorithms; and to study the complexity and correctness models for parallel algorithms. More specific: Learning the conceptual models for the specification and verification of parallel and distributed algorithms (P&D); Understanding the complexity models of P&D algorithms; Learning the development methods of efficient P&D algorithms using data partitioning and functional partitioning, parallelism exploitation for different process topologies, hiding communication latencies, load balancing; Development techniques for parallel algorithms; examples from the main classes: prefix algorithms, search, sort, selection, matrix processing, graph algorithms, lists, trees; Study of the main development techniques of classes of distributed algorithms: wave algorithms, mutual exclusion, topology establishment, termination, fault tolerance, leader election, genetic algorithms ...etc.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS410		Algorithmic Graph Theory		3	
Prerequisites		Data Structures and Algorithms (CS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
• Introduction to graphs: undirected graphs, directed graphs, weighted graphs, graph representation, and special classes of graphs (trees, planar graphs, etc.). • Applications of graphs (in computing, AI, telecommunications, networking, etc.). • Basic algorithmic techniques for graph problems: graph traversals (DFS and BFS), topological sorting, Euler tours. • Further algorithmic problems on graphs: minimum spanning trees, shortest path problems, and matching problems. • Planar graphs and their properties. Euler's formula, the planar separator theorem, and their algorithmic applications. • Further optimization problems on graphs, including graph coloring and graph questions in distributed systems. • Discussing practical applications of graphs and efficient algorithms for such practical problems. Approximation algorithms and heuristic algorithms. Applications to searching in massive graphs (e.g., page ranking); use of structural properties and algebraic properties.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS411		Advanced Genomics and Genetic Analysis		3	
Prerequisites		Bioinformatics (CS305)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course provides an in-depth understanding of genomes, focusing on genes within an organism's DNA. Students learn about genome sequencing technologies, genome assembly, and annotation. They explore the analysis of structural variations in genomes and comparative genomics to understand evolutionary relationships. The course covers functional genomics, including transcriptomics, proteomics, and epigenomics, to study gene function and regulation. Students gain skills in analyzing large-scale genomic datasets and interpreting genomics data using bioinformatics tools. The application of genomics in clinical settings, such as genomic medicine, is also discussed. Additionally, the course covers genomic variation in human populations and its impact on disease susceptibility. Students become familiar with genomic databases, resources, and tools. Throughout the course, practical exercises and data analysis projects offer hands-on experience with real genomic datasets and software.					
Taught in Program(s)					
As major in Program(s) Code(s): MI					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS412		Spoken Language Processing				3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
It covers topics in spoken language technology, including dialog and conversational systems, digital speech processing, automatic speech recognition, and speech synthesis. This includes recent end-to-end deep learning approaches for speech recognition and Spoken language products with modern toolkits. The course includes some spoken language processing applications, such as automatic dictation of speech for medical records, analysis of speech in language pathologies (e.g., in cerebral palsy, Parkinson's disease, and Alzheimer's disease), and assistive technologies such as text-to-speech (with and without brain-computer interfaces) for people with limited speech ability.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): CS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS413		Natural Language Processing with Deep Learning			3
Prerequisites		Deep Learning (CS309)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
In this course, students will gain a thorough introduction to applying Deep Learning for NLP. Students will become familiar with recent algorithms used to solve many NLP and Arabization problems. This includes word embeddings based on deep learning, such as GloVe and Word2Vec; contextual embeddings; self-supervised learning; recurrent neural networks; transformer models and their variations; large language models; dependency parsing; seq2seq; attention variants; neural machine translation.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

CS414		Introduction to Formal Languages and Automata Theory		3			
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		---		2			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
The course presents a study of formal languages and the correspondence between language classes and the automata that recognize them. Formal definitions of grammars and acceptors, deterministic and non-deterministic systems, grammar ambiguity, finite state and push-down automata, and normal forms will be discussed.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): CS and AI							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam		
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

CS415		Software Testing and Quality Assurance			3	
Prerequisites		Software Engineering (CS203)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides an understanding of software testing and quality assurance techniques. To help students to develop skills that will enable them to construct software of high quality, reliable software, and that is reasonably easy to understand, modify, and maintain. Software quality factors: the components of the software quality assurance system. SQA components in the project life cycle; fundamentals of Software Testing, types, and tools. Software metrics. Quality planning and quality control. Inspections and formal technical reviews. Blackbox and white-box testing, problem analysis, and reporting techniques. Verification and validation techniques. Process and product quality assessment. Process measurement. Software quality assurance standards.						
Taught in Program(s)						
As major in Program(s) Code(s): SE						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS416		Ontology Engineering			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces the methods and methodologies for building ontologies, their uses, and an overview of their application in semantically enabled systems. It also includes a knowledge construction of the domain using formal ontology representations. It includes topics such as Descriptive logic – RDF Ontology development – Ontology development – Ontology language – Web Ontology Language (OWL) – OWL API – Rule Interchange Format (RIF) – Query language – Semantic Portals.						
Taught in Program(s)						
As major in Program(s) Code(s): MI						
As minor in Program(s) Code(s): CS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS417		Autonomous Agents			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The goal of this course is to introduce students to the techniques needed to develop complete, integrated AI-based autonomous agents. Topics include architectures for intelligent agents; foundations of agent perception and Interpretations: visual perception, auditory perception, other sensor modalities; planning algorithms: discrete planning, motion planning; decision-theoretic planning, planning under sensing uncertainty.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS and AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS418	Reinforcement Learning and Control				3
Prerequisites	Machine Learning (CS302)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
This course introduces reinforcement learning, starting with basic concepts and building to current state-of-the-art algorithms, and provides an overview of methods for learning-based model generation, controller tuning, and learning control policies guided by reinforcement learning.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS and AI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

CS419		High-Performance Big Data Computing			3	
Prerequisites		High Performance Computing (CS402)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces an emerging field that brings together high-performance computing, big data, and deep learning to address the challenges of large-scale data computing. The course covers techniques for accelerating big data computing across computation, communication, memory and storage, codesign, workload characterization, and benchmarking. It presents a detailed overview of big data computing systems, high-performance deep learning over big data (DLoBD) stacks, and HPC cloud technologies.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS and AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS420		Computational Social Science		3	
Prerequisites		Introduction to Web Programming (IS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course serves as a multidisciplinary, multi-perspective, and multi-method guide to better understanding society and human behavior using modern research tools. This course gives students easy access to some of the exciting new possibilities for studying society and human behavior. An increasing share of human interaction leaves a massive digital footprint. Studying it can give unprecedented insights into what society is and how it works, including its intricate social networks that have long been obscure. Computational power can be used to detect hidden patterns through analytical tools like machine learning and natural language processing. Finally, computer simulations can be used to explore hypothetical situations that may not even exist, but that we would like to exist for a better world.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

CS421		Biometrics			3	
Prerequisites		Data Structures and Algorithms (CS201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The purpose of this course is to provide the student with a basic introduction to biometric technologies. Several types of biometric authentication can identify a person by using their unique physical or behavioral characteristics. These include fingerprint scanning, retinal scanning, facial recognition, and voice analysis. The student will gain an understanding of the sciences and methodologies used to measure and analyze unique biological traits, in order to safeguard identity.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): CS and MI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

CS422		Green Computing		3							
Prerequisites		Operating Systems (CS202)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
Reducing the energy usage of computers, whether they are handheld mobile devices, cloud computing platforms, or power-hungry supercomputers, is paramount today and will remain a continuous challenge in the future. This course is intended to provide an insight into: power and energy consumption modeling and analysis; energy aware computing, i.e., how various power reduction techniques can be used and orchestrated such that the best performance can be achieved within a given power budget, or the best power efficiency can be obtained under prescribed performance constraints; energy efficient software design, and green computing in the context of large scale computing systems or smart grid-aware computing. The course covers how to apply computing to sustainability to reduce our society's energy footprint.											
Taught in Program(s)											
As major in Program(s) Code(s): ---											
As minor in Program(s) Code(s): CS											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

IS101		Introduction to Web Programming			3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course is an introduction to programming for the World Wide Web. It will discuss the relationship between clients and servers, briefly How to design web pages for different applications, and how web pages are constructed using several technologies such as: • Hypertext Markup Language (HTML) for authoring web pages • Cascading Style Sheets (CSS) for applying stylistic information to web pages. • JavaScript (JS) for creating interactive web pages. • Asynchronous JavaScript and XML (Ajax) for enhanced web interaction and applications. • PHP web services for handling and responding to web service requests.						
Taught in Program(s)						
As major in Program(s) Code(s): CS, IS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS201		Data Analysis and Management		3	
Prerequisites		Probability and Statistics for Computer Science (BS105)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course is an introduction to the management and analysis of data, in which students will represent data graphically, describe and interpret graphical and numerical representations of data sets, compare data sets using data analysis, and draw relevant conclusions about the populations from which the data sets are drawn. This course is designed to provide students with a solid background in statistics, whilst developing their critical thinking and quantitative reasoning. Students will use appropriate computer software (Stata and R). Content includes: • Stata and R: The basics (importing and exporting data, recording data, formatting data, labelling variable names and data values; using dates, data display and summary presentation, and creating programs). • Stata and R: graphs, data management, and statistical quality assurance methods (including advanced graphics to produce publication-quality graphs). • Data management using Stata and R (using functions to generate new variables, appending, merging, transposing longitudinal data; programming skills for efficient and reproducible use of these packages, including loops and arguments).					
Taught in Program(s)					
As major in Program(s) Code(s): CS, IS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

IS202		Introduction to Database Systems			3	
Prerequisites		Data Structures and Algorithms (CS201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
In this course, students should study the following topics: characteristics and advantages of the database management systems (DBMS), database concepts and architecture; data models, database schemes and instances, DBMS and the concept of program-data independence, database languages and interfaces, database models, relational data model and relational algebra, relational model constraints; domains, keys, and integrity constraints, the structured query language (SQL); data definition, queries, update, statements, and views in SQL, database design; functional dependencies, normal forms.						
Taught in Program(s)						
As major in Program(s) Code(s): CS, IS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS203		Project Management			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides an introduction to management concepts, principles, techniques, and terminology with particular reference to IS projects. It addresses issues such as project definition, scope management, planning, organization, resources, scheduling, control, quality, cost estimation, time estimation, risk management, and project management tools and software. Covered project management tools include Work Breakdown Structure, Gantt charts, PERT, and the critical path method. Topics covered also include project management ethics and effective project manager skills such as people and leadership skills.						
Taught in Program(s)						
As major in Program(s) Code(s): CS, IS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS301	System Analysis and Design				3
Prerequisites	Software Engineering (CS203)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS		5.14
Course Content					
This course covers the fundamental knowledge, methods, and skills needed to analyze, design, and implement computer-based systems. It addresses the role of the systems analyst, the techniques and technologies used, and the ethical considerations in requirements specification. The structured software development life cycle, modeling techniques, and development phases are comprehensively discussed and reviewed. In the modeling techniques section, process models, information models, and object-oriented models are thoroughly described. A project is given to all students that should cover analysis and design phases of a relatively data-oriented business case, with emphasis on data modeling (ER diagrams) and process modeling (DFDs).					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS302		Advanced Database			3	
Prerequisites		Introduction to Database Systems (IS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course covers the following topics: DBMS architecture and administration; centralized and client-server approaches, system catalog, and data dictionary, transaction management; concepts, characteristics, and processing, recovery techniques, concurrency control techniques: serializability, deadlock, locking schemes, time-stamp ordering, multi-version, and optimistic techniques, DB security, Database Integrity object-oriented databases, introducing to new emerging DB technologies and applications; Web DB.						
Taught in Program(s)						
As major in Program(s) Code(s): IS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS303		Enterprise Architecture		3			
Prerequisites		Introduction to Database Systems (IS202)					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		---		2			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This Course presents the basic concepts and methodologies for Enterprise Architecture within a framework and structure. The objectives of this course are: Explain what Enterprise Architecture is and why it is important; Discuss the relationship between vision, strategy, and architecture; describe enterprise architectural IT frameworks. Describe the elements of an enterprise IT architecture. Describe a methodology for developing enterprise architecture. Address the challenges facing enterprise architects. The student will learn how to design enterprise architectures at scale, develop considerations for architecture in relation to an organization’s vision and strategy, and create a process for establishing an ongoing enterprise architecture.							
Taught in Program(s)							
As major in Program(s) Code(s): IS and SE							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

IS304		Advanced Web Programming		3	
Prerequisites		Introduction to Web Programming (IS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces advanced web programming technologies. Topics include basics of handling HTTP, MVC, IoC, AOP, ORM, DAO, DI design patterns, Client-side web development, RESTful web services, and cloud platforms. • Explain the concepts of web applications. • Identify and explain various components of the web application stack. • Explain the technologies and tools involved in web application development. • Design and implement a web application project that uses the components that are learnt. • Understand and identify the use of web services and cloud platforms in web programming.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): IT and SE					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

IS305		Information Theory		3	
Prerequisites		Probability and Statistics for Computer Science (BS105)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Information theory studies the fundamental limits of the representation and transmission of information. This course provides an introduction to information theory, studying fundamental concepts such as probability, information, and entropy, and examining their applications in data compression, coding, communications, and probabilistic inference. Lectures will focus on intuition, applications, and the ways in which communication and the representation of information manifest across various areas. Upon successful completion, students will have the knowledge and skills to: • Understand and apply fundamental concepts in information theory, such as probability, entropy, information content, and their interrelationships. • Understand the principles of data compression. • Compute entropy and mutual information of random variables. • Implement and analyze basic coding and compression algorithms. • Understand the relationship between information theoretical principles and Bayesian inference in data modelling. • Understand some key theorems and inequalities that quantify essential limitations on compression, communication, and inference. • Know the basic concepts regarding communications over noisy channels.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS306		System Development Methods		3	
Prerequisites		System Analysis and Design (IS301)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Successfully designing and developing information systems is complex and difficult. Several techniques and approaches have been developed. This course introduces students to various methodologies and equips them with the skills to identify their strengths and weaknesses in key areas. Upon successful completion of this subject, students should be able to:					
- Analyze key aspects of several Information System development methodologies to develop business Systems. - Explain that different methodologies target different aspects of Information System development, such as requirements gathering, project management, or cross-organizational development. - Evaluate development methodologies and articulate their underlying philosophies to solve organizational Information System problems. - Illustrate a methodology, or aspects of different methodologies, to substantiate the suitability of particular circumstances.					
Taught in Program(s)					
As major in Program(s) Code(s): IS and SE					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS307		Information Storage and Retrieval		3	
Prerequisites		Introduction to Database Systems (IS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Information storage and retrieval (ISR) concerns the representation, storage, organization, and access of information items. The representation and organization of the information items should provide the user with easy access to the information he is interested in. The main objective of this class is to study research methods and literature in information storage and retrieval systems, including analyzing, indexing, representing, storing, searching, retrieving, processing, and presenting information and documents using fully automatic systems. The information may be in text, hypertext, multimedia, or hypermedia formats. The systems are based on various models, e.g., Boolean logic, fuzzy logic, probability theory, and other approaches, and are implemented using inverted files, relational thesauri, special hardware, and other approaches. Core research skills of literature analysis, innovation, evaluation of new ideas, and communication are emphasized via homework and projects. Most students may want a broad overview of the research topics, methodologies, major results, open problems, and potential future research directions. The content includes the Information Systems Designer's view, models of information retrieval using the Boolean model, text analysis, retrieval effectiveness, output presentation and visualization, search effectiveness, document processing and storage, including search problems and solutions, and web information retrieval. Specifically, this course will help students: • To identify basic theories and analysis tools as they apply to information retrieval. • To develop an understanding of the problems and potentials of current IR systems • To learn and appreciate different retrieval algorithms and systems. • To apply various indexing, matching, organizing, and evaluating methods to IR problems • To become aware of current experimental and theoretical IR research.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)	Oral Exam (OE)		Final (FE)	

	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5		60
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-	-		-		-	

IS308	Data Mining			3	
Prerequisites	Data Analysis and Management (IS201), Introduction to Database Systems (IS202)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS	5.14	
Course Content					
This course introduces data mining concepts and techniques. The goal is for the students to have a solid foundation in data mining that allows them to apply data mining techniques to real-world problems and to conduct research and development in new data mining methods. Topics include data mining algorithms and methods, such as association analysis, classification, and cluster analysis, as well as emerging applications and trends in data mining.					
Taught in Program(s)					
As major in Program(s) Code(s): IS and MI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS309		Information System Quality		3	
Prerequisites		System Analysis and Design (IS301)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Quality Assurance is a part of quality management that ensures quality requirements are met. The requirements for high-quality, reliable, and predictable systems become increasingly important as we strive to meet the customer’s quality expectations. These activities start before the product is developed and continue through the product development and release cycles. In this course, the student will learn how to define and implement standards to improve the development life cycle and ensure QA processes are followed. The course also describes how these processes fit into the overall software development process. Students will learn how to deliver high-quality products, with predictability and reliability, on time and within budget. Students will learn how testing has adapted to incorporate Agile methodologies. Specific topics include test management phases, test metrics & reporting, implementing an organizational quality policy, building an effective SQA organization by hiring the right people, and operating in a global environment. Students will learn how test organizations use tools to increase productivity, and the growing role automation plays in achieving project goals. Students will also learn techniques and content of an SQA and how to conduct regular Software audits. Also, the Student will discover basic concepts of risk management and metrics collection for items such as rework, cost estimates, and code coverage.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS310		Economics of Information Systems				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
The course presents the fundamentals and emphasizes the importance of economic considerations and reasoning in information systems planning, design, and implementation. Students learn principles and techniques of I/S project cost and value estimation. Various project valuation techniques are also employed to guide IT investment decision-making. The principles and importance of IT Governance and project portfolio management in improving IT investment contribution to business value are emphasized.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): IS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

IS311		Business Intelligent systems		3	
Prerequisites		Data Analysis and Management (IS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Business Intelligence (BI) refers to technologies, applications, and practices for collecting, integrating, analyzing, and presenting business information. The purpose of business intelligence is to support better business decision-making. This course provides an overview of BI technology and its application to an organization’s strategies and goals.					
After completion of this course, the student should be able to:					
• Introduce the concepts and components of Business Intelligence (BI) • Evaluate the technologies that make up BI (data warehousing, OLAP) • Define how BI will help an organization. • Identify the technological architecture of BI systems. • Plan the implementation of a BI system.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS312		Risk and Crisis Management			3	
Prerequisites		Project Management (IS203)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course covers the basic concepts of software project risk and crisis management, and the important elements of identification, analysis, mitigation, and tracking that personnel need to be aware of when implementing a software project. This course will discuss Project management (overview), Project risk management, plan risk management, identify risks, qualitative risk analysis, quantitative risk analysis, plan risk response, and risk control.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS313		Web Application Technologies		3	
Prerequisites		Introduction to Web Programming (IS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The course focuses on designing and developing Web-based applications using a variety of programming languages and tools. Students will be exposed to the architecture of Internet application development. Class projects include developing business-to-consumer (B2C) and business-to-business (B2B) applications, among others. We will learn to create web pages using HTML, specifically XHTML. Topics also include tables, frames, and HTML form creation. We will learn design concepts using Cascading Style Sheets. We will then venture into web scripting with JavaScript, the primary client-side scripting language of the Internet. We will also review the object-oriented programming paradigm. We then jump into PHP, one of the world’s most popular programming languages for web development. It is a flexible, scalable, and easy-to-program language. PHP is currently used on more than 15 million websites, and the student will learn, as part of the curriculum, what this language is all about. With increased familiarity with PHP, we will introduce MySQL, an open-source and reliable database management system that is rapidly gaining a worldwide user base. The goals for the class are that, upon completion of the course, students will understand the challenges, technologies, and issues involved in developing and deploying Web-based applications. Also, the student will be able to apply the systems approach to business problems; be part of a team that helps solve a business problem involving information technology; and improve their oral, written, and group communication skills through in-class participation and a final group presentation.					
Taught in Program(s)					
As major in Program(s) Code(s):---					
As minor in Program(s) Code(s): IS and SE					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					

Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

IS401	Intelligent Information System			3	
Prerequisites	Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS		5.14
Course Content					
The objectives of this subject are to understand how intelligent techniques can be used in the construction of information systems to support management decision-making and to enable students to master the techniques for problem-solving across various application areas in business and finance, computing, and engineering. Specifically, this course will help students: - Understand and apply the latest techniques in artificial and computational intelligence that can be used to facilitate decision-making processes. - Apply knowledge management principles and techniques to solve practical business problems. - Critically review and consolidate existing knowledge to design the important components to support the decision-making processes in an Executive Support system. - Possess the knowledge to evaluate the different commercially available or public domain tools that can be used to tackle specific problems related to business decision making; and - Possess the ability to understand and decide if future new techniques in artificial and computational intelligence can be used to solve different practical business problems. - Get a deep knowledge and technical skills required to meet the surge of intelligence already being used in information systems for a wide range of domains, such as smart cities, advanced manufacturing, autonomous driving, and Internet of Things.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS402		Information Security and Privacy		3	
Prerequisites		Computer Networks (IT201), Advanced Database (IS302)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
The course is an introductory course in information security and privacy. The course will provide students with an in-depth understanding of technical and managerial issues of information security and privacy. The course achieves this by discussing foundational aspects of a wide range of activities, methods, methodologies, procedures, and best practices used in the field of information security and privacy. Upon successful completion of this course, the students will be able to:					
• Analyze the impact of systems security and information privacy on organizations. • Understand the broad role of information security and privacy in business processes. • Understand the technical foundations for information security and privacy. • Study the implications for information security and privacy in Enterprise Information Systems and Inter-Organizational business processes. • Understand the issues in managing the security and privacy of Information Systems. • Understand how to assess and mitigate the risks and vulnerabilities in systems. • Identify and prioritize information assets. • Identify and prioritize threats to information assets. • Define information security and privacy strategies and architecture. • Plan for and respond to intruders in an information system. • Describe legal and public relations implications of security and privacy issues.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					

Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

IS403	Big Data Processing			3	
Prerequisites	Information Storage and Retrieval (IS307)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS	5.14	
Course Content					
This course will cover a series of important Big-Data-related problems and their solutions. Specifically, we will introduce the characteristics and challenges of the Big Data, computing paradigm sand platforms (e.g., MapReduce), big data programming tools (e.g., Hadoop and MongoDB), big data extraction and integration, big data storage, scalable indexing for big data, big graph processing, big data stream techniques and algorithms, big probabilistic data management, big data privacy, big data visualizations, and big data applications (e.g., spatial, finance, multimedia, medical, health, and social data). Upon successful completion of this course, the students will be able to: • Understand the characteristics and challenges of big data. • Know the existing big data processing platforms/tools. • Understand big data collection, integration, and storage. • Know big data indexing. • Understand the basics of MapReduce paradigms. • Apply various queries over big data. • Understand the core techniques of processing big data. • Understand various real-world applications and the techniques that involve big data.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS404		Data Science Applications and Analysis		3	
Prerequisites		Data Analysis and Management (IS201), Data Mining (IS308)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Students will be able to analyze data and make data-driven business decisions, leveraging computer science and statistical analysis. Case studies will be used to demonstrate current best practices and common pitfalls. This course will help students develop data science skills while solving real-world problems. Also, Students will apply data skills to big data to gain business insights, such as hidden patterns, untold facts, unknown correlations, market trends, and customer preferences, extracting information to make informed business decisions related to understanding customer behaviors and financial investments. Case studies on big data analytics will be used to design and communicate concepts and techniques for business decision-making. Upon successful completion of this course, students will be able to: • Understand and employ tools used by real data Science, such as NumPy and Pandas. • Recommend methodologies for applying data science in business and modern organizations. • Practice predictive modeling and model selection. • Use unsupervised learning to explore data, engineer meaningful features, and select them. • Solve complex supervised learning problems using different models. • Learn to apply hyperparameter tuning and cross-validation strategies to improve model performance. • Discuss the key concepts, benefits, and challenges of big data analytics. • Critically evaluate data sources, data sets, and inferences made from big data analytics.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, SE, and MI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam

20	15	5	60
Student Activities Breakdown (Distribution for Study Blocks)			
Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

IS405		Distributed Database Systems			3
Prerequisites		Advanced Database (IS302)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course covers the fundamental issues of distributed databases with a focus on data fragmentation and allocation, query optimization, and transaction processing. Topics include Distributed database management systems architecture and design; data fragmentation, replication, and allocation; database security, authorization, and integrity control; query optimization; transaction management; distributed concurrency control and replica control; distributed object database management systems; multi-database systems.					
Taught in Program(s)					
As major in Program(s) Code(s): IS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

IS406	Selected Topics in Information Systems				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course intends to introduce selected topics of current trends in information systems.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS407		Enterprise Integration			3	
Prerequisites		Enterprise Architecture (IS303)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course is designed to provide students with an understanding of the theoretic and practical issues related to the application of enterprise systems within organizations. The main focus of this course is to demonstrate how enterprise systems integrate information and organizational processes across functional areas through a unified system comprising a single database and shared reporting tools. Enterprise systems, by their multi-dimensional, integrative nature, offer the depth of functionality and breadth of integration to demonstrate how organizations' global operations are managed.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS408		Enterprise Resource Planning		3	
Prerequisites		Enterprise Architecture (IS303), Introduction to Database Systems (IS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course provides students with an understanding of what Enterprise Systems (also commonly termed as Enterprise Resource Planning Systems, ERPs) are. The course is basically meant to introduce ERP Systems, which provide for integrating planning throughout the business cycle from raw materials to shop floor control to related human resource and finance processes. The course discusses the role of ERP Systems and software in reducing inventory, waste, scrap, and rework, and in utilizing resources efficiently. Functional modules will be given special emphasis to help users understand the practical world of enterprise planning and its impact on the bottom line of financial statements. This course systematically presents several conceptual and pragmatic methodologies, tools, and techniques for the various phases of enterprise implementation. After learning about these systems, we would discuss why they are useful to companies, and students would see the various jobs and positions associated with the use and deployment of ERPs. The course attempts to serve the practical needs of consultants, training personnel, industrial engineers, Supply Chain Planning managers, Finance managers, Shop floor managers (first-line managers), Human Resource Planning Managers, accountants, corporate planners, and administrators. In this course, we will cover important points to consider when shortlisting and purchasing an ERP, approaches to implementing an ERP System, and change management techniques to use during an ERP implementation. At the end of the course, the students will be able to: • Endow with practical knowledge and a deep understanding of ERP Systems. It would help them to address real-world business problems associated with ERP usage and implementation. • Develop an appreciation for the managerial aspects of selecting and implementing ERPs. • It will enhance learners' management and technical skills. Thereby enabling them to join advance careers in enterprise systems.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					

Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS409	Domain-specific Information Systems				3
Prerequisites	System Analysis and Design (IS301)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS		5.14
Course Content					
This course introduces students to the world of digital systems, including e-health, e-learning, e-business, etc. They will experience different examples of frameworks that enable the design, implementation, and optimization of such software systems. They will also be encouraged to explore new trends in domain-specific system architecture. This course is intended to be pure hands-on, targeting independent students who like to tackle challenges and dig deep to get things done. Enrolled students should expect to roll up their sleeves, become familiar with new frameworks, develop some code, conduct proper experimentation and analysis, and report their findings professionally.					
Taught in Program(s)					
As major in Program(s) Code(s):---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS410		Data Warehouse		3	
Prerequisites		Introduction to Database Systems (IS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course provides students with in-depth knowledge of data warehousing principles, techniques, and business intelligence systems. The course introduces data warehouse design, extract-transform-load (ETL), data cubes, and data marts. Classical notions of data warehousing and OLAP are developed: Architecture, conceptual and logical design, query processing, and optimization. Necessary techniques will be presented for designing, implementing, exploiting, and maintaining data warehouses, as well as for creating data integration workflows. Upon successful completion, students will have the knowledge and skills to: • Define, measure, and maintain data quality in the context of data warehousing. • Learn the steps involved in the development of an enterprise data warehousing solution. • Evaluate an organization for data warehouse maturity and business architecture alignment. • Create a data warehouse design and reflect on alternative design methodologies and design goals. • Create data integration workflows. • Create Business Intelligence using data warehouses with several OLAP and analytical tools. • Describe architecture and methods for the storage and provision of enterprise data.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS411		Advanced Operation Research			3	
Prerequisites		Operation Research (BS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to introduce advanced operations research approaches, i.e., advanced linear programming (Interior-Point Algorithm), integer programming (branch-and-cut, column generation), Dynamic Programming, Metaheuristics, Queuing theory, and the complexity analysis of algorithms. After completion of this course, the student should be able to: • Understand the Interior-Point Algorithm, Dynamic Programming • Apply exact and heuristic algorithms to solve the problem. • Analyze the complexity of the algorithm.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s):IS, CS, IT, and AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS412		Decision Support System		3	
Prerequisites		Data Analysis and Management (IS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Provides students with an understanding of the role of a decision support system in an organization, its components, and the theories and techniques used to construct them. After this course, students should be familiar with several decision-making methods in a business environment. Students will find that almost every type of problem in the business world that requires a quantitative solution has been explored, and methods for solving them have been devised. Students should be able to apply these methods to real-world situations they will one day encounter. Students should be able to: 1. Demonstrate an understanding of the theory of decisions and decision analysis, including probability, decisions under uncertainty, risk taking, and real-world problems. 2. Describe and understand the concept of decision support systems, including the collection of data, database management, modeling, group and organizational decision support systems, executive information systems, and expert systems. 3. Utilize commercial decision support and graphic presentation software packages to prepare typical business decision packages utilizing methodologies such as forecasting, linear programming, and decision trees.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS, CS, and SE					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS413		Data Visualization		3	
Prerequisites		Data Analysis and Management (IS201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Data visualization is one of the most powerful tools for exploring, understanding, and communicating patterns in data through readable graphics. It includes data modeling, data processing (such as aggregation and filtering), mapping data attributes to graphical attributes, and strategic visual encoding based on known properties of visual perception and the task(s) at hand. Upon successful completion of this course, students will be able to:					
• Generate powerful reports and dashboards using widely available tools. • Conduct exploratory data analysis using visualization. • Develop visual presentations of data for effective communication. • Apply data transformations for visualization. • Evaluate visualization design alternatives. • Students will create their own data visualizations and learn to use Open-Source data visualization tools.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS414		Spatial Database		3	
Prerequisites		Introduction to Database Systems (IS202)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Advanced topics in the storage, management, and retrieval of geospatial data using common proprietary and open-source relational database technologies. It will teach students how to create a spatial database and use it to answer geographic questions. Students will learn using both open-source and proprietary database technologies. At the successful completion of this course, students should be able to: • Retrieve data from an existing database using SQL Select queries. • Design a database schema from a set of requirements. • Implement that design by creating related tables. • Insert and update rows in a table. • Create spatially enabled tables in Postgres/PostGIS. • Work with PostGIS data using open-source desktop GIS software (QGIS). • Answer questions using PostGIS spatial functions. • Create an Esri file geodatabase. • Model real-world entities through subtypes, domains, topology rules, and relationship classes. • Set up and administer an enterprise (ArcSDE) geodatabase. • Manage vector and raster data in an enterprise geodatabase. • Understand enterprise geodatabase editing workflows. • Discuss the suitability of an open-source or proprietary approach to various project scenarios.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam

20	15	5	60
Student Activities Breakdown (Distribution for Study Blocks)			
Seminar-based (SB)	Self-learning-based (SL)	Project-based (PB)	Industry-based (IB)
-	-	-	-

IS415		Geographic Information Systems and Remote Sensing			3	
Prerequisites		Spatial Database (IS414)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Students study the fundamentals of Geographic Information Systems (GIS). of GIS; introduction to modern spatial data and structures; input of Geospatial data; functions of geographic information systems; spatial Analysis; coordinate transformation and map projection; interpolation techniques; relations between GIS and remote sensing, and applications of geographic information systems to a variety of of environmental and geologic issues.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS416		Usability and User Experience (UX)		3	
Prerequisites		System Analysis and Design (IS301)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
In the course, the student learns to evaluate the usability and user experience of products, understand how they affect product success, and design usable, enjoyable software products and services. After the course, the student can: - Evaluate the usability of products and services, e.g., using usability heuristics. - Estimate usage needs and formulate design requirements. - Do usability testing. - Understand quantitative analysis methods of usability and user experience. - Design user interfaces for simple applications and services. - Apply design patterns in UI design. - Understand the significance of accessibility in services.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IS417		Responsive Website Development and Design			3	
Prerequisites		Introduction to Web Programming (IS101)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Responsive web design is about creating the best user experience regardless of whether a user accesses the site from a desktop computer, a mobile phone, or a tablet. This course covers the basics of modern full-stack web development, from UX design to front-end coding to custom databases. Students will build foundational skills across a full range of technologies, including HTML/CSS, JavaScript, and MongoDB, and learn to implement a reactive, multi-user web application using the Meteor.js framework. They will also touch on more advanced topics, such as APIs, data visualization, and animation.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IS and IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IS418		Web of Data		3	
Prerequisites		Introduction to Web Programming (IS101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The Web became a way to exchange data between applications. Every day, the user consumes and produces this data through a growing variety of applications running on a growing variety of devices. This major evolution of the Web has applications in all domains of activity. So, in this course, learn how to publish, obtain, and use structured data directly from the Web. Learning the principles, languages, and standards for exchanging data on the Web will enable you to design and develop new applications, products, and services that leverage the volume and variety of data the Web provides.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS and CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

IT101		Fundamentals of Electronics and Electrical Circuits		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The Electrical and Electronic Fundamentals course focuses on the basic concepts and skills students need when working with electrical equipment. The course discusses the fundamental principles of electricity and electronics, as well as the operation of various devices, such as resistors, capacitors, amplifiers, and oscillators. Different types of circuits and their applications are also discussed. Students will use electrical test equipment and diagnostic tools. They will also learn about safety measures, codes, and rules and regulations that must be followed for preventing mishaps. The course covers the following topics: - Basics of Electricity, DC Circuit Analysis, Electrical Components and Devices, and AC Circuit Analysis. - Semiconductor Devices, Electronic Circuits, Digital Electronics, and Application of Digital Electronics. - Metering Devices, Regulatory Norms, and Protection of Electrical Equipment.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT102		Data Communication				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course provides an introduction to data communication protocols. The students will learn about the TCP/IP protocol stack, including the application, transport, network, and link layers. Specifically, students will learn application layer concepts and protocols (e.g., client-server model, HTTP, FTP, SMTP), transport layer concepts and protocols (e.g., reliable data transfer, TCP, UDP), network layer concepts and protocols (e.g., IP addressing, routing, forwarding) and link layer concepts and protocols (e.g., ARP, MAC addressing, MAC protocols).							
Taught in Program(s)							
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

IT103		Digital Logic Circuit		3	
Prerequisites		Fundamentals of Electronics and Electrical Circuits (IT101)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course aims to develop engineering skills in the design and analysis of digital logic circuits with applications to digital computers. It covers: Number systems, binary arithmetic and codes, logic gates, Boolean algebra and logic simplifications, Design and realization of combinational circuits, Functions of combinational circuits logic: Flip-Flops, analysis, design and realization of counters, analysis and realization of shift registers, Computer – aided engineering, analyzing synchronous and asynchronous machines. Upon completion of this course, students should be able to: • Know the basic differences between analog and digital systems. • Use binary numbers and codes. • Describe the operation of logic gates. • Apply Boolean algebra on a K-map. • Design combinational and sequential logic circuits.					
Taught in Program(s)					
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT201		Computer Networks			3	
Prerequisites		Data Communication (IT102)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides an introduction to networking and communication protocols. The students will learn about the TCP/IP protocol stack, including the application, transport, network, and link layers. Specifically, students will learn application layer concepts and protocols (e.g., client-server model, HTTP, FTP, SMTP), transport layer concepts and protocols (e.g., reliable data transfer, TCP, UDP), network layer concepts and protocols (e.g., IP addressing, routing, forwarding) and link layer concepts and protocols (e.g., ARP, MAC addressing, MAC protocols).						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT202	Internet Technology				3	
Prerequisites	Computer Networks (IT201)					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL	155		Equivalent ECTS		5.14	
Course Content						
Networking essentials, TCP/IP suite, Internet domains, Addressing, Internet infrastructure and information structure, Internet protocols, Internet hardware components, Internet access, Internet and Extranet, Video conferencing over the Internet, Mailing, Voice over IP, Multimedia communication over the Internet, Audio, Video streaming, Website design and application.						
Taught in Program(s)						
As major in Program(s) Code(s): IS, CS, and IT						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT301	Signal Processing			3	
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
The topics covered in this course aim:					
• Develop a good understanding of signals and their different types, analog-to-digital and digital-to-analog conversion processes, and digital filter design techniques. • Understand the types and structures of signal processing systems in the time domain. • Work with common analysis transforms (e.g., Fourier analysis, Laplace transform, z-Transform, difference equations, etc.) to develop an in-depth understanding of signals and systems. • Investigate techniques for efficient and real-time implementation of signal processing algorithms. • Study the algorithm's complexity to measure the viability of different implementation techniques with respect to real-time processing constraints. • Apply signal processing and real-time implementation techniques to formulate feasible alternatives. • Proposes solutions to design problems that will satisfy various design criteria. • Work with specialized engineering tools to develop DSP code, which is then used to obtain experimental results and correlate them with the theoretical understanding.					
Taught in Program(s)					
As major in Program(s) Code(s): IT					
As minor in Program(s) Code(s): CS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT302	Pattern Recognition				3
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS		5.14
Course Content					
This course covers the essentials of pattern recognition, with topics suitable for undergraduate-level courses in computer science and most engineering specialties. It provides a broad understanding of related topics, including pattern recognition systems, preprocessing and feature extraction, theories of supervised and unsupervised learning, object classification and recognition, artificial neural networks, and other topics.					
Taught in Program(s)					
As major in Program(s) Code(s): IT and NCS					
As minor in Program(s) Code(s): AI and MI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

IT303		Computer Graphics		3									
Prerequisites		Mathematics for Computer Science 3 (BS201)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
Introduction to Computer Graphics; Overview of Graphics systems; Line drawing algorithms; Circle drawing algorithms; Ellipse drawing algorithms; Area filling algorithms; Polygon filling algorithms; Line clipping algorithms; Polygon clipping algorithms; Two dimensional transformations; (translation – rotation – scaling –general transformations – composite transformations); Three dimensional object representation and Projections; Three dimensional modeling and transformations (translation – rotation – scaling – sheer – reflection – composite); Three dimensional Viewing and Camera Model.													
Taught in Program(s)													
As major in Program(s) Code(s): IT													
As minor in Program(s) Code(s): IS and CS													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

IT304		Wireless and Sensor Networks		3	
Prerequisites		Computer Networks (IT201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS	
				5.14	
Course Content					
The objective of this course is to provide an introduction to the fundamentals of wireless communication systems, wireless network architectures, protocols, and applications. Topics of study include an overview of wireless communications and mobile computing systems, signal propagation characteristics of wireless channels, wireless channel modelling, frequency reuse/cellular/microcellular concepts, spread-spectrum modulation for wireless systems, multiple access techniques, and wireless networking standards, Ad-hoc networks, MANets, Wireless sensor networks (Routing in ad-hoc networks).Wireless networking (MAC layer: Carrier sensing, CSMA/CA, RTS-CTS mechanisms, MACA, MACAW; IEEE 802.11 DCF and PCF (NAV, SIFS, DIFS, RTS/CTS, bakeoff, contention window, fragmentation, etc.); Power and rate control; Emerging technologies such as Bluetooth, WiMAX, new versions of 802.11, ZigBee).					
Taught in Program(s)					
As major in Program(s) Code(s): ITand NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT305		Microprocessor		3									
Prerequisites		Digital Logic Circuit (IT103)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
Introduction to Microprocessor; Basic microprocessor architecture; Segmented and paged memory models; Memory Read/Write bus cycles; Input/Output Bus; DMA bus cycles; Systems bus signals; Modes of operations (Real mode; Protected mode; and Virtual machine mode); Microprocessor registers; Interrupt system; DMA system; Memory addressing modes; Data movement instructions; Program control instructions; Arithmetic and logic Instructions; Building a complete system. Includes overviews of control applications using classical and fuzzy control techniques; reliability, redundancy, and multi-processor applications; selected programming techniques and issues; and ethics and professionalism in microcontroller design.													
Taught in Program(s)													
As major in Program(s) Code(s): IT													
As minor in Program(s) Code(s): ---													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

IT306	Intelligent and Quantum Computers				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course is concerned with understanding the concepts of intelligent computation, neural networks, neurocomputing, learning laws, data transformation structures, fuzzy logic, genetic algorithms, optical circuits, optical computers, and Quantum computers.						
Taught in Program(s)						
As major in Program(s) Code(s): IT						
As minor in Program(s) Code(s): AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT307		Digital Image Processing				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course include image fundamentals; image enhancement in the spatial domain: grey level transformation; Histogram processing; Spatial filters; Image enhancement in frequency domain: 2-D Fourier transform; Other transforms; Smoothing filters; Sharpening filters; Image restoration; Noise model; Estimating the degradation function; Wiener filter; Geometric transformations; Image segmentation: detection of discontinuities; edge linking and boundary detection; Thresholding; Region based segmentation; Morphological image processing: operation concepts; some basic algorithms; image compression.							
Taught in Program(s)							
As major in Program(s) Code(s): CS, IT, MI, and AI							
As minor in Program(s) Code(s): NCS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

IT308		Network Programming		3	
Prerequisites		Computer Networks (IT201)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course will cover the practical aspects of computer network programming, with emphasis on the Internet. The goal of this course is to introduce the students to the basics of computer networks and Internet programming. We will introduce students to the TCP/IP protocol stack and several of its key protocols. Students will also be introduced to multi-tier application development and RPC technologies, including: RMI, CORBA, EJB, and Web Services. We will also look at industry trends and discuss some innovative ideas that have recently been developed. Some of the course material will be drawn from the web, industry white papers, and Internet RFCs. We will introduce students to the TCP/IP protocol stack and to Sockets Programming: TCP Programming (TELNET, HTTP). UDP Sockets: TFTP, DNS, Web Programming: HTTP, CGI, Cookies, JavaScript, HTML, XML, Client/Server Programming, 3-tier architecture., Remote Method Invocation (RMI). Common Object Request Broker Architecture (CORBA). Simple Object Access Protocol (SOAP), UDDI, and Web Services. Enterprise Java Beans (EJB). Java Server Pages (JSP) and Java Server Faces (JSF). JavaMail and E-mail programming: SMTP, POP, IMAP.					
Taught in Program(s)					
As major in Program(s) Code(s): IT and NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT309		Computer and Network Security			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course present basic concepts of Computer and network security; Classical encryption techniques; Modern black ciphers and the data encryption standards; Block cipher cryptanalysis and usage; Modern stream ciphers; Number theory; Public key cryptography; Key certificates and management; Message authentication and hash functions; Hash algorithms; Digital signature and authentication protocols; Electronic mail security; IP security; Web security; Firewalls; Introduction to digital steganography and watermarking techniques.						
Taught in Program(s)						
As major in Program(s) Code(s): CS, IT, SE, MI, AI, and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT310	Speech Recognition				3
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
The Speech Recognition course provides students with a detailed look at the science of applying machine learning algorithms to process large amounts of speech data. Speech recognition is driving the growth of the AI market, and this course helps students develop the skills required to become speech recognition professionals.					
Taught in Program(s)					
As major in Program(s) Code(s): IT and AI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT311	Virtual Reality				3
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
In this course, learn all about virtual reality (VR) and the variety of virtual worlds you can experience using this technology. We will cover the history of VR, the relationship between VR and other mixed reality technologies, design considerations, communications, hands-on projects, as well as social and ethical implications of immersive technologies. While this course is focused on VR and virtual worlds, it is primarily a communication as well as a design theory course, so no prior technical experience is required.					
Taught in Program(s)					
As major in Program(s) Code(s): IT					
As minor in Program(s) Code(s): MI and AI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT312		Introduction to Accessibility and Inclusive Design		3									
Prerequisites		---											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
This course introduces some of the fundamental principles of accessibility and prepares students for further study in accessibility and inclusive design. Students will have an opportunity to explore the major types of disabilities (visual, hearing, motor, and cognitive), and related assistive technology and adaptive strategies, the most salient contours of the legal landscape, and the major principles that guide universal design and accessible content creation. The course content will highlight firsthand perspectives on disability, as well as topics like disability etiquette, universal design, and universal design for learning, accommodation in higher education, campus accessibility policy, and accessibility in a corporate setting.													
Taught in Program(s)													
As major in Program(s) Code(s): ---													
As minor in Program(s) Code(s): IT													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

IT313		Smart Mobile Systems			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course focuses on recent advances and developments in mobile sensing systems and their applications, especially those leveraging modern mobile devices and embedded sensors. Topics include conventional mote-class sensor networks, participatory sensing leveraging mobile devices, intelligent hardware and Internet-of-Things, location sensing, future information-centric networking, and applications in smart homes, buildings, transportation, environment, and health/fitness.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT and SE						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT314		Games Development			3	
Prerequisites		Fundamentals of Artificial Intelligence (CS204)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Upon completion of this course, the student should be able to define the principles of game design and development. gain programming capability to develop games. learn to structure and define the duties of the game development team. practice art creation, animations, and various tools needed to create assets. apply the mathematics and physics to game design. Apply basic artificial intelligence to developing computer games. Manage porting issues on various platforms.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT, SE, and AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT315		Computer Networks 2			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course includes data center networking, switch fabric technology, cloud data center network topologies, data center networking standards, network virtualization, storage networks, software-defined networking, and high-performance computing networks.						
Taught in Program(s)						
As major in Program(s) Code(s): --						
As minor in Program(s) Code(s): IT, IS, and NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT401	Computer Vision				3
Prerequisites	Digital Image Processing (IT307)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
This course provides an introduction to computer vision, including fundamentals of image formation, camera imaging geometry, feature detection and matching, stereo, motion estimation and tracking, image classification, and scene understanding. It will explore methods for depth recovery from stereo images, camera calibration, automated alignment, tracking, boundary detection, and recognition. The focus of the course is to develop the intuitions and mathematics of the methods in lecture, and then to learn about the difference between theory and practice in the projects.					
Taught in Program(s)					
As major in Program(s) Code(s): IT					
As minor in Program(s) Code(s): MI, AI, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT402		Robotics 1		3									
Prerequisites		Fundamentals of Artificial Intelligence (CS204)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
Robotic and Autonomous Systems are now used in areas such as autonomous vehicle guidance, health care, remote surgery, industrial manufacturing, and domestic assistance. This course is for creating and programming a robot to interact with its environment and perform basic tasks involving motion, collecting sensor data, and decision-making. The course teaches students basic methods in Artificial Intelligence, planning and search, localization, tracking, and mapping, in addition to an introduction to classical and modern control methods, including PID control, adaptive and hybrid control, reinforcement learning, cognitive robotics, modeling uncertainty, Markov decision processes, and partially observable Markov decision processes control.													
Taught in Program(s)													
As major in Program(s) Code(s): IT and AI													
As minor in Program(s) Code(s): ---													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

IT403		Multimedia		3	
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The creation of interactive multimedia products for cross-platform delivery. • Introduction to multimedia authoring and production. • The multimedia development process. • Introduction to multimedia scripting. • Types of lingo script/ behavior /handlers. • The sampling process: understanding audio/video. • Using lists and casts. • Understanding programming structures. • Graphics, audio, and movie file formats. • Databases, lists, and Shockwave. • Storage and delivery technologies. • Global development issues. • Legal issues, copyrights, taxes.					
Taught in Program(s)					
As major in Program(s) Code(s): IT					
As minor in Program(s) Code(s): IS and MI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT404		Embedded Systems		3	
Prerequisites		Digital Logic Circuit (IT103)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
Introduction to the unique characteristics and requirements of embedded systems. Modelling, design, and analysis of embedded systems. Modelling covers continuous dynamics, discrete dynamics, hybrid systems, composition of state machines, and concurrent models of computation. Design covers sensors and actuators, embedded processors, memory architecture, input and output, multitasking, and scheduling. Analysis covers invariants and temporal logic, equivalence and refinement, reachability analysis and model checking, quantitative analysis, and security and privacy. Some topics may include: - Models for embedded software: process-based, cyclic executives, OS-based & object-oriented. - Overview of specification and design techniques: Data Flow Diagrams (DFDs), State-charts, Tabular Languages, Traditional logics & Real-time logic. - Introduction to synchronous programming for control-dominated applications. Relevant QAA Computing Sections: Concurrency and Parallelism, Operating Systems, Software Engineering, Systems Analysis and Design.					
Taught in Program(s)					
As major in Program(s) Code(s): IT					
As minor in Program(s) Code(s): SE, MI, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT405	Cyber Security				3	
Prerequisites	Computer Networks (IT201)					
Number of weekly Contact Hours						
Lecture		Tutorial		Laboratory		
2		---		2		
Required SWL	155		Equivalent ECTS		5.14	
Course Content						
This course introduces various concepts such as ethical hacking, cryptography, computer networks & security, application security, IdAM (Identity & Access Management), vulnerability analysis, malware threats, sniffing, SQL injection, DoS, session hijacking, Firewalls, Security for wireless systems, and various security practices for businesses from scratch with hands-on demonstrations.						
Taught in Program(s)						
As major in Program(s) Code(s): IT and NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam	
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		
-		-		-		
				Industry-based (IB)		
				-		

IT406		Forensic Science		3							
Prerequisites		Cyber Security (IT405)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course aims to step into the role of a criminalist, the specialist who performs the scientific examination of evidence, as they process a case from start to finish. Therefore, the students will begin by learning a scientific approach to crime scene investigation, evidence collection, and transport. Then, they follow the collected evidence as it is disseminated throughout the crime lab for examination and analysis. Once the evidence has been processed, students will use critical thinking skills to interpret the evidence within the context of the case. Students will also explore the role of various scientific disciplines outside of criminalistics (anthropology, entomology, pathology, etc.) as they are used in modern forensics, as well as the role of forensic science in society and the criminal justice system.											
Taught in Program(s)											
As major in Program(s) Code(s): NCS											
As minor in Program(s) Code(s): IT											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

IT407		Selected Topics in IT-Security			3	
Prerequisites		Cyber Security (IT405)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course teaches the students the (1) symmetric and asymmetric encryption schemes, (2) classification and description of the vulnerabilities and protection mechanisms of popular network protocols, web protocols, software systems, and data, (3) analyzing the basic protection mechanisms for modern OSs, software, and hardware systems.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT408		Blockchain			3	
Prerequisites		----				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course covers the technical aspects of public distributed ledgers, blockchain systems, cryptocurrencies, and smart contracts. Students will learn how these systems are built, how to interact with them, and how to design and build secure distributed applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT, MI, AI, and NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT409		Computer Vision Applications			3	
Prerequisites		Computer Vision (IT401)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course will prepare students in both the theoretical foundations of computer vision as well as the practical approaches to building real Computer Vision systems. The course will examine data sources, features, and learning algorithms useful for understanding and manipulating visual data. Special emphasis will be placed on autonomous vehicles. The goal of this course is to give students the background and skills necessary to work in the field of computer vision and its application such as robotics, healthcare, and graphics.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT and AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT410		Internet of Things		3	
Prerequisites		Digital Logic Circuit (IT103)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The objective of this course is to understand the fundamentals of the Internet of Things (IoT) and Embedded Systems. This course covers the following topics: • Definition of an Embedded System and its characteristics. • Real-time systems and tasks. • The basics of processors and the difference between the General Processors in computers and the Embedded Processors. • The basics of Microcontrollers. • The concept of System-On-Chip and its examples. • The various components of the Embedded Systems. • The current components of typical IoT devices. • IoT design considerations, constraints, and interfacing between the physical world and the device. • How to make design trade-offs between hardware and software. • The key components of networking are to ensure that students understand how to connect their device to the Internet.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IS, CS, IT, MI, AI, and NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT411		Medical Image Analysis			3	
Prerequisites		Digital Image Processing (IT307)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course gives an overview of medical image formation, enhancement, analysis, visualization, and communication with many examples from medical applications. It starts with a brief introduction to medical imaging modalities and acquisition systems. Basic approaches to display one-, two-, and three-dimensional (3D) biomedical data are introduced. As a focus, image enhancement techniques, segmentation, texture analysis, and their application in diagnostic imaging will be discussed.						
Taught in Program(s)						
As major in Program(s) Code(s): MI						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT412	Cloud Computing				3
Prerequisites	---				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155		Equivalent ECTS		5.14
Course Content					
The course presents a top-down view of cloud computing, from applications and administration to programming and infrastructure. Its main focus is on parallel programming techniques for cloud computing and large-scale distributed systems, which form the cloud infrastructure. The topics include an overview of cloud computing, cloud systems, parallel processing in the cloud, distributed storage systems, virtualization, security in the cloud, and multicore operating systems.					
Taught in Program(s)					
As major in Program(s) Code(s): NCS					
As minor in Program(s) Code(s): IT, SE, and MI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT413	Robotics 2				3
Prerequisites	Robotics 1 (IT402)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
The course provides an overview of mathematical models and algorithms behind state-of-the-art robotic systems, especially in service robotics. The course focuses on manipulation, Locomotion, Kinematics, Inverse kinematics, optimal decision making, and algorithms for classical and modern control, and reinforcement learning, which are central components in more general-purpose robotic systems. In particular, mathematical methods and models that help cope with the inherent uncertainty of the environment are considered.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IT and AI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT414		Multimedia Communication		3	
Prerequisites		Multimedia (IT403)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
The course introduces students to multimedia communication systems and applications. The curriculum includes background knowledge in the areas of design and development of multimedia communication systems (digitizing, encoding, compression, transmission, analysis, and mining of multimedia content) and the corresponding Internet technologies for streaming and Quality of Service for real-time multimedia communications. During the course, case studies will be presented, and there will be projects assigned to students. Students will be able to: • Understand basic methodologies of design and development of multimedia communication and management systems. • Know the stages of creating and editing multimedia content (digitization, editing, coding, compression, transmission, analysis, and retrieval of multimedia information). • Choose the best Internet and streaming technologies and quality of service mechanisms for real-time online multimedia systems.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): IT and IS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

IT415		Mobile Computing			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course introduces the mobile computing domain with a strong emphasis on application development and a basic understanding of the role of mobile cloud computing in the context of everyday living. This course will cover: Introduction to mobile computing, Mobile technology and terminology, Location-based services, Mobile learning, Mobile health, Constraints and challenges in the mobile domain, cellular networks, Mobile Web vs native applications, Developing mobile applications, Mobile system architecture, Mobile information architecture, Mobile UI and usability, embedded sensor systems, the energy efficiency of mobile devices, wearable and vehicular mobile systems, mobile security, virtual reality and augmented reality, trends in mobile computing.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT416		System-on-Chip (SoC) Programming			3	
Prerequisites		Introduction to Programming and Problem Solving (CS101)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course presents hardware and software aspects of (1) integrating heterogeneous components into a complete system; (2) evaluating designs in a multi-objective optimization space; and (3) designing new components that are reusable across different systems, product generations, and implementation platforms. The course presents the technology and economic trends, design and programming methodologies and supporting CAD tools for system-level design and verification, software simulation and virtual platforms, FPGA prototyping, models of computation, the System C language, transaction-level modeling, hardware-software partitioning, high-level synthesis, memory organization, device drivers, on-chip communication architectures, power management and optimization, integration of programmable cores and specialized accelerators—case studies of modern SoC platforms for various classes of applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT and MI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT417		Planning and Designing of Information Networks			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Concepts of planning and design of information networks; Planning and design standards; Integrated planning and design approach; Planning and design tools; Computer aided Network planning; Computer-aided network design; design forms and specifications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT418	Computer Animation				3	
Prerequisites	Computer Graphics (IT303)					
Number of weekly Contact Hours						
Lecture		Tutorial		Laboratory		
2		---		2		
Required SWL	155		Equivalent ECTS		5.14	
Course Content						
3D Modeling; Rendering Techniques; Key framing; Interpolations; Hierarchical animation; Camera animation; Light animation; Special effects; Digital animation techniques; Recording and production planning.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT419		Computer Graphics – 2			3	
Prerequisites		Computer Graphics (IT303)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Visible surface detection algorithms; Reflection and illumination models; Rendering algorithms for 3-D objects; Parametric representation of 3-D objects; Shadows algorithms; 2-D texture mapping; 3-D texture mapping; Ray tracing; Volume rendering; Anti-Aliasing; Introduction to fractals; 3-D computer animation; Color Space in Computer Graphics.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT420		E-Learning			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Development and deployment of successful e-learning systems; the advantages as well as the possible pitfalls of e-learning; Different means of delivering e-learning and the advantages and disadvantages associated with each; intelligent tutoring systems; adaptive hypermedia; and user modeling; evaluation and maintenance of an e-learning system.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT421		Digital Libraries				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
Objectives of digital library; Services provided; Components of digital library; Storage requirements and specifications; Digital library software; Building digital library; Tools for designing digital libraries.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): IT							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

IT422		Computer Interfaces			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Review of Logic design basics (Flip-Flops, decoders, multiplexers, and counters). Opto-couplers and opto-isolators; Relays and driving circuits; Introduction to stepper and servo motors; Analog components for computer interfacing; Signal conditioning; Memory Read/Write bus cycles; Input/Output Bus power characteristics; Hardware interrupts and interrupt controllers; Direct memory access; Addressing of memory I/O; Timers and counters; RS 232 Interface; USB Interface; Introduction to assembly language; Microprocessor basic instruction set.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT423		Digital Image Processing – 2			3	
Prerequisites		Digital Image Processing (IT307)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Digitizing images; Color image processing; Image compression; Wavelet and multiresolution processing; Representation and description; Chain codes; Signatures; boundary descriptors; regional descriptors; Image classification; statistical-based classification; Verification of image classification – 3D Imaging – Application Field.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT and MI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT424	Fault Tolerant Computer Systems				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial		Laboratory		
2		---		2		
Required SWL	155		Equivalent ECTS		5.14	
Course Content						
Systems and their failures; Errors and failures; computer structure and faults; concepts of redundancy; Damage assessment and confinement; Error recovery and faults diagnosis; Reconfigurable computer systems; Design of fault-tolerant computer systems.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam	
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-	-		-		-	

IT425		Automatic Control			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Mathematical modeling of dynamic systems; State variable analysis; Stability of linear control systems; Time domain analysis of control systems; Design of control systems; Digital control systems; Process control; Distributed computer control systems; Applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT426	Selected Topics in Information Technology				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course intends to introduce special topics regarding current trends in information technology. Such possible topics include: metaverse, Holography Technology, and Digital Marketing.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

IT427	Selected Topics in Information Technology - 2				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course intends to introduce special topics regarding current trends in information technology.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): IT						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

SE201		Business Process Management				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course delves into the systematic approach to making an organization's workflow more effective, efficient, and adaptable to changing environments. Students will learn to model, analyze, design, implement, and monitor business processes, aligning them with organizational goals. Topics include process mapping, workflow analysis, business process modeling (using BPMN), and the use of business process management systems (BPMS) to optimize processes and drive continuous improvement.							
Taught in Program(s)							
As major in Program(s) Code(s): SE							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

SE301		Software Design and Architecture				3	
Prerequisites		Software Engineering (CS203)					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course covers advanced concepts in designing and architecting robust software systems. Students will explore various architectural patterns, design principles, and best practices for creating scalable and maintainable software. Topics include architectural styles (e.g., microservices, layered architecture), design patterns (e.g., Singleton, Factory), architectural documentation, and using UML for design modeling. The course emphasizes the impact of design choices on system attributes like performance, security, and maintainability.							
Taught in Program(s)							
As major in Program(s) Code(s): SE							
As minor in Program(s) Code(s):---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

SE302		Software Requirement Analysis			3	
Prerequisites		Software Engineering (CS203)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course focuses on advanced techniques for eliciting, analyzing, specifying, and managing software requirements. Students will learn to work with stakeholders to gather detailed requirements, create use cases, and develop functional and non-functional requirements. Topics include requirements elicitation methods (e.g., interviews, surveys), requirements modeling (e.g., use cases, user stories), requirements validation and verification, and managing requirements change.						
Taught in Program(s)						
As major in Program(s) Code(s): SE						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

SE401		Software Construction			3	
Prerequisites		Software Design and Architecture (SE301)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course emphasizes the practical aspects of software development, focusing on the construction phase of the software lifecycle. Students will learn best practices for writing high-quality code, integrating components, and ensuring software reliability. Key topics include advanced coding techniques, automated testing, debugging strategies, refactoring, version control, and continuous integration and delivery (CI/CD) pipelines.						
Taught in Program(s)						
As major in Program(s) Code(s): SE						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

SE402		Cost Estimation of Software Projects' Development and Maintenance		3							
Prerequisites		Software Design and Architecture(SE301)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course provides comprehensive training on the techniques and methodologies used to estimate the costs associated with software development and maintenance projects. Students will learn how to predict resource requirements, timeframes, and financial costs for various stages of software projects. Key topics include estimation models (e.g., COCOMO, function point analysis), expert judgment techniques, parametric estimation, and the use of software tools for cost estimation. The course also covers risk management in estimation, dealing with uncertainty, and strategies for refining estimates throughout the project lifecycle. Emphasis is placed on practical application through case studies and real-world project scenarios.											
Taught in Program(s)											
As major in Program(s) Code(s): SE											
As minor in Program(s) Code(s): ---											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

SE403		Open-Source Software Development				3	
Prerequisites		Software Design and Architecture (SE301)					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course introduces the principles and practices of contributing to and managing open-source software projects. Students will learn about open-source licenses, community engagement, and collaborative development practices. Topics include version control with Git, code review processes, issue tracking, and best practices for contributing to open-source projects. The course also covers strategies for building and maintaining successful open-source communities and projects.							
Taught in Program(s)							
As major in Program(s) Code(s): SE							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

SE404		Software Reengineering			3	
Prerequisites		Software Construction (SE401)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course focuses on techniques for analyzing and improving existing software systems. Students will learn methods for reverse engineering, code refactoring, and system migration to enhance software performance and maintainability. Topics include legacy system analysis, software metrics, automated reengineering tools, and strategies for managing reengineering projects. The course emphasizes minimizing disruption and maintaining continuity during the reengineering process.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): SE						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

SE405		Global Software Development				3	
Prerequisites		Software Engineering (CS203), Distributed Systems (CS306)					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course explores the strategies and challenges of developing software in globally distributed teams. Students will learn about communication, coordination, and collaboration in a global context, and how to effectively manage distributed software projects. Key topics include distributed agile methodologies, cross-cultural communication, time zone management, global project management tools, and version control in distributed environments. The course also addresses risk management and quality assurance practices in global software development.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): SE							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

SE406		Selected Topics in Software Engineering -1				3	
Prerequisites							
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course aims to explore novel topics in Software Engineering and its applications.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): SE							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

SE407		Selected Topics in Software Engineering -2			3	
Prerequisites		---				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Software Engineering and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): SE						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

MI201		Genetics		3							
Prerequisites		Fundamentals of Biology (UN202)									
Number of weekly Contact Hours											
Lecture		Tutorial		Laboratory							
2		---		2							
Required SWL		155		Equivalent ECTS		5.14					
Course Content											
This course provides a comprehensive overview of genetics with a focus on its applications in medical informatics. Students will explore the principles of heredity, genetic variation, and the molecular basis of genetic diseases. Key topics include DNA structure and function, gene expression, genetic mapping, and modern techniques in genomics and bioinformatics. The course emphasizes the role of genetic information in personalized medicine, the ethical considerations of genetic testing, and the integration of genetic data into healthcare informatics systems.											
Taught in Program(s)											
As major in Program(s) Code(s): MI											
As minor in Program(s) Code(s): ---											
Targeted Sustainable Development Goals:											
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights											
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)			
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment ☒ Exam	
20				15				5		60	
Student Activities Breakdown (Distribution for Study Blocks)											
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)		
-			-			-			-		

MI301		E-Health Record			3	
Prerequisites		Introduction to Database Systems (IS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course delves into the design, implementation, and management of electronic health records (EHR) systems. Students will learn about the standards and regulations governing EHRs, data interoperability, and patient privacy and security. Key topics include EHR system architecture, health information exchange (HIE), clinical data standards (e.g., HL7, FHIR), and the impact of EHRs on clinical workflows and patient care. The course also covers strategies for successful EHR adoption and optimization in healthcare settings.						
Taught in Program(s)						
As major in Program(s) Code(s): MI						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

MI302		Hospital Management Systems				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course focuses on the development and implementation of integrated hospital management systems that support administrative, clinical, and financial operations. Students will learn about the components of hospital management systems, including patient administration, billing, scheduling, and resource management. Key topics include system integration, data analytics, decision support systems, and the use of information technology to improve hospital efficiency and patient care quality. The course emphasizes real-world case studies and the practical challenges of managing hospital information systems.							
Taught in Program(s)							
As major in Program(s) Code(s): MI							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

MI303		Health Information Systems			3
Prerequisites		Healthcare Informatics (CS323)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course provides an in-depth exploration of health information systems (HIS) and their role in supporting healthcare delivery and management. Students will learn about the architecture and components of HIS, including patient records, laboratory information systems, radiology information systems, and public health information systems. Key topics include data standards, interoperability, health informatics infrastructure, and the use of HIS for decision support, clinical research, and public health surveillance. The course also covers strategies for ensuring data quality and security in health information systems.					
Taught in Program(s)					
As major in Program(s) Code(s): MI					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

MI304		Quality Management in Hospitals				3	
Prerequisites		Hospital Management Systems					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course covers the principles and practices of quality management in healthcare, focusing on improving patient outcomes and operational efficiency in hospitals. Students will learn about quality improvement methodologies, performance measurement, and regulatory requirements. Key topics include the use of data and analytics for quality monitoring, patient safety initiatives, accreditation standards, and the role of healthcare information systems in supporting quality management. The course emphasizes the implementation of continuous quality improvement (CQI) programs and the use of evidence-based practices in hospital settings.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): MI							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

MI305		Telemedicine and Mobile Health		3			
Prerequisites		---					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		---		2			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course explores the technologies and strategies for delivering healthcare services remotely through telemedicine and mobile health (mHealth) solutions. Students will learn about the design, implementation, and evaluation of telemedicine systems, as well as the use of mobile devices for health monitoring and patient engagement. Key topics include telehealth platforms, remote patient monitoring, mobile health applications, regulatory and reimbursement issues, and the impact of telemedicine on healthcare access and outcomes. The course also covers the integration of telemedicine and mHealth into existing healthcare infrastructures and the challenges of ensuring data security and patient privacy.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): MI							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

MI401	Selected Topics in Medical Informatics -1				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Medical Informatics and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): MI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

MI402		Selected Topics in Medical Informatics -2			3
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course aims to explore novel topics in Medical Informatics and its applications.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): MI					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

AI401		Selected Topics in Artificial Intelligence -1				3	
Prerequisites		---					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course aims to explore novel topics in Artificial Intelligence and its applications.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): AI							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)		Oral Exam (OE)		Final (FE)	
		☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20			15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

AI402	Selected Topics in Artificial Intelligence -2				3	
Prerequisites	---					
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Artificial Intelligence and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): AI						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS301	Cryptography				3
Prerequisites	Cyber Security (IT405), Mathematics 3 for Computer Science (BS201)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
Introduction to Cryptology – Elementary number theory – Group Theory – Fields and Finite Fields – Polynomial finite fields and Galois fields – Discrete Logarithm Problem and Diffie-Hellman Key Exchange – Integer Factorization and RSA – Stream and block Ciphers – Public key cryptography – Digital Signatures – Hash functions and message authentication codes – Key establishment.					
Taught in Program(s)					
As major in Program(s) Code(s): NCS					
As minor in Program(s) Code(s): ---					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS302		Operating Systems Security			3	
Prerequisites		Operating Systems (CS202)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
The course covers fundamental concepts and mechanisms for enforcing security and defining secure operating systems, defining an ideal secure operating system, the first OS designed for security goals, systems development approaches applied to build a secure operating system, access control fundamentals, formal security goals and corresponding security models proposed for secure operating systems, security kernels, secure virtual machine systems, surveying a variety of approaches applied to the development of secure operating systems, system assurance methodologies.						
Taught in Program(s)						
As major in Program(s) Code(s): NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS303		Secure Software Development		3			
Prerequisites		Object-Oriented Design and Programming (CS102)					
Number of weekly Contact Hours							
Lecture		Tutorial		Laboratory			
2		---		2			
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course explores the implementation of security controls within web applications, mobile applications, utility applications, and traditional applications. Students will explore secure coding and testing techniques as well as application security configuration techniques. Specific review of secure coding techniques will include Data Validation, Session Management, Exception Handling, and Data Encryption. Specific review of application security configuration techniques will include the secure configuration management of the application web server, middleware, and database. Students will also review policy-specific requirements necessary to implement a secure development program within enterprise organizations.							
Taught in Program(s)							
As major in Program(s) Code(s): NCS							
As minor in Program(s) Code(s): ---							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

NCS304		Security of Distributed Systems			3	
Prerequisites		Wireless and Sensor Networks (IT304), Cyber Security (IT405)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course focuses on a broad range of topics related to securing distributed systems. Students will gain an understanding of the principles and techniques behind the design of distributed systems along with their associated cybersecurity issues and remediation techniques. Cloud computing and the IoT will be some of the major types of distributed environments examined.						
Taught in Program(s)						
As major in Program(s) Code(s): NCS						
As minor in Program(s) Code(s): ---						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS305	Social Engineering				3
Prerequisites	Cyber Security (IT405)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
This course introduces the psychological principles that underlie social engineering, some of the tools commonly used in social engineering attacks, and how to prevent and mitigate social engineering attacks. Therefore, the students will be able to use open-source intelligence techniques to gather information, understand the principles necessary for pretexting and impersonation, plan a social engineering attack, and recommend steps to prevent social engineering attacks.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS306		Information Security Management				3	
Prerequisites		Cyber Security (IT405)					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
Information security management is an organization’s approach to ensuring the confidentiality, availability, and integrity of IT assets and safeguarding them from cyberattacks. A Chief Information Security Officer, IT Operations Manager, or Chief Technical Officer, whose team comprises Security Analysts and IT Operators, may carry out the tasks involved in information security.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): NCS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

NCS307		Social Networks Computing		3	
Prerequisites		Computer networks (IT201), Cryptography (NCS301)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course introduces the fundamental concepts, models, and technologies underlying social networks from a computing and cybersecurity perspective. Students study the structure and dynamics of social networks, including graph-based representations, network metrics, community detection, and information diffusion. The course emphasizes data collection and analysis of social network platforms, privacy preservation, and security challenges such as fake accounts, misinformation, social engineering, and cybercrime propagation. Practical components include hands-on analysis using real-world social network datasets and tools to assess vulnerabilities, detect malicious behavior, and design secure and privacy-aware social networking systems.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS308		Human Security			3	
Prerequisites		Cyber Security (IT405)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Focuses on protecting individuals’ data and privacy in the context of organizations (i.e., as employees) and personal life, in addition to the study of human behavior as it relates to cybersecurity. Humans have a responsibility to ensure the confidentiality, integrity, and availability (CIA) of their organizational and personal computer systems.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS309	Cyber Law and Cyber Security				3	
Prerequisites	Cyber Security (IT405)					
Number of weekly Contact Hours						
Lecture		Tutorial		Laboratory		
2		---		2		
Required SWL	155		Equivalent ECTS		5.14	
Course Content						
This course provides students with an understanding of the development of law and policy related to cyberspace and cybersecurity. It examines social and legal cyber governance rules, processes, and structures with an emphasis on Egyptian and international laws related to cybersecurity, data security, law enforcement surveillance, cybercrime, privacy, and related issues.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam	
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-	-		-		-	

NCS310	Grid Computing				3
Prerequisites	Computer Networks (IT201)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
Network computing: internet computing, cluster, grid, meta computing, and middleware computing – grid monitoring: tasks, types, architecture, components –web and grid services – grid information system – grid resource management – grid scheduling – grid security system – grid-enabled workflow system – semantic grid – knowledge grid.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS311		Proactive Computer Security		3									
Prerequisites		Object-Oriented Design and Programming (CS102), Cyber Security (IT405)											
Number of weekly Contact Hours													
Lecture		Tutorial		Laboratory									
2		---		2									
Required SWL		155		Equivalent ECTS		5.14							
Course Content													
This course teaches how to proactively test what the student has put in place to protect the data. Be able to discuss the basics of deterrents and how to “trick” attackers into believing they have hit a goldmine of data away from your real systems. Be able to understand and discuss the steps of penetration testing methodology. Be able to understand and apply what the student has learned about the systems to test whether the systems are secure or not. Discuss planning for the methodology that the student can apply to systems. Finish up with a project that will allow the student to test skills in a safe environment.													
Taught in Program(s)													
As major in Program(s) Code(s): ---													
As minor in Program(s) Code(s): NCS													
Targeted Sustainable Development Goals:													
☐ GOAL 1: No Poverty		☐ GOAL 2: Zero Hunger		☐ GOAL 3: Good Health and Well-being		☒ GOAL 4: Quality Education		☐ GOAL 5: Gender Equality		☐ GOAL 6: Clean Water and Sanitation			
☐ GOAL 7: Affordable and Clean Energy		☒ GOAL 8: Decent Work and Economic Growth		☒ GOAL 9: Industry, Innovation and Infrastructure		☒ GOAL 10: Reduced Inequality		GOAL 11: ☒ Sustainable Cities and Communities		☐ GOAL 12: Responsible Consumption and Production			
☐ GOAL 13: Climate Action		☐ GOAL 14: Life Below Water		GOAL 15: ☐ Life on Land		☒ GOAL 16: Peace and Justice Strong Institutions		☒ GOAL 17: Partnerships to achieve the Goal					
Assessment Weights													
Student Activities (SA) / Practical Exam (PE)		Midterm (MT)			Oral Exam (OE)			Final (FE)					
		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam		☐ Assessment		☒ Exam	
20				15				5				60	
Student Activities Breakdown (Distribution for Study Blocks)													
Seminar-based (SB)			Self-learning-based (SL)			Project-based (PB)			Industry-based (IB)				
-			-			-			-				

NCS312		Software Security Engineering		3	
Prerequisites		Software Engineering (CS203), Secure Software Development (NCS303)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155		Equivalent ECTS 5.14	
Course Content					
This course mainly deals with engineering techniques for developing secure software systems. It covers Security fundamentals of Information assurance concepts (confidentiality, integrity, and availability); Nature of threats (e.g., natural, intentional, and accidental); Encryption, digital signatures, message authentication, and hash functions; Common cryptographic protocols (applications, strengths, and weaknesses) relevant to buffer overflows, SQL injections, format string bugs, integer overflows, heap attacks, race conditions, concurrency vulnerabilities, access control, and malicious code; and Nontechnical security issues (e.g., social engineering). Furthermore, the course focuses on developing secure software by covering topics such as building security into the software development life cycle, security in requirements analysis and specification, secure design principles and patterns, and secure software construction techniques.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS313		Advanced Cryptography		3	
Prerequisites		Cryptography (NCS301)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
As a next level of Cryptography (NCS301) course, this course explores more complex primitives and protocols, which typically combine cryptography with some form of general-purpose computation, like zero-knowledge proof systems, functional encryption, forms of verifiable computation, secure two-party and multi-party computation, and fully homomorphic encryption.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS314		Cyber Security for Internet of Things			3
Prerequisites		Internet of Things (IT410), Cyber Security (IT405)			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course will introduce tools and methodologies for cyber security of IoT systems. The topics covered include the basics of cybersecurity for IoT systems, threats and vulnerabilities, tools and techniques for detecting attacks, and mitigation strategies. In addition to the fundamental concepts, students will be exposed to hands-on training for analyzing IoT and cyber-physical system data for detecting cyber-threats and attacks.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

NCS315		Incident Response, Disaster Recovery, and Business Continuity Systems			3
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course is designed to provide the student with an understanding of the concepts and practices of contingency operations, including the administration of the planning process for incident response, disaster recovery, and business continuity planning.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS316	Selected Topics in Cyber Security - 1				3
Prerequisites	Cyber Security (IT405)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course aims to explore novel topics in Cyber Security and its applications.					
Taught in Program(s)					
As major in Program(s) Code(s): NCS					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

NCS317		Selected Topics in Cyber Security -2				3	
Prerequisites		Cyber Security (IT405)					
Number of weekly Contact Hours							
Lecture			Tutorial			Laboratory	
2			---			2	
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
This course aims to explore novel topics in Cyber Security and its applications.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): NCS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

NCS318		Selected Topics in Network Technology-1			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Network Technology and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS319		Selected Topics in Network Technology-2			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course aims to explore novel topics in Network Technology and its applications.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS401		Wireless Networks and Mobile Security			3	
Prerequisites		Computer Networks (IT201)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course explores the impediments of the mobile and wireless environments, problems and limitations due to such impediments, various network layers solutions, location management techniques, mobile IP, wireless TCP, wireless LANs, 802.16/WIMAX, Wireless Mesh Networks, ad-hoc networks, routing and power optimization, performance, and mobile security issues.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS402		Data Integrity and Authentication			3
Prerequisites		---			
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL		155	Equivalent ECTS		5.14
Course Content					
This course focuses on how data can be authenticated, including ensuring data integrity, using various cryptographic primitives, especially hash functions and Message Authentication Codes (MACs). The course will cover the types of attacks related to breaching data integrity and authentication during information transferring. In turn, different mechanisms that are available to prevent/detect such attacks will be discussed.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)	
-		-		-	
				Industry-based (IB)	
				-	

NCS403	Computer Security and Privacy				3
Prerequisites	Cryptography (NCS301)				
Number of weekly Contact Hours					
Lecture		Tutorial		Laboratory	
2		---		2	
Required SWL	155	Equivalent ECTS		5.14	
Course Content					
This course examines the fundamentals of computer security including human factors; attack detection, measurements, and models; cryptography and communications security; system design and implementation; and side channels.					
Taught in Program(s)					
As major in Program(s) Code(s): ---					
As minor in Program(s) Code(s): NCS					
Targeted Sustainable Development Goals:					
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal	
Assessment Weights					
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment ☒ Exam
20		15		5	60
Student Activities Breakdown (Distribution for Study Blocks)					
Seminar-based (SB)	Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-	-		-		-

NCS404		Cryptanalysis			3	
Prerequisites		Cryptography (NCS301)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course helps to understand how cryptanalysis is used to break cryptography. The course covers topics such as: introduction to attacks using human-computable crypto, factoring attacks, Attacks on RSA, discrete logarithm attacks, symmetric system constructs, generic attacks, linear cryptanalysis, differential cryptanalysis, and integral cryptanalysis on reduced AES.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS405		Cyber Security Risk Management			3	
Prerequisites		Cyber Security (IT405)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course will provide students with a comprehensive understanding of the foundational principles, standards, and best practices related to cybersecurity, as well as the operational aspects of managing and protecting information systems and networks. This course aims to equip students with the knowledge and skills needed to effectively secure and manage cyber environments, and to understand the importance of adhering to established cybersecurity standards and practices.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS406		Cloud Computing Security				3	
Prerequisites		Cloud Computing (IT412), Cryptography (NCS301)					
Number of weekly Contact Hours							
Lecture		Tutorial			Laboratory		
2		---			2		
Required SWL		155		Equivalent ECTS		5.14	
Course Content							
The course aims to address security issues in cloud computing to ensure a secure and efficient cloud environment for dynamic business environments. The discussion includes common cloud computing threats and attacks, analysis of security architecture in various cloud services models and deployment models, various software-based security tools to monitor and protect the flow of information into and out of the cloud, identity and access management, patch management, data security controls and requirements, etc.							
Taught in Program(s)							
As major in Program(s) Code(s): ---							
As minor in Program(s) Code(s): NCS							
Targeted Sustainable Development Goals:							
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation		
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production		
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal			
Assessment Weights							
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)		
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	
20		15		5	60		
Student Activities Breakdown (Distribution for Study Blocks)							
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)	
-		-		-		-	

NCS407		Social Networks Analytics			3	
Prerequisites		Social Network Computing (NCS307)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Discuss how social networks concepts, theories, and visual-analytic methods are being used to map, measure, understand and design a wide range of phenomena such as social networking sites (e.g., Facebook, Twitter), recommender systems (e.g., Amazon, Netflix, Pandora), trust and reputation systems (e.g., eBay, Epinions, Slashdot), search engines (e.g., Google, Bing), P2P file-sharing (e.g., BitTorrent), user-generated content (e.g., Flickr, Wikipedia, Yelp), social news sites (e.g. reddit) and virtual worlds (e.g., Second Life, EverQuest, World of Warcraft).						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS408		Enterprise Server Management and Security			3	
Prerequisites		Computer Security and Privacy (NCS403)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
Manage local and remote users and security groups – Configure network interfaces and routing for security – Installation and configuration of DHCP service and cache – only DNS service – Installation and configuration of FTPS and HTTPS services – Installation and configuration of MAIL service and authentication mechanisms – Linux router/firewall & installation and configuration of a network – Configure advanced security mechanisms based on SELinux – Introduction to cloud systems – OpenStack and cloud security.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-

NCS409		Reverse-Engineering Malware: Malware Analysis Tools and Techniques			3	
Prerequisites		Cyber Security (IT405)				
Number of weekly Contact Hours						
Lecture		Tutorial			Laboratory	
2		---			2	
Required SWL		155		Equivalent ECTS		5.14
Course Content						
This course provides a strong foundation in reverse-engineering malicious software using various system and network monitoring utilities, disassemblers, debuggers, and other freely available tools. It also aims to build essential malware analysis skills crucial for deriving threat intelligence, responding to cybersecurity incidents, and enhancing enterprise defenses.						
Taught in Program(s)						
As major in Program(s) Code(s): ---						
As minor in Program(s) Code(s): NCS						
Targeted Sustainable Development Goals:						
☐ GOAL 1: No Poverty	☐ GOAL 2: Zero Hunger	☐ GOAL 3: Good Health and Well-being	☒ GOAL 4: Quality Education	☐ GOAL 5: Gender Equality	☐ GOAL 6: Clean Water and Sanitation	
☐ GOAL 7: Affordable and Clean Energy	☒ GOAL 8: Decent Work and Economic Growth	☒ GOAL 9: Industry, Innovation and Infrastructure	☒ GOAL 10: Reduced Inequality	GOAL 11: ☒ Sustainable Cities and Communities	☐ GOAL 12: Responsible Consumption and Production	
☐ GOAL 13: Climate Action	☐ GOAL 14: Life Below Water	GOAL 15: ☐ Life on Land	☒ GOAL 16: Peace and Justice Strong Institutions	☒ GOAL 17: Partnerships to achieve the Goal		
Assessment Weights						
Student Activities (SA) / Practical Exam (PE)	Midterm (MT)		Oral Exam (OE)		Final (FE)	
	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam	☐ Assessment	☒ Exam
20		15		5	60	
Student Activities Breakdown (Distribution for Study Blocks)						
Seminar-based (SB)		Self-learning-based (SL)		Project-based (PB)		Industry-based (IB)
-		-		-		-